

Islamic Republic of Pakistan

Punjab Clean Air Program (P508222)

**Environmental and Social Systems Assessment
(ESSA)**

March 3, 2025

Document of the World Bank



Contents

Abbreviation and Acronyms	v
EXECUTIVE SUMMARY	vii
1. INTRODUCTION	1
1.1. Environmental and Social Systems Assessment: Purpose and Objectives.....	1
1.1.1. Exclusions.....	2
1.2. ESSA Methodology.....	3
2. PROGRAM DESCRIPTION.....	5
2.1. Program Context.....	5
2.2. Relationship to Country Partnership Strategy	7
2.3. The Program.....	7
2.4. Program Institutional and Implementation Arrangements.....	11
2.5. Program DLIs	11
3. ENVIRONMENTAL AND SOCIAL EFFECTS OF THE PROPOSED PROGRAM	13
3.1. Environmental Benefits and Risks	13
3.1.1. RA - 1 (AQM governance and awareness raising) – Environmental Effects	13
3.1.2. RA – 2 (Transport sector abatement measures) – Environmental Effects	14
3.1.3. RA – 3 (Agriculture Sector Abatement Measures) – Environmental Effects ...	16
3.2. Social Benefits and Risks	23
3.2.1. RA-1 (AQM governance and awareness raising) – Social Effects	23
3.2.2. RA – 2 (Transport sector abatement measures) – Social Effects.....	24
3.2.3. RA – 3 (Agriculture Sector Abatement Measures) – Social Effects	25
4. POLICY AND LEGAL FRAMEWORK	34
4.1. Environmental Legislation:	34
4.2. Social Legislation:.....	34
5. COMPARATIVE ANALYSIS OF BORROWER ENVIRONMENTAL AND SOCIAL MANAGEMENT SYSTEMS AND BANK POLICY CORE PRINCIPLES	36
5.1. Introduction	36
5.2. Environmental Systems Capacity Assessment.....	37
5.2.1. Core Principle 1 – Environmental and Social Management	38
5.2.2. Core Principle 2 – Natural Habitats and Physical Cultural Resources	41
5.2.3. Core Principle 3 – Public and Worker Safety	42
5.3. Recommendations to Strengthen Environmental Management Systems	54
5.4. Social Management Systems Capacity Assessment	54

5.4.1.	Core Principle 1 – Environmental and Social Management	55
5.4.2.	Core Principle 3 – Workers and Public Safety.....	56
5.4.3.	Core Principle 4 – Land Acquisition	57
5.4.4.	Core Principle 5 – Indigenous Peoples and Vulnerable Groups	59
5.4.5.	Core Principle 6– Social Conflict	60
6.	DISCLOSURE AND CONSULTATION	71
6.1.	Disclosure.....	71
6.2.	Stakeholder Consultations	71
6.3.	Summary of the Multi-Stakeholder Workshop	72
7.	RECOMMENDATIONS AND PROPOSED ACTIONS	76
7.1.	Introduction	76
7.2.	Recommendations to be Included in the Program Action Plan	77
ANNEX 1: POLICY AND LEGAL FRAMEWORK FOR MANAGING ENVIRONMENTAL AND SOCIAL IMPACTS OF THE PROGRAM		83
ANNEX-2: E-WASTE MANAGEMENT IN PAKISTAN AND THE GLOBAL E-WASTE MANAGEMENT OPTIONS.....		113
ANNEX-3 CAPACITY ASSESSMENT FOR MANAGING ENVIRONMENTAL AND SOCIAL EFFECTS		115
ANNEX 4: PROCEEDINGS OF THE CONSULTATION WORKSHOP		132
ANNEX 5: PROPOSED/TENTATIVELY IDENTIFIED PRIVATE LANDS FOR BUS DEPOTS		139

List of Tables

Table 1: Program DLIs

Table 2: DLIs and Responsible Agencies

Table 3: Assessment of Environmental Benefits and Risks for PCAP

Table 4: Assessment of Social Effects and Risks for PCAP

Table 5: Environmental Systems Capacity Assessment – Core Principles 1, 2 and 3

Table 6: Social Systems Capacity Assessment – Core Principles 1, 3, 4, 5 and 6

Table 7: Recommended Input for Program Action Plan (PAP)

Abbreviation and Acronyms

µg/m³	Microgram per cubic meter
AD	Agriculture Department
IAs	Implementing Agencies
ADB	Asian Development Bank
AQM	Air Quality Management
ASI	Avoid-Shift-Improve
Bop	Bank of Punjab
CCDR	Country Climate and Development Report
CEMP	Continuous Emission monitoring program
CO₂e	Carbon Dioxide Gas Equivalent
CPS	Country Partnership Strategy
DLIs	Disbursement Linked Indicators
DLRs	Disbursement Linked Results
E-2W	Electric Two Wheelers
E-3W	Electric Three Wheelers
e-buses	Electric Buses
EFF	Extended Fund Facility
EIA	Environmental Impact Assessment
EPA	Environmental Protection Agency
EPCCD	Environmental Protection and Climate Change Department
E&S	Environmental and Social
ESRI	Eco System Restoration Initiative
ESSA	Environmental and Social Systems Assessment
EV	Electric Vehicles
FY	Fiscal Year
GDP	Gross Domestic Product
GHG	Greenhouse Gas
Gop	Government of Punjab
GRM	Grievance Redress Mechanism
GST	Goods and Services Tax
HFC	Hydro Floro Carbon
ICISD	Industries, Commerce, Investment & Skills Department
ICS	Improved Cookstoves
IEE	Initial Environmental Examination
IGP-HF	Indo-Gangetic Plain and Himalayan Foothills
IMF	International Monetary Fund
IoT	Internet of Things
IPF	Investment Project Financing
ITS	Intelligent Transportation System
LFS	Life and Fire Safety
MCC	Ministry of Climate Change
MVEs	Motor Vehicle Examiners
NCCP	National Climate Change Policy
NEQS	National Environmental Quality Standards

NMT	Non Motorized Traffic
NOC	No Objection Certificate
NSDS	National Sustainable Development Strategy
OP/BP	Operational Procedure/Bank Procedure
P&D	Planning and Development
PAHs	Polycyclic Aromatic Hydrocarbons
P-CAP	Punjab Clean Air Program
PCAP-CMU	Program Coordination and Monitoring Unit
P&D	Planning and Development
PDO	Program Development Objective
PEECA	Punjab Energy Efficiency and Conservation Authority
PEPA	Pakistan Environmental Protection Act
PEQS	Punjab Environmental Quality Standards
PforR	Program for Results
PGDP	Punjab Green Development Program
PM_{2.5}	Particulate Matter of 2.5 micron or less diameter
POPs	Persistent Organic Pollutants
PPP	Public Private Partnership
PSC	Program Steering Committee
PSVs	Public Service Vehicles
RA	Result Area
SDGs	Sustainable Development Goals
SEP	Stakeholder Engagement Plan
SLCP	Short Lived Climate Pollutants
SMART	Strengthening Markets for Agriculture and Rural Transformation
SOPs	Standard Operating Procedures
TEPA	Traffic Engineering and Transport Planning Agency
T&MD	Transport and Masstransit Department
UET	The University of Engineering and Technology
VICS	Vehicle Inspection and certification Services
WASH	Water, Sanitation and Hygiene
WHO	World Health Organization

EXECUTIVE SUMMARY

Introduction

This Environmental and Social Systems Assessment (ESSA) has been prepared by the World Bank Pakistan Team for the proposed Punjab Clean Air Program (P-CAP). The proposed program will be supported through the World Bank's Program-for-Results (PforR) financing. World Bank Policy on PforR Financing requires conducting a comprehensive ESSA to evaluate the borrower's environmental and social management systems function. The ESSA has been carried out to assess gaps, risks, and opportunities in the country-level systems and to suggest measures to strengthen the system's capacity for managing environment and social risks. The program has been assessed to have Substantial E&S risks.

Objectives of the ESSA

The primary objectives of this ESSA are to:

- Identify the Program's environmental and social effects.
- Assess the legal and policy framework for environmental and social management, including relevant legislation, regulations, procedures, and institutional responsibilities.
- Evaluate the institutional capacity and performance of implementing agencies in managing (avoid, minimize, or mitigate) potential environmental and social risks.
- Recommend specific actions to address gaps in the Program's environmental and social management system, particularly concerning policy, legal frameworks, and institutional capacity.

Program Description and Environmental & Social Effects

The Punjab Clean Air Policy and Action Plan (2023) and the Punjab Smog Mitigation Action Plan (2024) form the basis of the government's air quality program, covering air quality improvement objectives, policy interventions, implementation mechanisms, coordination, and monitoring. The Smog Mitigation Action Plan (2024) serves as the short-term implementation strategy, with budgeted activities extending until Q1 2025. The Government of Punjab (GoPb) will update this plan annually, incorporating priority activities and budgets.

World Bank's support for GoPb's program will be structured through sequential operations to support Air Quality Management (AQM), with an initial focus on road transport emissions. The P-CAP will support government initiatives to curb emissions across priority sectors transport and agriculture while conducting assessments in the industrial sector to lay the foundation for future interventions.

The P-CAP aims to strengthen AQM and reduce emissions from targeted sectors. It will achieve these objectives through three results areas, each targeting specific pollution sources:

- Air Quality Management Governance and Awareness Raising
- Transport Sector Abatement Measures
- Agriculture Sector Abatement Measures

P-CAP will use a hybrid approach that combines the PforR and a complementary Investment Project Financing (IPF). The IPF component will support technical assistance, capacity building, and necessary assessments to facilitate Program implementation. This component will also design interventions in industry, which will be implemented under a subsequent operation to enhance air quality.

The Program Development Objective (PDO) is to strengthen air quality management and reduce emissions from targeted sectors in Greater Lahore.

Program for Results

The following section gives a summarized description of the three results areas of the Program to incentivize GoPb.

Results Area 1 (RA1): AQM Governance and Awareness Raising. This area strengthens EPCCD's capacity to enhance Punjab's regulatory, institutional, and technical capacity for AQM, as well as encourages robust data-driven communications and awareness raising to bring together diverse actors towards the goal of improving air quality:

- a. **Strengthen AQM infrastructure.** The Program will support the expansion of air quality monitoring regulatory grade stations across Punjab, with real-time data reporting on a publicly available website. The Program will also enhance fuel quality testing by adding two new laboratories (one mobile, one fixed), with testing quality aligned with the American Society for Testing and Materials (ASTM) standard.
- b. **Data, inventory, and source attribution.** The Program will facilitate the development of an integrated pollutant and greenhouse gas (GHG) emissions inventory system, which will provide a comprehensive and systematic approach to track and manage emissions data. The inventory will be shared with the Minister for Environment to inform policy making, including future revisions to Punjab's Smog Mitigation Action Plan.
- c. **Strengthening public communications and raising awareness.** The public communications campaign will be informed by a data-driven stock take early in the Program, drawing on lessons learned from previous campaigns, monitoring data, and existing inventories. The campaign will then use different channels (including five "roadshow" vehicles and 450 awareness sessions) to inform stakeholders about air pollution sources, health impacts, and available reduction measures, with special attention to vulnerable populations. This comprehensive outreach effort builds upon the existing health advisory system supported by PGDP.

Results Area 2 (RA2): Transport Sector Abatement Measures will focus on measures to curb road transport emissions through a combination of shift and improve measures:

Expanding and encouraging modal shifts to mass transit. This incentivizes the deployment of 648 e-buses in Lahore and supporting infrastructure viz. depots and charging facilities. Public-private partnership (PPP) options will be explored under a study to identify business model(s). GoPb will be incentivized to implement a distance-based fare policy for e-buses using updated payment systems, which could increase fare revenue and improve the financial sustainability of buses. A Public Transport Service Contract (PTSC) between GoPb and the Punjab Mass transit Authority (PMA) that improves government funding commitment in exchange for greater accountability from PMA will also be incentivized. Measures to boost female labor force participation in e-

bus and depot operations will be promoted. Complementary interventions to improve access to public transport while tackling road dust will form part of the results framework.

Fostering the transition to e-2/3Ws. GoPb will be incentivized to adopt a provincial electric vehicle (EV) policy with clear EV adoption targets, institutional arrangements and action plan, and an associated charging infrastructure roadmap for key cities in Punjab. A provincial EV Cell will be established to spearhead policy implementation. A financing facility will be established via a financial intermediary (Bank of Punjab) to catalyze the flow of affordable commercial financing to e-2/3Ws across Punjab.

Expanding vehicle inspections and retiring polluting vehicles. The combination of an expanded vehicle inspection and maintenance regime and a buyback scheme will reduce the emissions intensity of the existing vehicle stock. Two Disbursement-Linked Indicators (DLIs) will incentivize the expansion of vehicle inspections to 2Ws and a buyback scheme for aged and polluting vehicles, with associated amendments to the Provincial Motor Vehicles Ordinance (1965). Financial incentives to encourage cleaner alternatives will be provided, with the possibility of linking these to the EV financing facility described in the para above.

Results Area 3 (RA3): Agriculture Sectors Abatement Measures. This RA will finance abatement measures in the agriculture sector. Activities in this RA will focus on immediate alternative measures to address rice crop residue burning, secondary pollution contribution from fertilizer used, and awareness raising among farmers on air quality.

- a. **Alternative to crop residue burning.** This activity will incentivize the scale-up of GoPb's initiative on the provision of 5,000 super seeders and other straw collection and management machinery to small farmers as an alternative to rice crop residue burning.
- b. **Digital access to straw management machinery and crop residue management service.** This activity will focus on digital solutions using proven models to improve access to super seeders, balers, and rice harvesters to small farmers through rental subsidies to reduce crop residue burning.
- c. **Awareness raising and enforcement.** This activity will run rigorous campaigns on awareness raising on alternative options for crop residue management, digital machinery rental services and subsidies, off-farm management options for straw management, and banning of burning to improve air quality in Punjab. These efforts will include targeted awareness campaigns for women in agriculture, enabling them to adopt climate-smart practices and advocate for better residue management, supporting emissions reduction goals.
- d. **Research and development.** This activity will explore promotion of soil testing and slow-release fertilizer alternatives, and policy reforms for the introduction and enforcement of slow-release fertilizer manufacturing and use. This intervention will be included in the long-term measures.

Environmental Risks

The environmental risk rating of the PforR activities is 'Substantial'.

Besides the numerous environmental benefits of the Program, the activities under RAs 1, 2 and 3 will also pose certain environmental risks summarized as below:

RA-1: . The Program activities, under expansion of air quality monitoring regulatory grade stations across Punjab, with real-time data reporting will require procurement and installation of IoTs (Internet of Things). By virtue of IoTs, electronic waste (e-waste) is expected to be generated by the end of life of the procured equipment. The environmental and health and safety impacts occur as a result of improper recycling and disposal of the e-waste. The environmental risks of fuel testing laboratories primarily stem from the use of hazardous chemicals and the generation of chemical waste, including potential air, water and soil pollution including Occupational Health and Safety (OHS) as well as Community Health and Safety risks (CHS), if not properly managed. Finally, the operation of roadshow vehicles can pose limited environmental risks, linked with the release of air emissions and leakage of chemicals, if properly maintained vehicles are not utilized.

RA-2: Primary environmental risks of the Program are associated with RA-2 involving transport-related activities. The deployment of e-buses and promotion of e-two/three wheelers uptake, can contribute towards greenhouse gas emissions indirectly, by battery charging through grid electricity supply (grid supply in the Punjab is mostly fossil fuel based), during the operation of these e-vehicles. However, the quantum of greenhouse gas thus emitted from e-buses and e-two/three wheelers charged through grid electricity as compared to diesel, petrol, and other fuels is expected to be substantially low.

Overall, the e-buses are safer because of advanced safety technology features in their designs. However, Lithium Iron Phosphate (LiFePO₄) battery used in e-buses and e-2/3 wheelers are uncertain in terms of their fire risk and safety. When a battery is damaged, it presents a significant fire hazard that cannot be extinguished through conventional fire extinguishing systems. Battery charging stations and battery storage areas will also be potential fire-risk prone areas.

After a few years, lithium batteries will expire and require proper disposal. The battery recycling facilities in Punjab are in the informal sector where recycling practices are not environmentally friendly, and these facilities are not regulated under any legislation¹. Environmental, Occupational Health and Safety (OHS) and Community Health and Safety (CHS) impacts are expected due to the discharge of heavy metals, dust and hazardous materials and emissions.

Construction related activities to develop bus depots, bus stops, repair roads, improve road shoulders, develop footpaths and cycling lanes and installation of battery charging stations are assessed to be temporary, localized and reversible in nature, including OHS/CHS, noise and dust as well as construction and general waste generation. Solarization of battery charging stations poses the risks of structural safety (due to rooftop solar panels), OHS risks (specially electrical hazard during installation and operations), spent solar batteries , and fire hazards during operations. These are localized risks, manageable through compliance with existing regulations and good industry practices. The parallel IPF component will also support

¹ World Bank is supporting the Federal Government for the development of E-waste Management Policy, under Islamic Republic of Pakistan: Digital Economy Enhancement Project (P174402), included as part of Environmental and Social Commitment Plan (ESCP). The policy development process has, however, been delayed, but is expected to set the path for the development and adoption of subsequent provincial E-waste Management Rules, which are absent thus far.

the EPCCD in the development of -battery recycling and management rules.

Bus washing activities at bus depots will consume a substantial amount of water , generate wastewater and sludge from suspended solids and soap suds. As Punjab is recently experiencing an extensive water shortage, bus washing activities will contribute to additional stress on the current scarce water resources. Improper disposal of wastewater sludge will be another environmental concern at the bus depots. Furthermore, whilst EVs eliminate risks associated with fuel and motor oil leaks, they can still pose environmental risks in parking areas due to potential chemical leaks. Battery coolant leaks from damaged or aging thermal management systems can contaminate soil and water. Hydraulic brake fluid leaks, though less frequent due to regenerative braking, remain a potential hazard. Certain EV drivetrains and hybrid systems may also release transmission fluids or gear oil if seals degrade. In rare cases, battery damage or thermal runaway can result in electrolyte leakage, which contains toxic and corrosive chemicals. Additionally, washer fluid and air conditioning refrigerants may leak, contributing to environmental pollution. Without proper drainage, impermeable flooring, and regular maintenance, these leaks could lead to contamination and hazardous conditions in the bus depots/parking areas.

The buy-back scheme will produce stock of old/polluting vehicles in the province whose recycling/disposal will be an environmental challenge. There is also potential risk of redeployment of retired vehicles on the far-flung rural roads of the province where enforcement of rules and standards is typically more difficult. These risks will be addressed by formulating and implementing environmentally sound recycling/disposal strategies as per the requirements of Punjab Environmental Protection Act 2012 (Amendment 2017), Punjab Hazardous Substances Rules 2022 including other related local rules and regulations along with the Core Principles, by the recycling facilities in the province.

Vehicle inspections is done through digital equipment and sensors. At the end of the life of these equipment and sensors, e-waste will be generated. At the operational level, low environmental impacts are expected. These impacts can be mitigated through good housekeeping practices.

RA-3: The use of super-seeders and other machinery may lead to impacts like soil disturbance, suspended dust and air emissions especially on rural dust roads, fuel consumption, biodiversity impacts due to increased traffic in a rural environment and promotion of monoculture, which can result in more frequent pest and disease outbreaks in crops, increased use of chemicals, and potential financial losses for farmers. The operation of super-seeders, through diesel-based tractors, will contribute about 150,000 tons of CO_{2e} greenhouse gases every year, in case, if the total 7 million acres of rice crop fields of Punjab is mechanized. The Program will compute the reduction in greenhouse gases due to stoppage of crop burning in a unit area to establish the benefit of super-seeders in reduction of greenhouse gases.

Social Risks

The social risk rating of the PforR activities is 'Substantial'.

Besides the many social benefits of the Program, social risks under each result area are described below:

RA-1: Government employees with low or zero IT literacy, especially older and female staff may face redundancy. Increased monitoring may shutdown small industries, leading to losses of jobs and worker protests. Enforcement of stricter fuel quality and emissions regulations may disproportionately affect small-scale vendors, transporters, and industries, leading to potential economic displacement or loss of livelihoods. The operations of “roadshow” vehicles and public awareness sessions pose community health and safety risks, especially for children and women. There is also a risk of exclusion of women, illiterate, persons with disabilities, rural communities and those without access to digital mediums from accessing awareness materials.

RA-2: The transition to e-buses could displace existing informal transport operators (e.g., rickshaw drivers), leading to economic losses and opposition. Traditional mechanics and workers dependent on internal combustion engine vehicles may face job displacement as EVs require different skill sets. A sudden shift towards EVs may reduce the resale value of traditional 2/3-wheelers, affecting small-scale vehicle owners. A distance-based fare policy may increase costs for low-income commuters, disproportionately impacting daily wage earners.

Construction of e-bus routes and depots pose health and safety risks for workers and public. Risks include forced removal of encroachers from land identified for the e-bus depots and bus corridors. Exclusion risks exist for persons with disabilities in using the e-buses and for women, students and low-income individuals from accessing financing facilities for electric two and three wheelers. Risks of Sexual Exploitation and Abuse (SEA)/Sexual Harassment (SH) for women employed in e-bus operations and public transport users.

Vehicles inspections may disproportionately penalize poor (who may not afford upgrades). Sudden bans may impact the livelihoods of individuals in ridesharing or delivery services. A buyback and retrofitting scheme without a clear transition plan could lead to sudden unemployment among informal transporters and small mechanics. Improper recycling of scrapped vehicles could pose health and safety risks for workers due to toxic material exposure.

RA-3: Small and marginalized farmers, especially women, may face barriers in accessing super seeders and straw management machinery due to lack of awareness, affordability, or digital access. Risks include loss of income for small farmers due to crop burning banning measures; unequal penalization of small farmers for crop burning; and exclusion of women, illiterate, persons with disabilities, rural communities and those without access to digital mediums from benefiting from digital solutions to mechanization. New machinery (e.g., super seeders) poses occupational hazards for farmers and laborers due to lack of proper training and safety measures. The availability of machinery may favor larger farmers, leaving smallholders at a disadvantage. Awareness campaigns may not effectively reach rural and marginalized farmers, particularly women and those with limited education. Small-scale farmers may be unable to afford soil testing and slow-release fertilizers, limiting adoption.

Policy and Legal Framework for Environment and Social Management

The following legal framework exists in the Punjab for environment and social risks management of the PCAP:

Environmental Legislation

- The Punjab Clean Air Policy and Action Plan (2023)
- Punjab Smog Mitigation Action Plan (2024)
- The Smog Mitigation Action Plan (2024)
- Punjab Environmental Policy (2015)
- Punjab Environmental Protection Act 2012
- Punjab Hazardous Substances Rules 2020
- Environmental Impact Assessment (EIA) Regulations
- Provincial Motor Vehicle Ordinance, 1965 & Motor Vehicle Rules, 1969. In addition, recently promulgated Provincial Motor Vehicles Amendment Act, 2023. PCAP will further contribute to the amendments in the Ordinance.
- Punjab Environmental Quality Standards (PEQS)

Social Legislation

- Punjab Local Government Land Use Plan (Classification, Reclassification, and Redevelopment) Rules 2020
- Punjab Local Government Act 2019
- Punjab Occupational Safety and Health 2019
- The Factories Act, 1934
- Punjab Restriction of Employment of Children Act 2015
- Land Acquisition Act, 1894
- Punjab Land Acquisition Rules, 1983
- Punjab Labour Policy, 2018
- Punjab Antiquities Amendment Act 2012
- Guidelines for Public Consultation, 1997
- Punjab Minimum Wages Act, 2019
- Punjab Industrial Relations Act, 2010
- Punjab Right to Public Services Act, 2019
- Punjab Transparency and Right to Information Act, 2013
- Punjab Protection of Women Against Violence Act, 2016
- Punjab Empowerment of Persons with Disabilities Act, 2022
- Punjab Women Development Policy, 2018

Assessment of Environmental and Social Management Systems and Implementation Capacity

Environmental Management Systems and Capacity Assessment

This ESSA assesses Program's Systems for managing environmental and social impacts,

considering, among other things, the capacity to plan, implement, monitor, and report on the environmental and social mitigation measures, the scope for improvements, and the risks and related mitigation measures. Summary is as below:

*The current available systems and associated gaps under **Core Principle 1 - Environmental and Social Management**:*

- Legal and regulatory framework exists at national and provincial level to assess the environmental and social impacts of the projects. The guidelines for the preparation and review of environmental reports include the assessment of environment, social, health, and economic and fiscal impacts. IEE/EIA studies are prepared and submitted to the environmental agencies for acquiring NOC to construct and operate projects. However, these assessments are only required for those projects which are listed in Schedules I and II of the IEE/EIA regulations. Development of bus depots requires an environmental and social impact assessment, as informed by The Punjab Environmental Protection (Amendment) Act, 2017. Under the Program, ESIA of the bus depots will be carried out and its recommendations will be implemented while designing, construction and operational phases.
- Generally, the hazardous materials are regulated under PEPA 97 and Punjab EPA 97, and Pakistan Penal Code but there are no specific provisions for e-waste and waste batteries management in these regulations. Hazardous Substance Rules do not mention e-waste as hazardous substance. There are no SOPs for e-waste handling in the implementation departments.
- Institutional responsibilities and resources for preparation, implementation, monitoring, and inspection of the environmental plans are clearly spelled out by relevant regulations (PEPA 2012, Review of IEE/EIA Regulations, 2000). The mandate for monitoring the implementation of IEE/EIA lies with EPCCD.
- Key implementation agencies of the Program activities will be the Environmental Protection and Climate Change Department (EPCCD), Transport and Mass-transit Department (T&MD) (in collaboration with PMA and PTC) and Agriculture Department (AD), whereas Planning and Development (P&D) Board will be responsible for the overall coordination among all the implementing agencies and monitoring the Program performance.
- EPCCD has developed its capacity to manage environmental and social aspects under the World Bank PforR financed 'Punjab Green Development Program (PGDP). However, they have limited experience in implementing the World Bank's ESF policy and staffing for environmental specialists under PGDP has been a challenge. The EPCCD is lacking in infrastructure, capacity and resources (human, financial and technical) to enforce environmental regulations.
- The PGDP's experience of implementing E&S policy and regulatory requirements by the Implementing Agencies (IAs) was deficient. The E&S responsibilities of each IA were not spelled out in the program, leading to confusion amongst the IAs on effective E&S policy implementation, monitoring and reporting. Learning from this, The Program should clearly define E&S responsibilities of each IA.
- P&D has established P-CAP Coordination and Monitoring Unit (PCAP-CMU). The E&S staff will be appointed in the Unit. EPCCD will designate E&S focal persons from its existing staff to ensure compliance with E&S safeguard requirements.

- Transport and Agriculture departments are also the implementing agencies under PGDP and have experience with the E&S requirements, but their institutional capacity for E&S management is low. Currently, there is no E&S position within the organogram of the department. The Transport department will be responsible for larger share of project implementation. Accordingly, a Program Implementation Unit (PIU) will be established in the Transport department. E&S specialists will be part of the PIU team, whereas E&S focal points will be appointed in PMA and PTC. The scope of Agriculture department activities is relatively small. It will designate focal E&S persons from its existing staff.
- A complaint cell is active at EPCCD to address complaints by communities related to environmental and social aspects. There are many governmentally operated and department specific GRMs such as GRM at PM and CM portals operational in the Punjab province. The public grievances related to environmental and social issues are effectively addressed and resolved by the specific department.

*The current available systems and associated gaps under **Core Principle 2 – Natural Habitats and Physical Cultural Resources:***

- The scope of IEE/EIA covers the profiling of natural habitats, flora and fauna, national parks, important ecological areas, and impacts on downstream freshwater bodies. Local EIA guidelines emphasize to consider cultural resource impacts including changes in archaeological, historical and cultural artifacts and structures and environmental features with religious and ritual significance and measures to avoid, minimize or mitigate these impacts.
- There are regulations on biotic habitats and critical ecosystems. However, these are not relevant as most physical interventions under the program will be in and around the established urban areas of Lahore.
- For conservation of trees in and around Lahore, there exist specific legal instruments prohibiting tree cutting and establishing strict protocols where it is unavoidable. These include Punjab Local Government Act 2019, Cantonments Act, and Lahore Canal Heritage Act 2013.
- The main legislation on conservation of archaeological heritage is the Pakistan Antiquities Act of 1975. This Act was adopted by Punjab in 1985. The Antiquities Act 1975 only focused on conservation of a monument whereas the Punjab Law also includes conservation of the area surrounding the monument.

*The current available systems and associated gaps under **Core Principle 3 – Public and Worker Safety:***

- For the safety of the buildings and structures, with respect to fire and earthquake, Building Code of Pakistan-Fire Safety Provisions-2016 and Seismic Provisions-2007 are available. Local government and development authorities building bylaws and safety standards are applied while extending approval of the infrastructures.
- The Punjab Occupational Safety and Health Act, 2019 focuses on the safety of the construction workers.
- Hazardous Substance Rules 2019 do not specify e-waste and waste batteries. Hence recycling facilities for such waste are not regulated and are a potential safety risk for

the workers and community. The program will mobilize EP&CCD to develop Waste Battery Management Rules under the Hazardous Substance Rules, and adopt e-waste management SOPs (existing under other EP&CCD projects) at IA level.

- The ESIA of the bus depots under the program will contain workers and public safety related recommendations (including Occupational and Public Health & Safety Plans) that will be implemented while designing, construction and operational phases.
- The safety of e-buses concerning lithium battery-specific fires is an important risk. The Pakistan Standards and Quality Control Authority (PSQCA) standards does not include lithium-specific fires safety standards. The Program will only procure e-buses equipped with lithium battery-specific fire suppression systems to avoid fire hazards and protect public and government assets.

Social Management Systems and Capacity Assessment

The social management under the Program will be largely based on the existing legal, regulatory, and institutional system in Pakistan and in the province of Punjab. The existing social management systems need to be improved to address the underlying social risks. The ESSA identified gaps in some areas, as summarized below, which will need to be addressed through the Program Action Plan (PAP).

*The current available systems and associated gaps under **Core Principle 1 - Environmental and Social Management** and **Core Principle 3 – Public and Worker Safety** are summarized below:*

- Generally, environment and social management is not included in the mandate of the IAs. EPCCD has a Social Specialist on staff under PGDP who has also been covering as an environmental specialist for an extended time. T&MD, AD, P&D and Industries Department do not have any staff dedicated to the management of environmental and social impacts.
- Stakeholder consultations by law are only mandated for activities requiring an EIA. These requirements are weak and lack mandates for inclusion and transparency as per Bank standards.
- Information dissemination is governed by the Punjab Transparency and Right to Information Act. Public Information Officers (PIOs) as mandated by the Act are present in the IAs, and the IAs have functioning websites with detailed information on services, structures and grievances.
- Regional Transport Authorities (RTAs) have a performance matrix which includes indicators for public engagement through advertisements and information sharing.
- The Pakistan Citizens Portal and the Punjab Portal are functioning as GRMs at the national and provincial level to address public queries and grievances. All IAs also have their own department level GRMs which are functional. However, the accessibility and performance of the provincial and departmental GRMs for vulnerable groups is not known.
- Workers' health and safety is governed by the Punjab Occupation Health and Safety Act.
- Legislation regarding community health and safety is weak. SEA/SH risks for women at the workplace are governed by the Punjab Protection Against Harassment of Women at Workplace (Amendment) Act 2012, and use of child labor is restricted by the Punjab Restriction on Employment of Children Act, 2016.
- Community health and safety in IAs is most often considered for activities under donor financed projects requiring specific measures to be in place as per donor requirements.

- Overall implementing agencies are limited by human resources availability, capacity and infrastructure to target and respond to risks faced by communities during construction activities.

*The current available systems and associated gaps under **Core Principle 4 – Land** are summarized below:*

- The main legal tool used for land acquisition in Punjab is the Land Acquisition Act (LAA) 1894 and Punjab Land Acquisition Rules 1983. However, there are certain gaps, as in, compensation is only for titleholders, excluding non-titleholders, and there are no provisions for livelihood restoration, leading to financial instability and legal disputes.
- Emergency land acquisition (Section 17) allows dispossession without consultation or prior compensation, increasing vulnerability.
- Anti-encroachment drives (AED) are commonly carried out in urban areas by local and district governments.

*The current available systems and associated gaps under **Core Principle 5 – Indigenous Peoples and Vulnerable** are summarized below:*

Punjab does not have the presence of any recognized Indigenous People. The assessment for this Core Principle focuses on equal access for and consultations with vulnerable groups.

Government departments are not required to consult vulnerable groups for their feedback into any interventions. Equitable access is governed through the Punjab Women Development Policy 2018, Punjab Agriculture Policy 2018, Punjab Social Protection Policy 2022 and Empowerment of Persons with Disabilities Act 2022. Generally, there is weak implementation of inclusion policies despite existing frameworks (Punjab Women Development Policy, Punjab Social Protection Policy, Disability Act). Moreover, IAs lack dedicated staff/resources to target and support vulnerable groups (encroachers, minorities, persons with disabilities). Donor-funded projects include specific measures for vulnerable people, but only within project scope.

*The current available systems and associated gaps under **Core Principle 6 – Social Conflict** are summarized below:*

Punjab government has put in place GRM systems online and through the Provincial Ombudsman. However, local authorities have a history of dealing with protests and conflict situations with force. Experience by the EPCCD shows that effective communication on Green Lockdowns has been helpful in managing public expectations. Therefore, ensuring accessible, effective and continued communications on the air pollution abatement measures will be critical in keeping the public engaged and informed, and will reduce the risk of conflicts and protests.

Disclosure and Consultations

The draft ESSA will be disclosed in-country and on the World Bank’s website before the PforR appraisal for discussion and feedback. After incorporating stakeholder input, the revised ESSA will be re-disclosed.

Institutional consultations were held virtually on Dec 24, 2024, followed by in-person meetings with P&D, TMD, EPCCD, AD, PMA, PTC and Industries Department in Lahore from Jan 15–17 & Feb 18-20, 2025. A multi-stakeholder workshop was organized on Jan 30, 2025, attended by 35 representatives from government, academia, private sector, and civil society (including women and persons with disabilities). The stakeholders provided feedback on program design, ESSA recommendations, stakeholder engagement, and inclusion. The feedback was incorporated into the final ESSA report.

Exclusion Criteria

The Program will exclude activities that do not meet the World Bank’s PforR eligibility criteria (Sept 2020), including those with significant, sensitive, diverse, or unprecedented environmental impacts or requiring significant land acquisition, displacement, or resettlement.

The Program will support activities after being screened against the following Exclusion Criteria:

- Significant conversion or degradation of critical natural habitats or critical cultural heritage sites.
- Air, water, or soil contamination leading to significant adverse impacts on the health or safety of individuals, communities, or ecosystems.
- Workplace conditions that expose workers to significant risks to health and personal safety.
- Land acquisition and/or resettlement of a scale or nature that will have significant adverse impacts on affected people, or the use of forced evictions.
- Acquisition/use of lands with encumbrances or lands that have been cleared in anticipation of program activities.
- Acquisition/use of lands which have legacy anti-encroachment drives (AED) starting from the approval date of the Program Concept (December 9, 2024).
- Large-scale changes in land use or access to land and/or natural resources.
- Adverse E&S impacts covering large geographical areas, including transboundary impacts, or global impacts such as greenhouse gas (GHG) emissions.
- Activities that involve the use of forced or child labor.
- Marginalization of, discrimination against, or conflict within or among, social (including ethnic and racial) groups.

Recommendations and Inputs to the Program Action Plan (PAP)

The overall recommendations from the ESSA have been captured in eight actions in the PAP. The summary of recommendations and PAP is as follows:

Recommendation 1: Install fire suppression systems in e-buses and charging/storage areas. Train drivers and staff to manage fires and evacuate passengers early. Separate fire-prone

areas and recycle wastewater for bus washing to conserve water.

Recommendation 2: Program will develop SOPs e-waste management at the departmental level (for project IAs). The program will also mobilize EPCCD to develop and enforce waste batteries recycling management rules and certify recycling facilities.

Recommendation 3: Program should conduct ESIA's for all bus depots, integrating site alternative analysis, environment and social impacts, measures for identification of biodiversity and cultural resources of sites, sites screening for archaeologically important places, LFS aspects, occupational and public health and safety aspects, and climate resilience, into design, construction, and operations.

Recommendation 4: Program should develop and implement strategy for safe disposal of buy-back or retired vehicles.

Recommendation 5: Program will build the capacity of institutions on updating/amending the existing rules/ordinance on Motor Vehicles emissions.

Recommendation 6: The fuel testing laboratories establishing under the Program will require to prepare and follow the ESMP and get ISO/IEC 17025 certification.

Recommendation 7: The institutional arrangement for E&S, as described in the previous section (PIU and E&S focal persons) to be put in place from the outset of the program. IAs to develop internal institutional procedures, appoint staff, and build their capacity on institutional procedures for environmental and social risk management. PCAP-PMU to clearly describe E&S roles and responsibilities of all relevant departments under the program; develop and oversee monitoring and reporting procedures.

Recommendation 8: Ensure E&S considerations in Procurement & Program Operations. Conduct E&S assessments for super-seeders & agricultural machinery and develop SOPs for deployment of super-seeders including awareness raising for its safe E&S performance. Develop buses procurement, and operator manuals with E&S provision.

Recommendation 9: Workers and Community Health and Safety:

- Prepare an OHS and CHS Plan. These plans are to be prepared as part of the ESIA by the T&MD and implemented by PMA, PTC and construction contractors. T&MD to ensure all construction contracts include community health and safety, labor safety child labor and SEA/SH prevention clauses.
- Prepare a Gender Inclusion and SEA/SH Action Plan for the Program activities related to ensuring inclusion of women in accessing program benefits, and SEA/SH prevention in program activities, especially for women employed in e-bus operations and women using public transport.

Recommendation 10: Fair Land Pricing and Displacement Mitigation: The Program will apply two key exclusion criteria regarding land acquisition and resettlement. Sites identified for e-bus depots will exclude: (i) areas displacing 200 or more individuals per site; and (ii) areas which have legacy anti-encroachment drives (AED) starting from the approval date of

the Program Concept (December 9, 2024). The SOPs on land acquisition and resettlement from PGDP will be tailored for PCAP, addressing gaps in areas affected by anti-encroachment drives (AEDs). Key provisions will include:

- Land acquisition and resettlement screening checklists for each e-bus depot against the exclusion criteria and to identify resettlement impacts;
- Compensation for affected landowners, tenants, and non-titleholders, aligned with market rates; and
- Preparation of Abbreviated Resettlement Action Plans (ARAP) or Resettlement Action Plans (RAP) before commencing work (if required).

Recommendation 11: Stakeholder Engagement, Accessibility of Information and Inclusion of Vulnerable Groups

- Prepare a Program Wide Stakeholder Engagement Plan (P-SEP) to ensure consistent, transparent and inclusive consultations throughout the program life. This will build upon the SEP prepared for the TA component, will identify all relevant program stakeholders, engagement activities, methods of engagement and timelines. As part of the SEP, conduct annual Citizen Feedback Surveys on program activities and publish results.
- Conduct a social assessment to establish a baseline of access of women and persons with disabilities to public transport and e-vehicles. The social assessment will focus on establishing a baseline of the use of public transport by women and persons with disabilities, and cultural acceptance of women driving e-2 wheelers. This will allow the Program to monitor and establish a midline and an endline on the same indicators.
- Prepare Disability Inclusion Plan for implementation of The Punjab Empowerment of Persons with Disabilities Act 2022 across RA 2 activities to ensure that public transport infrastructure including buses, bus stops and pedestrian access to bus stops such as pavements and road shoulders are accessible.
- Prepare SOPs for ensuring accessibility of communications materials and awareness campaigns for vulnerable groups. These SOPs should ensure that all communications and awareness campaigns conducted through the Program are inclusive and accessible to vulnerable groups including women, illiterate, persons with disabilities, rural communities and those without access to digital mediums.
- Prepare SOPs for inclusive communication, ensure awareness campaigns and materials are accessible to women, illiterate, persons with disabilities, and rural populations, including those without digital access.

Recommendation 12: Strengthening Grievance Mechanisms

- Review the accessibility of the existing provincial and department wise Grievance Redressal and Right to Information forums for vulnerable groups. Punjab government has existing forums on grievance redressal and RTI, whereas all IAs also have forums for receiving and responding to grievances. However, the accessibility of these forums for vulnerable groups including women, gender minorities, persons with disabilities, illiterate and rural communities is lesser known. By reviewing their accessibility, the Program can ensure grievance redressal and information sharing is more inclusive and responsive towards vulnerable groups.
- Prepare SOPs for IAs to track, collate, respond to and report on program related grievances received through different national and provincial channels.

Recommended Inputs for Program Action Plan

No	Action Description	Source	Responsibility	Timing	Completion Measurement
1	Appoint one Environmental Specialist, and one Social/Gender Specialist at the PCAP-CMU and T&MD PIU, and one Environmental/Social Focal Point within each IA	Environmental and Social Systems	Specialists – PCAP-CMU and T&MD PIU E&S Focal Points - AD, EPCCD, Industries Dept, PSIC, PMA, PTC and other agencies as relevant	Within 90 days of Program effectiveness	Listed specialists and focal points appointed and maintained throughout Program implementation
2	Prepare & operationalize E&S technical manual	Environmental and Social Systems	PCAP-CMU & T&MD PIU	Within 90 days of Program effectiveness	E&S technical manual acceptable to WB and approved (based on PSC minutes)
3	Tailor and adopt PGDP land acquisition and resettlement standard operating procedures.	Environmental and Social Systems	Developed and implemented by T&MD Overseen by PCAP-CMU	Within 90 days of Program effectiveness and before civil works start.	Tailored standard operating procedures acceptable to WB and approved (based on PSC minutes)
4	Develop and implement i) Gender Inclusion and SEA/SH Action Plan, ii) P-SEP including GM, iii) Disability Inclusion Plan	Environmental and Social Systems	i) & ii) PCAP-CMU, to develop and all IAs to implement; iii) T&MD to develop, and PMA and PTC to implement; iv) PCAP-CMU in coordination with IAs	i) , ii) and iii) within 12 months of Program commencement and prior to the start of any Program activities	Gender Inclusion and SEA/SH Action Plan, Disability Inclusion Plan, P-SEP including GM prepared and submitted to Bank, disseminated to stakeholders and operationalized
5	Environmental and Social Impact Assessment (E SIA) prepared for bus infra and operations, incl. OHS and CHS plans.	Environmental and Social Systems	T&MD to develop PMA and PTC to implement EPCCD to monitor	Prior to the start of civil works	ESIA acceptable to WB and approved by EPA.
6	Tailor and adopt SOP for e-waste management	Environmental and Social Systems	Tailored and adopted by PCAP-CMU	Within 6 months of	SOPs acceptable to WB and approved

			All IAs to implement	Program effectiveness	(based on PSC minutes).
7	E&S checklists for deployment of super seeders	Environmental and Social Systems	Developed and implemented by Agriculture Department. Overseen by PCAP-CMU	Prior to the deployment of super-seeders	E&S checklists acceptable to WB and approved (based on PSC minutes).
8	Develop and implement trainings for IAs on E&S management, including CHS, child labor, SEA/SH, e-waste and lithium battery fire safety	Environmental and Social Systems	Developed by PCAP-CMU Implemented by IAs	Within 90 days of Program effectiveness and before civil works start	E&S training module developed and acceptable to the WB. E&S training completed
9	Conduct annual citizen feedback surveys and publish results	Environmental and Social Systems	PCAP-CMU	Annual	Citizen feedback surveys implemented and results disseminated
10	Exclude from consideration as a Pro ram area any areas subject to AED on or after December 9, 2024, or that may involve the displacement of 200 or more individuals	Environmental and Social Systems	PCAP-CMU & IAs	Continuous	Regular reporting by PCAP-CMU

1. INTRODUCTION

1.1. Environmental and Social Systems Assessment: Purpose and Objectives

1. This Environmental and Social Systems Assessment (ESSA) has been prepared by the World Bank for the proposed 'Punjab Clean Air Program (P-CAP)', which will be supported by the World Bank's Program for Results (PforR) financing instrument. In accordance with the requirements of the World Bank Program-for-Results Financing instrument, PforR relies on country-level systems for the management of environmental and social effects². The PforR financing requires that the Bank should conduct a comprehensive ESSA to assess the degree to which the relevant program's systems, under PforR financing, promote environmental and social sustainability and to ensure that effective measures are in place to identify, avoid, minimize, or mitigate adverse environmental, health, safety, and social impacts. ESSA based on comprehensive assessment makes recommendations to enhance environmental and social management within the program as part of the overall management action plan.
2. The main purposes of this ESSA is to: (i) identify the program's environmental and social effects, (ii) assess the legal and policy framework for environmental and social management, including a review of relevant legislation, rules, procedures, and institutional responsibilities that are being used by the program; (iii) assess the implementing institutional capacity and performance to date to manage potential adverse environmental and social issues; and (iv) recommend specific actions to address gaps in the program's environmental and social management system, including with regard to the policy and legal framework and implementation capacity.
3. The ESSA Team assesses the arrangements, within a program, for managing environmental and social effects in a manner consistent with Bank Policy: Program for Results Financing (2024). This policy sets out core principles and key planning elements intended to ensure that PforR operations are designed and implemented in a manner that maximizes potential environmental and social benefits, while avoiding, minimizing, or otherwise mitigating environmental or social harm. The Core Principles are listed below. These are further defined in detail through corresponding Key Planning Elements that are included under each Core Principle in Section V.
 - a) **Core Principle 1: *Environmental and Social Management***: Environmental and social management procedures and processes are designed to (a) promote environmental and social sustainability in program design, (b) avoid, minimize, or mitigate against adverse impacts; and (c) promote informed decision making relating to a program's environmental and social effects.
 - b) **Core Principle 2: *Natural Habitats and Physical Cultural Resources***: Environmental and social management procedures and processes are designed to avoid, minimize, and mitigate any adverse effects (on natural habitats and physical cultural resources) resulting from the program.
 - c) **Core Principle 3: *Public and Worker Safety***: Program procedures ensure adequate

²"Effects" is used throughout this report to refer collectively to benefits, impacts, and risks. The subsumed terms are used where necessary to focus on specific topics or issues. The term "benefits" include positive impacts, and the term "impacts" refer to adverse or negative consequences.

measures to protect public and worker safety against the potential risks associated with (a) construction and/or operations of facilities or other operational practices developed or promoted under the program and (b) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials.

- d) **Core Principle 4: *Land Acquisition*:** Land acquisition and loss of access to natural resources are managed in a way that avoids or minimizes displacement, and affected people are assisted in improving, or at least restoring, their livelihoods and living standards.
 - e) **Core Principle 5: *Indigenous Peoples and Vulnerable Groups*:** Due consideration is given to cultural appropriateness of, and equitable access to, program benefits, giving special attention to the rights and interests of indigenous peoples and to the needs or concerns of vulnerable groups.
 - f) **Core Principle 6: *Social Conflict*:** Avoid exacerbating social conflict, especially in fragile states, post conflict areas, or areas subject to territorial disputes.
4. An additional purpose of this ESSA is to enable informed decision making by the relevant authorities in the borrower country and to aid the Bank's internal review and decision process associated with the P-CAP. The findings, conclusions and opinions expressed in this document are those of the World Bank. The recommended actions that flow from this analysis have been discussed and agreed with the Government of Punjab counterpart and will become legally binding agreements under the conditions of the new loan.

1.1.1. Exclusions

5. Under the Bank's PforR Policy, activities that are "assessed to likely to have significant adverse impacts that are sensitive, diverse, or unprecedented on the environment and/or affected people are not eligible for financing and are excluded from the Program." More specifically, PforR financing should not be used to support programs, or activities within programs, that in the Bank's opinion involve the following:
- Significant conversion or degradation of critical natural habitats or critical cultural heritage sites.
 - Air, water, or soil contamination leading to significant adverse impacts on the health or safety of individuals, communities, or ecosystems.
 - Workplace conditions that expose workers to significant risks to health and personal safety.
 - Land acquisition and/or resettlement of a scale or nature that will have significant adverse impacts on affected people, or the use of forced evictions;
 - Anti-encroachment drives conducted within the last 5 years from program approval;
 - Large-scale changes in land use or access to land and/or natural resources;
 - Adverse E&S impacts covering large geographical areas, including transboundary impacts, or global impacts such as greenhouse gas (GHG) emissions;
 - Significant cumulative, induced, or indirect impacts;
 - Activities that involve the use of forced or child labor;
 - Marginalization of, discrimination against, or conflict within or among, social (including

- ethnic and racial) groups; or
- Activities that would (a) have adverse impacts on land and natural resources subject to traditional ownership or under customary use or occupation; (b) cause relocation of Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities from land and natural resources that are subject to traditional ownership or under customary use or occupation; or (c) have significant impacts on Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities' cultural heritage that is material to the identity and/or cultural, ceremonial, or spiritual aspects of the affected communities.

1.2. ESSA Methodology

- The World Bank undertook the following actions as part of the assessment from January – March 2025: (a) a comprehensive review of government policies, legal frameworks and program documents, and other assessments of the environmental and social management systems of the implementing agencies such as Environmental Protection and Climate Change Department (EPCCD), Transport and Mass Transit Department (T&MD), Agriculture Department (AD), Planning and Development (P&D) Board, Punjab Transport Company (PTC), Punjab Mass Transit Authority (PMA) and Bank of Punjab (BoP), (b) consultations were done with relevant experts and officials from the EPCCD, T&MD, AD, PMA, PTC and P&D Board.
- This report is prepared by the World Bank staff and consultants through a combination of reviews of existing program materials and available technical literature, interviews with government staff, and consultations with key stakeholders and experts. Findings of the assessment will be used in the formulation of an overall Program Action Plan (PAP) with key measures to improve environmental and social management outcomes of the program. The findings, conclusions, and opinions expressed in the ESSA document are those of the World Bank. Recommendations contained in the analysis were presented and discussed during the meetings with the concerned program implementing agencies before finalization and disclosure of the ESSA.
- The ESSA review process seeks to describe and assess the systems for managing environmental and social effects of a proposed program. In between January-February 2025, the World Bank drew on a wide range of data, sources, and inputs during the ESSA review process, including the following actions:
 - Assessment of the environmental and social effects of the program:** The ESSA Team assessed the potential for the program to cause adverse environmental and social effects, either due to its design and program components or due to gaps in program systems.
 - Comprehensive desk review of policies, legal framework, program documents, and other assessments of environmental and social management systems:** The review examined the set of national/provincial policy and legal requirements related to environment and social management associated with the activities described under the program components. The review also examined technical and supervision documents from previous and ongoing World Bank projects and programs whereby ESSA's were produced.
 - Institutional analysis:** An institutional analysis was carried out to identify the roles, responsibilities, and structure of the relevant institutions responsible for implementing the P-CAP funded activities, including coordination between different entities at the

national, provincial and local levels. The assessment of the capacity of key institutions to implement required environmental and social management actions was assessed. An important input for this assessment was an evaluation of these institutions' previous track record in management of such risks in the context of previous projects and programs.

- **Consultations and interviews:** Consultations were done with relevant experts and officials from the EPCCD, T&MD, AD, PTC, PMA and P&D Board through one-to-one meetings and through stakeholder consultation workshop during January 2025.
9. This draft ESSA has been updated based on the feedback received from stakeholders and experts prior to finalization of the program. The final ESSA will be disclosed prior to World Bank Board consideration of the program. It will be updated after Board approval, if needed, to reflect any important changes agreed by the Board to the program that affect environmental and social issues.

2. PROGRAM DESCRIPTION

2.1. Program Context

10. Punjab, the most populous province of Pakistan, has been facing air pollution as the most pressing environmental issue lately. It is part of the Indo-Gangetic Plain and Himalayan Foothills (IGP-HF), which is a global air pollution hotspot. IGP-HF has the most people exposed to hazardous air in the globe, including Bangladesh, Bhutan, India, Nepal, and Pakistan. In Pakistan, Bangladesh, India, and Nepal, 93–97% of the population is exposed to air pollution.
11. PM_{2.5}, which can be inhaled, is the biggest public health risk from air pollution. Air pollution has major transboundary impacts. In 2021, 53% of PM_{2.5} air pollution came from Punjab, 9% from other Pakistani provinces, and 13% from neighbouring nations.
12. Recent World Bank analysis found five anthropogenic sources of air pollution in the Punjab Province airshed that account for 75% of air pollution. Along with natural sources (mostly mineral dust and sea salt), the province's major contributors to population-weighted yearly average PM₂ include: Residential (23 percent), mostly solid fuels for cooking; transportation (16 percent), mostly heavy-duty vehicles (trucks and buses), two- and three-wheelers, and road dust; industrial (16 percent), mostly heavy industry and brick kilns; agriculture and livestock (13 percent), including crop residue burning, fertilizer, and manure; and municipal solid waste (7 percent). Greater Lahore and Punjab have different sectoral contributions, but the picture is comparable. Residential contributes the most (28%), but transport contributes more (24%). The other sectors are comparable, but natural sources affect them less (9%).
13. National and provincial priorities in Punjab include air pollution, a chronic health threat. The Punjab Government (GoPb) is laying the groundwork for better air quality management (AQM) planning and governance, which reduce air pollution. The 2017 Smog Control Policy, 2022 Health Advisory System for Critical Air Pollution Events, 2023 Clean Air Policy and Action Plan, and 2024 Smog Mitigation Action Plan show GoPb's commitment to air quality. The plans aim to reduce PM_{2.5} in transport, agriculture, industry, energy, and municipalities while raising public awareness. Limited institutional capability and money make policy enforcement difficult despite these attempts. The Punjab Environment Protection and Climate Change Department (EPCCD), which manages air quality, lacks manpower, infrastructure, and systems.
14. Punjab AQM governance requires technological features. GoPb is boosting air quality monitoring under the Punjab Green Development Program. This extension is essential for air quality compliance and public education. Additional monitoring stations, operating processes, data validation, and quality assurance strategies are needed. To make informed judgements and evaluate actions, the province must strengthen its analytical framework, including emissions inventory, scenario planning, and decision support systems. Public awareness initiatives help turn monitoring data into useful information and promote pollution reduction strategies. These campaigns should make data available and involve residents in air quality improvement.
15. Road travel is Punjab's second-largest air polluter, and emissions will rise without action. Heavy vehicles (trucks and buses), two- and three-wheelers, and cars contribute most PM_{2.5}. Rapid private vehicle growth indicates a problem that could increase emissions if not handled. Private automobiles in Lahore have doubled in the last decade to 7.4 million, with 82% being two-wheelers.⁷ Income growth, urbanization, and public transportation shortages are to blame. With 27 km of metro, a BRT line, and feeder buses, Lahore has a

small mass transit system. In 2026, Lahore will need over 3,000 buses, yet 800 presently operate. Public buses are overloaded and a last resort. Lahore's public buses cost PKR 15.00 to 30.00 each journey, requiring US\$15 million in annual subsidies, straining fiscal resources. GoPb has commissioned many integrated transport studies, including the Lahore Transport Study with thorough stakeholder consultations, to improve urban transit.

16. Outdated emission standards increase road transport emissions. The EURO II emissions standard is far dirtier than current ones. EURO II diesel buses generate 16 times higher PM_{2.5} per kilometer than EURO VI. Unclean diesel with 0.5 percent Sulphur is still used. Expand fuel quality inspections to reduce adulterated gasoline and fuel quality testing facilities to verify emission regulations.
17. Road transport pollution can be reduced utilizing Avoid-Shift-Improve (ASI). "Avoid" means lowering motorized travel, "shift" means boosting public and non-motorized transport, and "improve" means reducing car emissions, partially through low-emission automobiles. World Bank investigations have found cost-effective ways to cut emissions, such as increasing heavy-duty vehicle inspection and maintenance and introducing electric two- and three-wheelers (e-2/3Ws). Electric buses (e-buses) can reduce emissions by combining "shift" and "improve" dimensions, especially when replacing private automobiles. E-buses lower life cycle PM_{2.5} emissions by 88% compared to EURO VI diesel buses and over 98% compared to vehicles and two-wheelers of equal capacity, emphasizing the need for modal shifts to maximize emissions reduction.
18. Punjab's e-bus initiative is gaining steam. The total cost of ownership for e-buses in Pakistan is greater than EURO II buses but lower than hybrid and EURO VI diesel buses. E-buses cost more upfront but save on operations and maintenance. GoPb purchased 27 e-buses for a Lahore pilot under the PGDP, preparing for scale-up. Under its Annual Development Program in FY25, GoPb wants to install 500 e-buses across Punjab, expanding public transport. Punjab mass transit agencies need e-bus operations and contract management training to roll out e-buses successfully. Public-private partnership (PPP) e-bus deployment may become more appealing as macroeconomic conditions improve.
19. E-2/3Ws provide favorable cost economics for electrification, but high upfront costs, limited charging infrastructure, and the lack of accessible commercial financing are hampering implementation. In high-utilization commercial applications, E-2Ws offer a total cost ownership benefit. Analyses suggest that commercial e-2Ws can recover the greater upfront cost in two years and commercial e-3Ws in four years through decreased operations and maintenance costs. Increasing risk sharing for e-2/3Ws can boost cheap commercial finance. Risk sharing strategies like partial credit guarantees have been shown to accelerate EV adoption in India and are being tested in Punjab under the Chief Minister's Youth Initiative. Such procedures can provide financial institutions confidence to lend for EVs, improving their risk assessment and management. Risk sharing methods can be reduced, creating a self-sustaining EV financing environment.
20. Besides electricity, off-road polluting vehicles are needed. Punjab is implementing car inspections. It was introduced for PSVs in 2015 and expanded to private vehicles under PGDP. Two-wheelers aren't inspected. PSVs that fail emissions inspections lose their route permits, but the government cannot dismantle or confiscate them. Inspection noncompliance fines are low. The Smog Mitigation Action Plan considers buying back polluting vehicles for environmental reasons. Peshawar¹⁰, Pakistan, Latin America, and China have shown success using such incentives.
21. Agriculture is Punjab's fourth biggest air polluter. Six to seven million tonnes of rice straw

are burned in Punjab annually. Secondary pollutants include improper nitrogenous fertilizer and manure management. A comprehensive short-, medium-, and long-term approach by the GoPb may minimize air pollution, solve sectoral development issues, and expand on the Strengthening Markets for Agriculture and Rural Transformation (SMART, P162446) Program. In the short run, the GoPb might use super seeders and other machinery to handle rice straw instead of burning and raise awareness. Digital solutions can help small farmers access super seeders by providing rental subsidies using established worldwide models like Hallo tractor. ADB's Low Carbon and Adaptive Agriculture Project intends to provide efficient rice harvesters to gather and value rice straw to reduce burning and raise awareness in the medium term. Policy reform that prioritizes research and development, diversified cropping from rice-wheat monocropping, livestock soil fertility management, agriculture marketing, and value chain infrastructure development could benefit the GoPb in the long run.

22. Punjab's first and third largest air polluters, residential cooking and industry, need further analysis before intervention. Traditional cookstoves that pollute home and ambient air must be replaced. In the meanwhile, GoPb can encourage the adoption of upgraded cookstoves and remove barriers to cleaner fuels. Industrial technology must become greener and more efficient. The enabling environment for market-based green finance must be strengthened and enforcement customized to each subsector. The Smog Mitigation Action Plan prioritizes brick kilns, boilers, including alternative fuels, emissions control systems in critical industries, green investments, and cleaner technology in industry. Better regulation would assist heavy industries by requiring tougher emissions standards, continuous stack emissions monitoring, and end-of-pipe stack emissions controls.

2.2. Relationship to Country Partnership Strategy

23. The Program is consistent with the World Bank Group's (WBG) Pakistan Country Partnership Strategy (CPS) FY15–FY19 discussed by the Board on May 1, 2014 (Report No. 84645-PK) and subsequently extended to FY21 under the corresponding May 2017 Performance and Learning Review (Report No. 113574). The Program will support Crosscutting Area 3 (Climate change adaptation and mitigation in public and private sectors) by improving air quality in Lahore Division. The Program will also contribute to Results Area 2 (Private sector development) by creating the conditions for private sector investments across sectors. The Program will support Results Area 3 (Inclusion) by generating employment opportunities for women. Lastly, the Program will support Crosscutting Area 2 (Deepening engagement at the province level) by strengthening air quality governance and institutional capacity. The Program also fits in with World Bank's AQM IGP-HF program and aligns with the World Bank South Asia Regional Integration, Cooperation, and Engagement Approach pillar on Reducing Vulnerabilities and Increasing Resilience.

2.3. The Program

24. The Punjab Clean Air Policy and Action Plan (2023) and the Punjab Smog Mitigation Action Plan (2024) form the basis of the government program ('p', the program) for this PforR. The Punjab Clean Air Policy and Action Plan has a longer time horizon (through 2030) and covers air quality improvement objectives, policy interventions across the key sectors, implementation mechanisms, coordination and monitoring. The Smog Mitigation Action Plan (2024) is the short-term implementation mechanism, with

budgeted activities for one year (up to Q1 2025). GoPb is expected to update the Smog Mitigation Action Plan annually with priority activities and budgets. Both these plans identify the need for action across multiple sectors and emphasize the need for awareness raising and behavior change to reduce air pollution.

25. World Bank's support for GoPb's program will be offered through sequential operations to support AQM, the first of which will have a stronger focus on managing road transport emissions. This operation entitled the "Punjab Clean Air Program (P-CAP)" will support the government's plans to curb emissions across priority sectors, viz. transport and agriculture, with assessments in the industry and residential cooking sectors to lay the foundation for subsequent operations.
26. The proposed P-CAP PforR Program ('P', the Program) will support selected activities of GoPb's program where World Bank would add the most value. The 'Program' aims to strengthen AQM and reduce emissions from targeted sectors. It will achieve these objectives through three results areas across sectors for a first series of operations to tackle air pollution sources by 2035: (i) Air Quality Management Governance and Awareness Raising, (ii) Transport Sector Abatement Measures, and (iii) Agriculture Sector Abatement Measures. A complementary IPF component will support the technical assistance and assessments required to facilitate Program implementation, and design interventions in industries sector, to be implemented under a subsequent operation under the series of operations to support air quality improvement.
27. P-CAP will use a hybrid approach that combines the PforR and IPF instruments. The World Bank's strategic, technical, and financing support will help GoPb tackle the challenges of a large and complex air quality program, while developing capacity for the long term. The bulk of AQM governance and sectoral interventions will be supported under a PforR instrument. PforR will provide financial incentives to a broad range of stakeholders to foster alignment towards improving air quality. The IPF component will cover technical assessments required to implement the transport sector abatement measures, as well as design the industrial sector interventions to be supported in a subsequent operation.
28. The Program Development Objective of the Program is to strengthen air quality management and reduce emissions from targeted sectors in Greater Lahore.
29. The following section describes the three results areas of the Program.
30. **RA1: AQM Governance and Awareness Raising.** This incentivizes EPCCD to enhance Punjab's regulatory, institutional, and technical capacity for AQM, as well as encourages robust data-driven communications and awareness raising to bring together diverse actors towards the goal of improving air quality.
 - a. **Strengthen AQM infrastructure.** The Program will support the expansion of air quality monitoring regulatory grade stations across Punjab, with real-time data reporting on a publicly available website. The Program will also enhance fuel quality testing by adding two new laboratories (one mobile, one fixed), with testing quality aligned with the American Society for Testing and Materials (ASTM) standard.
 - b. **Data, inventory, and source attribution.** The Program will facilitate the development of an integrated pollutant and greenhouse gas (GHG) emissions inventory system, which will provide a comprehensive and systematic approach to track and manage

emissions data. The inventory will be shared with the Minister for Environment to inform policy making, including future revisions to Punjab's Smog Mitigation Action Plan.

- c. **Strengthening public communications and raising awareness.** The public communications campaign will be informed by a data-driven stock take early in the Program, drawing on lessons learned from previous campaigns, monitoring data, and existing inventories. The campaign will then use different channels (including five "roadshow" vehicles and 450 awareness sessions) to inform stakeholders about air pollution sources, health impacts, and available reduction measures, with special attention to vulnerable populations. This comprehensive outreach effort builds upon the existing health advisory system supported by PGDP.

31. **RA2: Transport Sector Abatement Measures** will focus on measures to curb road transport emissions through a combination of shift and improve measures:

- a. **Expanding and encouraging modal shifts to mass transit.** This incentivizes the deployment of 648 e-buses in Lahore and supporting infrastructure *viz.* depots and charging facilities. Public-private partnership (PPP) options will be explored under a study to identify business model(s). GoPb will be incentivized to implement a distance-based fare policy for e-buses using updated payment systems, which could increase fare revenue and improve the financial sustainability of buses. A Public Transport Service Contract (PTSC) between GoPb and the Punjab Mass transit Authority (PMA) that improves government funding commitment in exchange for greater accountability from PMA will also be incentivized. Measures to boost female labor force participation in e-bus and depot operations will be promoted. Complementary interventions to improve access to public transport while tackling road dust will form part of the results framework.
- b. **Fostering the transition to e-2/3Ws.** GoPb will be incentivized to adopt a provincial electric vehicle (EV) policy with clear EV adoption targets, institutional arrangements and action plan, and an associated charging infrastructure roadmap for key cities in Punjab. A provincial EV Cell will be established to spearhead policy implementation. A financing facility will be established via a financial intermediary (Bank of Punjab) to catalyze the flow of affordable commercial financing to e-2/3Ws across Punjab.
- c. **Expanding vehicle inspections and retiring polluting vehicles.** The combination of an expanded vehicle inspection and maintenance regime and a buyback scheme will reduce the emissions intensity of the existing vehicle stock. Two Disbursement-Linked Indicators (DLIs) will incentivize the expansion of vehicle inspections to 2Ws and a buyback scheme for aged and polluting vehicles, with associated amendments to the Provincial Motor Vehicles Ordinance (1965). Financial incentives to encourage cleaner alternatives will be provided, with the possibility of linking these to the EV financing facility described in the para above.

32. **RA3: Agriculture Sector Abatement Measures.** This RA will finance abatement measures in the agriculture sector. Activities in this RA will focus on immediate alternative measures to address rice crop residue burning, secondary pollution contribution from fertilizer used, and awareness raising among farmers on air quality.

- a. **Alternative to crop residue burning.** This activity will incentivize the scale-up of GoPb's initiative on the provision of 5,000 super seeders and other straw collection and management machinery to small farmers as an alternative to rice crop residue burning.

- b. **Digital access to straw management machinery and crop residue management service.** This activity will focus on digital solutions using proven models to improve access to super seeders, balers, and rice harvesters to small farmers through rental subsidies to reduce crop residue burning.
- c. **Awareness raising and enforcement.** This activity will run rigorous campaigns on awareness raising on alternative options for crop residue management, digital machinery rental services and subsidies, off-farm management options for straw management, and banning of burning to improve air quality in Punjab. These efforts will include targeted awareness campaigns for women in agriculture, enabling them to adopt climate-smart practices and advocate for better residue management, supporting emissions reduction goals.
- d. **Research and development.** This activity will explore promotion of soil testing and slow-release fertilizer alternatives, and policy reforms for the introduction and enforcement of slow-release fertilizer manufacturing and use. This intervention will be included in the long-term measures.

33. An IPF component of US\$20 million will support studies and capacity building to facilitate PCAP implementation and identify industrial abatement measures for SOP 2. The list of activities is outlined below.

Table 1: IPF Studies

Study to determine business model for e-buses	To determine appropriate business models for e-bus deployment in Punjab. Technical feasibility studies are already underway under the government program.
Public transport satisfaction survey	To gather regular feedback on the experience of passengers on public transport.
Buyback scheme	Consultancy service to design buyback scheme for polluting vehicles, with detailed stakeholder consultations.
Battery waste management	Consultancy service to develop battery waste management rules and identify strategies for ecosystem development.
Study on technical and legal component for 2Ws inspections	Consultancy services to develop legal, financial, and technical framework for inspection and emission certification of 2Ws.
E-2/3W adoption	Technical assistance (TA)/consultancy to design e-2/3W financing facility.
Carbon credits	End-to-end process for monetizing GHG emissions reductions from the shift to cleaner vehicles in the form of carbon credits.
Institutional strengthening (transport)	The component will support institutional strengthening for transport sector agencies with a focus on curbing road transport emissions, bus procurement and contract management, and road safety.
Industry detailed study	Detailed study of cleaner technologies and alternative less polluting combustion fuels for boilers, furnaces, and brick kilns. The study will first map the problem in Greater Lahore and 3 districts, and then propose a set of solutions, considering many different scenarios. The study will be used to inform the design of SOP 2.
Other studies	Any other studies to support Program implementation as necessary.
Staffing and miscellaneous Program implementation costs	Staff salaries and miscellaneous expenses, aligned with the institutional arrangements described below, will also be supported via the IPF component.

2.4. Program Institutional and Implementation Arrangements

34. GoPb has established a P-CAP Program Coordination and Monitoring Unit (PCAP-CMU) under P&D Board. The PCAP-CMU will be responsible for overall Program coordination across the different implementing agencies responsible for respective DLIs and results areas, as well as monitoring and reporting of Program progress, DLIs/DLRs. The PCAP-CMU will be headed by a Program Director.
35. A Program Steering Committee (PSC) will oversee Program implementation and provide strategic guidance to the PCAP-CMU and implementing agencies. The PSC will be led by Chairman P&D and include senior management (secretary or director general level) representation from T&MD, EPCCD, Agriculture, and Industries departments.
36. A dedicated Program Implementation Unit (PIU) will be established in T&MD given the scope and scale of transport sector activities under the Program. Remaining implementing agencies, i.e., EPCCD and Agriculture will execute the Program using their existing staff. The departments will involve agencies under their charge for selected Program activities or studies, viz. the Punjab Mass Transit Authority (PMA), the Punjab Transport Company (PTC), the Punjab Provincial Transport Authority (PTA), and EPA. A study on industries will be procured by PCAP-CMU with technical reviews from the Department of Commerce and Industries.

2.5. Program DLIs

37. The details of the Disbursement Linked Indicators (DLIs) under which the Program will measure progress towards the PDO, are given below:

Table 2: Program DLIs

Purpose of DLI	DLI	Agency responsible	Financing (US\$ mil)
Results Area 1: Air Quality Management Governance and Awareness Raising			
Strengthen air quality monitoring on a continuous basis	DLI 1: Additional regulatory grade monitors added to air quality monitoring network (Number)	EPCCD/EPA	9
Strengthen air quality management governance	DLI 2: Integrated air pollutant/SLCP/GHG emissions inventory system is developed and routinely updated, used for assessment of Smog Action Plan and informs annual development plans (Yes/No)	EPCCD/EPA	6
Strengthen air quality monitoring infrastructure	DLI 3: Fuel quality testing laboratories with mobile and fixed facility operational (Number)	EPCCD/EPA	3
Enhance public awareness campaign and citizen engagement	DLI 4: Interactive public information and awareness sessions held (Number)	EPCCD/EPA	2
Results Area 2: Transport Sector Abatement Measures			
Expand public transport provisions with electric buses	DLI 5: Expand Cleaner Public Transport in Lahore Division (Electric Buses) (Number)	T&MD via PMA/PTC	140

to encourage modal shifts away from private vehicles			
Catalyze a transition towards cleaner 2/3Ws	DLI 6: Transition to Electric Two- and Three-Wheelers (new vehicles)	T&MD; EPCCD	50
Expand vehicle inspections and strengthen nexus with vehicle licensing and retirement	DLI 7: Expand vehicle inspections	T&MD	10
Scrap polluting vehicles with appropriate financial incentives	DLI 8: Buyback of aged polluting vehicles	T&MD	40
Results Area 3: Agriculture Sector Abatement Measures			
Improve mechanized rice straw management to reduce crop residue burning	DLI 9: Expanding mechanized rice straw management (Number)	Agriculture Department	20

3. ENVIRONMENTAL AND SOCIAL EFFECTS OF THE PROPOSED PROGRAM

38. Consistent with the requirements of the Bank PforR financing, the proposed PforR operation does not support activities that pose high environmental or social risks. The activities to be supported by the Program are likely to provide environmental and social benefits, and substantial environmental and social impacts and risks.

39. The environmental and social risk rating of the Program is 'Substantial'.

3.1. Environmental Benefits and Risks

3.1.1. RA - 1 (AQM governance and awareness raising) – Environmental Effects

- a) **Environmental Benefits:** The overall environmental benefit of the Program activities will be the improved air quality of Punjab in general and specifically Greater Lahore and reduction in disease burden attributed to air pollution. Fuel testing laboratories will facilitate in controlling fuel adulteration practices and contribute towards the use of standard/cleaner fuel by the vehicles. Use of standard/clean fuel will ultimately improve the urban air quality due to reduction in air pollutants in the vehicular tailpipe emissions. Improved air quality governance will enhance air emission monitoring and tracking of the industrial units and management through enforcement.
- **Environmental Risks:** Under RA-1, the air quality governance activities under the program will generate electronic waste (e-waste). The Program activities, under expansion of air quality monitoring regulatory grade stations across Punjab, with real-time data reporting, will require procurement and installation of IoTs (Internet of Things) i.e., the devices with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the internet or other communication networks. By virtue of IoTs, e-waste is expected to be generated by the end of life of the procured equipment. The environmental and health and safety impacts occur during improper recycling and disposal of the e-waste. The improper recycling arrangements result in polluting air, water and soil due to release of pollutants such as persistent organic pollutants (POPs) including flame retardants (poly brominated biphenyl), dioxins (polychlorinated dibenzodioxins and dibenzofurans), perfluoroalkyls, polycyclic aromatic hydrocarbons (PAHs), and heavy metals (lead, chromium or hexavalent chromium, cadmium, mercury, zinc, nickel, lithium, beryllium) during dismantling, burning, chemical processing and disposal activities. The e-waste management situation in Pakistan is described in annex 2.
 - The environmental risks of fuel testing laboratories primarily stem from the use of hazardous chemicals and the generation of chemical waste, including potential air, water and soil pollution if not properly managed. From the mobile laboratories, these risks also potentially become diffused-source pollution hazards.
 - The operation of roadshow vehicles may pose some environmental risks, related to air emissions and the potential leakage of chemicals like lubricants if vehicles are not properly maintained. Additionally, there could be public disturbance due to excessive horn use in sensitive areas such as hospitals and schools and licensed drivers are not used.

40. The details of environmental assessment for RA-1 are in Table – 3.

3.1.2.RA – 2 (Transport sector abatement measures) – Environmental Effects

- **Environmental Benefits:** Deployment of e-vehicles will also result in a reduction in fossil fuels consumption and contribute to less reliance on fossil fuels of the province. Improved public transport in the cities will reduce traffic congestion and improve ambient conditions due to less vehicles (2/3 wheelers and private cars) on the road. The environment of the cities will be improved due to improved governance, and e-vehicles operation and retiring/scraping of old/polluted vehicles.
- **Environmental Risks:** Under RA-2, most of the environmental risks of the Program are related to transport related activities. Under Program component of deploying e-buses and promoting e-two/three wheelers uptake, the operation of these e-vehicles will generate greenhouse gas emissions indirectly by battery charging through grid electricity supply (grid supply in the Punjab is mostly fossil fuel based). The electricity consumed to travel one kilometer by e-bus is about 1 kWh which will generate about 0.4 kg CO₂e greenhouse gases. In comparative terms the amount of greenhouse gas emissions due to battery charging through grid electricity will be substantially lower than the diesel buses and petrol run two/three wheelers.
- Overall, the electric buses and e-two/three wheelers are safer because of advanced safety technology features in their designs. However, Lithium Iron Phosphate (LiFePO₄) battery used in e-buses and e-2/3 wheelers poses risks in terms of safety, since buses that use this battery for power can represent a higher fire risk. When the battery is damaged, it presents a significant fire hazard that is not solved by traditional fire extinguishing system. When a lithium battery is exposed to extremely high temperatures, mechanical failure, overcharging, or sustains a physical impact, its internal cells can short circuit, producing excessive heat. Excessive heat can trigger a chemical reaction that is known as "thermal runaway," in which the heat is continually produced. Resultantly, the overheated battery can ignite, explode, and/or emit toxic gases. Subsequently, the associated risks are worsened since the lithium battery produces its own oxygen inside its cells after entering the thermal runaway stage. Thus, in event of a fire breakout, the oxygen continues to fuel the flames and traditional fire suppression systems are not able to handle fires that cannot be starved of oxygen. Overall, battery fires are less common than traditional engine fires, but when they occur, they pose a much higher safety risk. Battery charging stations and battery storage areas are also assessed as the potential fire prone areas.
- The main components of LiFePO₄ battery are cathode materials i.e., LiFePO₄, anode materials (graphite), current collectors i.e., aluminum (Al) and copper (Cu), electrolyte salts such as lithium hexafluorophosphate (LiPF₆), organic solvents (ethylene carbonate, diethyl carbonate, ethyl methyl carbonate, dimethyl carbonate, etc. All these different components contain hazardous materials. Similarly, batteries to be used for solar power storage at the bus depots and charging stations will contain hazardous material. When these batteries are recycled and disposed of under improper disposal arrangements, the environment and the health and safety impacts will occur in the form of discharge of heavy metals in the environment, metal dust, organics, and fluorine contaminations and risks to the health of the associated workers and communities. Once the electrode materials enter the environment, metal ions from the cathode, carbon dust from the anode, strong alkali, and heavy metal ions from the electrolyte may cause severe environmental pollutions, hazards, etc., including raising the pH value of the soil, and producing the toxic gases (HF, HCl, etc.). In addition, the metals and electrolytes in batteries can harm human health.

- After a few years, large numbers of batteries (both from e-buses and from solar systems) will expire and will accordingly require disposal. The battery recycling facilities in Punjab are in the informal sector where recycling practices are not environmentally friendly, and these facilities are not regulated under any legislation. When these batteries are improperly recycled and disposed of, the environmental, Occupational Health and Safety (OHS) and Community Health and Safety (CHS) safety impacts to the associated workers and the community will occur due to discharge of heavy metals, dust and hazardous material and emissions.
- In Punjab, there is no secure landfill site where hazardous waste could be disposed. There are very limited incineration facilities in the country where hazardous waste could be incinerated. At some places, paid incineration facilities exist where hazardous waste is incinerated on a payment basis. Mostly the operation of the incineration facilities is not up to the required standard. The combustion temperatures are not maintained as high as required to completely ignite the hazardous waste. These facilities are not equipped with air pollution control devices. The disposal of the ash produced from these facilities is another issue.
- Conventional bus washing activities at bus depots consume substantial amounts of water, generating wastewater and sludge from suspended solids and soap suds. As Punjab is recently experiencing extensive water shortage, bus washing activities will contribute to additional stress on the current scarce water resources. Improper disposal of wastewater sludges will be another environmental concern at the bus depots. Furthermore, whilst EVs eliminate risks associated with fuel and motor oil leaks, they can still pose environmental risks in parking areas due to potential chemical leaks. Battery coolant leaks from damaged or aging thermal management systems can contaminate soil and water. Hydraulic brake fluid leaks, though less frequent due to regenerative braking, remain a potential hazard. Certain EV drivetrains and hybrid systems may also release transmission fluids or gear oil if seals degrade. In rare cases, battery damage or thermal runaway can result in electrolyte leakage, which contains toxic and corrosive chemicals. Additionally, washer fluid and air conditioning refrigerants may leak, contributing to environmental pollution. Without proper drainage, impermeable flooring, and regular maintenance, these leaks could lead to contamination and hazardous conditions in the bus depots/parking areas.
- Construction related activities to develop bus depots, bus stops, road repairs, improve road shoulders, develop footpaths and cycling lanes and install battery charging stations are assessed to be temporary, localized and reversible in nature, including OHS/CHS, noise, and dust as well as construction and general waste generation.
- Solarization of battery charging stations poses the risks of structural safety, especially if the solar panels are installed on the rooftops of existing structures without due consideration of their load bearing capacity. The charging stations could also pose OHS risks, especially electrical hazard during installation and operations, and fire hazards during operations. These are localized risks, manageable through compliance with existing regulations and good industry practices.
- The buyback scheme will produce a stock of old/polluting vehicles in the province whose recycling/disposal will be an environmental challenge. There could be further risk of redeployment of these vehicles, especially in rural and far-flung areas of the province, where typically enforcement of rules and standards is more difficult. These risks will be addressed by ensuring that the retired vehicles are not resold for potential continual use through formulating and implementing environmentally sound recycling/disposal strategies as per the requirements of Punjab/local rules and regulations including

Environmental Protection Act and in accordance with the World Bank's Core Principles, by the recycling facilities in the province.

Vehicle inspections is done through digital equipment and sensors. At the end of the life of these equipment and sensors, e-waste will be generated. At the operational level, low environmental impacts are expected. These impacts can be mitigated through good housekeeping practices.

41. The details of environmental assessment for RA-2 is in Table – 3.

3.1.3.RA – 3 (Agriculture Sector Abatement Measures) – Environmental Effects

- **Environmental Benefits:** Agriculture mechanization will reduce and finally eliminate the rice stubble burning situation and contribute to air quality improvement. Super-seeders will help to increase the fertility of the soil due to the mixing of stubble into the soil instead of burning it and causing air pollution. They will also avoid killing of good microorganisms of the soil, loss of soil nutrition due to burning and biodiversity loss of the fields. The mechanization will also save time as a single pass will be required for land preparation, fertilizer placement and seed sowing. The rice straw management will enhance biomass uptake as an alternative to fossil fuels by the industries under environmentally controlled technologies for PM2.5 capture.
- **Environmental Risks:** Under RA-3, the use of super-seeders may lead to soil disturbance, suspended dust and air emissions due to fuel consumption, biodiversity impacts due to promotion of monoculture, which can result in more frequent pest and disease outbreaks in crops, increased use of chemicals, and potential financial losses for farmers. The operation of super-seeders, through diesel-based tractors, will contribute about 150,000 tons of CO₂e greenhouse gases every year, in case, if the total 7 million acre of rice crop fields of Punjab is mechanized³. The Program will compute the reduction in greenhouse gases due to stoppage of crop burning in a unit area to establish the benefit of super-seeders in reduction of greenhouse gases.

42. The details of environmental assessment for RA-3 are in Table – 3.

³ Diesel consumption in the tractors to operate Super Seeders = 8 liter/acre, total diesel consumption for 7 million acre = 56,000,000 liters, diesel emission factor = 2.7 kgCO₂e/lit, total emissions = (56,000,000 x 2.7)/1,000 = 151,200 tons CO₂e

Table 3: Assessment of Environmental Benefits and Risks for PCAP

Result Area and Activities	Core Principle 1: Environmental and social management procedures and processes are designed to (a) avoid, minimize, or mitigate against adverse impacts; (b) promote environmental and social sustainability in program design; and (c) promote informed decision making relating to a program's environmental and social effects.	Core Principle 2: Environmental and social management procedures and processes are designed to avoid, minimize, and mitigate adverse impacts on natural habitats and physical cultural resources resulting from the program.	Core Principle 3: Environmental and social management procedures and processes are designed to protect public and worker safety against the potential risks associated with (a) construction and/or operations of facilities or other operational practices developed or promoted under the program; (b) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials; and (c) reconstruction or rehabilitation of infrastructure located in areas prone to natural hazards.
Results Area 1: Air Quality Governance and Awareness Raising			
DLI 1: Additional regulatory grade monitors added to air quality monitoring network DLI 2: Integrated air pollutant/SLCP/GHG emissions inventory system is developed and routinely updated, used for assessment of Smog Action Plan and informs	Positives Impacts: - Improved air quality of Punjab in general and specifically Greater Lahore. - Reduction in diseases burden attributed to the air pollution. Environmental Risks: Generation of electronic waste (e-waste) in the form of waste IoTs from air emission monitoring equipment after the end of their life and their improper recycling practices will pose environmental, health and safety risks	Positives Impacts: - Improved air quality will positively impact natural habitats by minimizing damage to plants and animals, preserving biodiversity, maintaining healthy ecosystems, and reducing the negative effects of air pollution like acid rain, which can disrupt soil chemistry and aquatic life. - Improved air quality will enhance structural life of the physical cultural resources. Environmental Risks: No considerable environmental risks on natural habitats and physical resources are associated with the air quality monitoring network.	Positive Impacts: Improved air quality of Punjab in general and specifically Greater Lahore will improve community health and reduce diseases burden attributed to the air pollution. Environmental Risks: Improper management and / or recycling of e-waste will pose environmental, health and safety risks to the workers associated with the recycling activities and the communities around the informal unregulated recycling facilities.

annual development plans			
DLI 3: Fuel quality testing laboratories with mobile and fixed facility established	Positive Impacts: Improved urban air quality due to reduction in air pollutants in the vehicular tailpipe emissions by using clean/standard fuel. Environmental Risks: The mobile and fixed laboratories for fuel quality testing pose no significant environmental risks. The environmental risks of fuel testing laboratories primarily stem from the use of hazardous chemicals and the generation of chemical waste, including potential air, water and soil pollution if not properly managed.	Positive Impacts: As above Environmental Risks: Environmental pollution associated with the improper disposal of hazardous chemicals/waste from the fuel testing laboratories such as water pollution, air pollution and soil pollution will pose serious environmental risks. Significant risks to the natural habitats and the biodiversity of the province and physical cultural resources are not foreseen associated with the mobile and fixed laboratories for fuel quality testing.	Positive Impacts: As above Environmental Risks: Lack of availability and use of PPEs and poorly designed laboratories can pose environmental, OHS and LFS risks for workers and the nearby communities associated with the establishment of mobile and fixed laboratories for fuel quality testing.
DLI 4: Interactive public information and awareness sessions held	Positive Impacts: Enhance public understanding of air pollution sources, health impacts, and mitigation measures. By raising awareness, the program will empower communities to adopt cleaner practices and support air quality improvement initiatives. This will lead to a reduction in air pollutants and greenhouse gas emissions, contributing to better air quality and public health. Environmental Risks:	Positive Impacts: As above Environmental Risks: Leakage of fuels, lubricants, or other hazardous substances from poorly maintained road-show vehicles, could lead to soil and water contamination, posing risks to ecosystems and groundwater resources.	Positive Impacts: As above Environmental Risks: Inadequately trained/unlicensed drivers can cause public nuisance due to unnecessary use of horn, especially in sensitive zones such as residential areas, school zones, hospitals, and other noise-sensitive environments.

	OHS/CHS risks due to utilization of poorly maintained vehicles for the “road-shows” resulting in release of air emissions including carbon monoxide (CO), hydrocarbons (HC), and particulate matter (PM), exacerbating local air quality issues.		
Results Area 2: Transport Sector Abatement Measures			
DLI 5: Expanding Cleaner Public Transport in Lahore (Electric Buses) DLI 6: Transition to Electric Two- and Three-Wheelers (new vehicles)	Positives Impacts: - Improved air quality of Punjab in general and specifically Greater Lahore. - Reduction in diseases burden attributed to the air pollution. - Improved environment of the cities due to e-vehicles operation. - Reduction in fossil fuels consumption in the province due to deployment of e-vehicles. - Improved public transport in the cities will reduce traffic congestion and improve ambient conditions due to less vehicles (ICE 2/3 wheelers and private cars) on the road. Environmental Risks: - Potential fire hazards due to lithium-ion batteries of the e-vehicles. - Generation of waste batteries (from e-buses and solarized charging stations) and their improper recycling	Positives Impacts: - Improved air quality due to deployment of e-vehicles will positively impact natural habitats by minimizing damage to plants and animals, preserving biodiversity, maintaining healthy ecosystems, and reducing the negative effects of air pollution like acid rain, which can disrupt soil chemistry and aquatic life. - Improved air quality will enhance structural life of the physical cultural resources. Environmental Risks: Environmental pollution associated with the improper waste batteries recycling practices such as water pollution, air pollution and soil pollution will pose serious risks to the natural habitats and the biodiversity of the province.	Positive Impacts: Improved air quality of Punjab in general and specifically Greater Lahore due to e-vehicles deployment will improve community health and reduce diseases burden attributed to the air pollution. Environmental Risks: Improper recycling of waste batteries will pose environmental, health and safety risks to the workers associated with the recycling activities and the communities around the informal unregulated recycling facilities.

	will pose environmental health and safety risks		
DLI 7: Expand vehicle inspections	Positive Impacts: Improved urban air quality due to operation of environmentally sound vehicles on the roads. Environmental Risks: No considerable environmental risks are associated with the vehicle inspection activities.	Positive Impacts: • Improved air quality due to operation of environmentally sound vehicles on the roads will positively impact natural habitats by minimizing damage to plants and animals, preserving biodiversity, maintaining healthy ecosystems, and reducing the negative effects of air pollution like acid rain, which can disrupt soil chemistry and aquatic life. • Improved air quality will enhance structural life of the physical cultural resources. Environmental Risks: No considerable environmental risks to the natural habitats, biodiversity and the physical cultural resources are associated with the vehicle inspection activities.	Positive Impacts: Improved air quality of Punjab in general and specifically Greater Lahore due to the operation of environmentally sound vehicles on the roads will improve community health and reduce diseases burden attributed to the air pollution. Environmental Risks: No considerable environmental risks are associated with the vehicle inspection activities on workers and the communities.
DLI 8: Buyback of aged polluting vehicles	Positive Impacts: Improved urban air quality due to reduction in old/unfit/polluted vehicles from the roads. Environmental Risks: The buyback scheme will produce stock of old/polluted vehicles in the province whose recycling/disposal will be an environmental challenge. Improper recycling/disposal will pose environmental risks such as water, air and soil pollution.	Positive Impacts: • Improved air quality due to removal of old/unfit/polluted vehicles from the roads will positively impact natural habitats by minimizing damage to plants and animals, preserving biodiversity, maintaining healthy ecosystems, and reducing the negative effects of air pollution like acid rain, which can disrupt soil chemistry and aquatic life. • Improved air quality will enhance structural life of the physical cultural resources. Environmental Risks: Environmental pollution associated with the improper recycling/disposal of old/polluted vehicles such as water, air, and soil pollution	Positive Impacts: Improved air quality of Punjab in general and specifically the Greater Lahore due to the removal of old/unfit/polluted vehicles from the roads will improve community health and reduce diseases burden attributed to the air pollution. Environmental Risks: Improper recycling of old/polluted vehicles will pose environmental, health and safety risks to the workers associated with the recycling activities and the communities around the informal unregulated recycling facilities.

		will pose serious risks to the natural habitats and the biodiversity of the province.	
Results Area 3: Agriculture Sector Abatement Measures			
DLI 9: Expanding mechanized rice straw management	Positives Impacts: • Agriculture mechanization will reduce and finally eliminate the rice stubble burning situation and contribute to air quality improvement. • Super Seeders will help to increase the fertility of the soil due to the mixing of stubble into the soil instead of burning it and causing air pollution. • Super Seeders will avoid killing of good microorganisms of the soil, loss of soil nutrition due to burning and biodiversity loss of the fields. • The mechanization will save time as a single pass will be required for land preparation, fertilizer placement and seed sowing. • The rice straw management will enhance biomass uptake as an alternative to fossil fuels by the industries under environmentally controlled technologies for PM_{2.5} capture from the stacks. Environmental Risks: Super Seeders' operation through diesel-based tractors, will contribute about 150,000 tons of CO₂e greenhouse gases every year, in case,	Positives Impacts: • Improved air quality due to crop burning stoppage will positively impact natural habitats by minimizing damage to plants and animals, preserving biodiversity, maintaining healthy ecosystems, and reducing the negative effects of air pollution like acid rain, which can disrupt soil chemistry and aquatic life. • Improved air quality will enhance structural life of the physical cultural resources. Environmental Risks: Operation of Super Seeders through tractors will disturb the natural habitats of the agricultural fields due to noise, promotion of monoculture and generation of tailpipe emissions.	Positive Impacts: Improved air quality of Punjab in general and specifically the Greater Lahore due to crop burning stoppage will improve community health and reduce diseases burden attributed to the air pollution. Procurement of fuel-efficient technology will have further positive environmental impacts in terms of reduction of fuel consumption and release of air emissions. Environmental Risks: No considerable environmental risks to the workers and the communities are associated with the provision of Super Seeders for rice management.

	if the total 7 million acre of rice crop fields of Punjab is mechanized. Though these emissions will be much lower than the emissions resulting due to crop burning.		
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3.2. Social Benefits and Risks

3.2.1. RA-1 (AQM governance and awareness raising) – Social Effects

43. This RA incentivizes EPCCD to enhance Punjab’s regulatory, institutional, and technical capacity for AQM, as well as robust communications and awareness raising including roadshows to corral diverse actors towards the goal of improving air quality.

44. A summary of social benefits and risks under this RA is provided below, followed by a DLI wise social benefits and risk assessment for the relevant Core Principles.

- **Social Benefits:** Activities for strengthening air quality monitoring and enforcement capacity will directly benefit the citizens of Punjab in general and the Greater Lahore region specifically by helping improve the air quality. The awareness raising and communications campaigns will help inform the public on the health impacts of air pollution and abatement measures, allowing them to take steps to safeguard their well-being poor air-quality events. Sharing of information by the government on air pollution abatement measures will also help build citizen-state trust.
- **Social Risks:** There is a risk of redundancy of government employees with low or zero IT literacy, especially older and female staff. Increased monitoring may also result in stricter actions taken on smaller/less influential industries, leading to possible closures and loss of jobs and workers protests. The operations of “roadshow” vehicles and public awareness sessions carries a risk of community health and safety, especially for children and women attending these events. There is also a risk of exclusion of women, illiterate, persons with disabilities, rural communities and those without access to digital mediums from accessing awareness materials.
- **The expansion of air quality monitoring stations may prioritize urban centers, potentially leaving rural or marginalized communities without adequate air quality data and regulatory oversight.** If external calibration and real-time data reporting are not effectively enforced or made publicly accessible, there is a risk of misinformation or lack of transparency in decision-making. Enforcement of stricter fuel quality and emissions regulations may disproportionately affect small-scale fuel vendors, transporters, and industries that rely on outdated fuel sources, leading to potential economic displacement or loss of livelihoods. Stricter monitoring and regulatory enforcement could trigger resistance from businesses and industries, leading to social unrest or lobbying efforts to weaken enforcement. The development of an emissions inventory system requires careful handling of data to ensure it is not misused for punitive actions that disproportionately affect certain businesses or communities. If data collection methodologies do not adequately capture informal or small-scale polluters (e.g., small brick kilns, informal transporters, or waste burning), policies may unfairly target only large industrial actors while neglecting other significant sources. Communities might distrust the inventory system if they perceive it as favoring industrial interests or failing to account for their lived experiences with air pollution. Public messaging on health risks from air pollution could cause anxiety, mistrust, or even resistance, especially if communities feel they are being blamed for pollution rather than supported in mitigation efforts. Awareness campaigns that do not specifically target women, low-income workers, and other vulnerable populations may fail to address their unique exposures and mitigation needs. In areas where pollution

is linked to livelihoods (e.g., agriculture-related burning, transport workers, or small factories), there may be resistance to communication campaigns if alternative solutions are not provided alongside awareness-raising efforts.

45. The activities supporting this results area are expected to produce social benefits and social risks, summarized below and expanded upon in the table – 4.

3.2.2.RA – 2 (Transport sector abatement measures) – Social Effects

46. This RA will focus on measures to curb road transport emissions through expanding and encouraging modal shifts to mass transit; fostering the transition to e-2/3Ws; expanding vehicle inspections and retiring polluting vehicles.

47. A summary of social benefits and risks under this RA is provided below, followed by a DLI wise social benefits and risk assessment for the relevant Core Principles.

- **Social Benefits:** Activities under this Results Area will have numerous social benefits including contributing to better air quality; increased access to public transport; safe, comfortable and accessible e-buses; increased female workforce; and access for women and youth to financing for electric two and three wheelers.
- **Social Risks:** Construction activities for the e-bus depots and operations of e-buses and maintenance at depots carry health and safety risks for workers as well as the public. There is also a risk of land acquisition and forced removal of encroachers from lands identified for the e-bus depots. A distance-based fare policy may increase costs for low-income commuters, disproportionately impacting daily wage earners. Poor road infrastructure and lack of inclusive design may hinder persons with disabilities from fully benefiting from the mass transit system. Risks of sexual exploitation, abuse and sexual harassment (SEA/SH) exist for women employed in the operations of e-buses as well as women using public transport. The transition to e-buses could displace existing informal transport operators (e.g., rickshaw drivers), leading to economic losses and opposition. Traditional mechanics and workers dependent on internal combustion engine vehicles may face job displacement as EVs require different skill sets.
- Risks of exclusion exist for women, students and low-income individuals from accessing financing facilities for electric two and three wheelers. Unequal deployment of charging stations could exclude rural and low-income urban areas, limiting adoption.
- Stricter vehicle inspections could disproportionately penalize small vehicle owners who cannot afford upgrades, risking livelihood losses for informal transport workers. There is a risk that wealthier or politically connected individuals could bypass regulations, while enforcement falls disproportionately on lower-income groups. There is a risk of exclusion of lower income individuals with polluting vehicles who are unable to retrofit or change their vehicles through the buyback scheme. A buyback and retrofitting scheme without a clear transition plan could lead to sudden unemployment among informal transporters and small mechanics. Improper recycling of scrapped vehicles could pose health and safety risks for workers due to toxic material exposure.

48. The activities supporting this results area are expected to produce social benefits and social risks, summarized below and expanded upon in the table – 4.

3.2.3.RA – 3 (Agriculture Sector Abatement Measures) – Social Effects

49. This result area will incentivize abatement measures in the agriculture sector. Activities under the agriculture sector in this RA will focus on immediate alternative measures to address rice crop-residue burning, secondary pollution contribution from fertilizer used and awareness raising among farmers on air quality.
50. A summary of social benefits and risks under this RA is provided below, followed by a DLI wise social benefits and risk assessment for the relevant Core Principles.
- **Social Benefits:** Activities under this Results Area will have numerous social benefits including contributing to better air quality; improved mechanisms to manage rice straw; and increased access to and subsidies for farm machinery rentals.
 - **Social Risks:** Small and marginalized farmers, especially women, may face barriers in accessing super seeders and straw management machinery due to lack of awareness, affordability, or digital access. The transition from crop burning to mechanized alternatives may increase input costs for small farmers, impacting their financial stability. New machinery (e.g., super seeders) poses occupational hazards for farmers and laborers due to lack of proper training and safety measures. Illiterate, persons with disabilities, rural, and women farmers may struggle to use digital platforms for machinery rentals and subsidies. The availability of machinery may favor larger farmers, leaving smallholders at a disadvantage. Farmers lacking access to alternative solutions may face unequal enforcement of crop burning bans, disproportionately affecting smallholders. Awareness campaigns may not effectively reach rural and marginalized farmers, particularly women and those with limited education. Small-scale farmers may be unable to afford soil testing and slow-release fertilizers, limiting adoption.
51. The activities supporting this results area are expected to produce social benefits and social risks, summarized below and expanded upon in the table – 4.

Table 4: Assessment of Social Effects and Risks for PCAP

RA and Activities	Core Principle 3: E&S management systems are designed to protect public and worker safety against the potential risks associated with the construction and/or operation of facilities or other operational practices under the Program	Core Principle 4: Program E&S systems manage land acquisition and loss of access to natural resources in a way that avoids or minimizes displacement and assists affected people in improving, or at the minimum restoring, their livelihoods and living standards	Core Principle 5: Due consideration is given to cultural appropriateness of, and equitable access to, Program benefits, giving special attention to rights and interests of indigenous peoples and to the needs or concerns of vulnerable groups.	Core Principle 6: Avoid exacerbating social conflict, especially in fragile states, post-conflict areas, or areas subject to territorial disputes.
Results Area 1: Air Quality Governance and Awareness Raising				
DLI 1: Additional regulatory grade monitors added to air quality monitoring network DLI 2: Integrated air pollutant/SLCP/GHG emissions inventory system is developed and routinely updated, used for assessment of Smog Action Plan and informs annual development plans DLI 3: Fuel quality testing laboratories	Positives Impacts: Activities for strengthening air quality monitoring and enforcement capacity will directly benefit the citizens of Punjab in general and the Greater Lahore region by helping improve the air quality and safeguarding the well-being of the public. Social Risks: Government staff with limited digital literacy face a risk of redundancy and exclusion due to the introduction of innovative/artificial	N.A.	Positives Impacts: Improved air quality will help improve the well-being of vulnerable groups including the elderly, school children, and those at greater risk for health impacts due to air pollution. Social Risks: Increased monitoring may disproportionately impact smaller or less influential industrial units who are easier to penalize and act against.	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. Improved air quality monitoring and enforcement is likely to improve the air quality and enhance citizen-state trust. Social Risks: - The closure of polluting industrial units may result in loss of jobs, especially for labor, resulting in possible protests and conflicts. - There is a risk of reinforced existing inequalities in monitoring/penalizing of certain smaller industries

with mobile and fixed facility established	intelligence technologies for data validation and analysis, and emissions inventory. This risk may be greater for women and elderly staff members. OHS risks for staff working in the fuel testing laboratories resulting from exposure to fuel fumes, and insufficient OHS measures in mobile facilities.			over others and SMEs over larger influential factories due to easier access by enforcement agencies.
DLI 4: Interactive public information and awareness sessions held	Positive Impacts: Greater public awareness on air pollution, abatement measures and poor air-quality events will help improve public well-being by equipping them with information to take appropriate measures. Social Risks: Possible community health and safety risks due to accidents and crowds that may gather for 'roadshow' vehicles, especially children and women.	N.A.	Positive Impacts: Communications campaigns will include a focus on women and youth to provide them with information to improve their well-being Social Risks: This activity carries a risk of social exclusion due to lack of accessibility of communications and awareness materials by women, illiterate, persons with disabilities, rural communities and those without access to digital mediums. Communications materials used by the government rely heavily on social media, digital platforms and written communications, which may not	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. Improved air quality monitoring and enforcement is likely to improve the air quality and enhance citizen-state trust. Social Risks: Lack of clarity in communications materials may result in misunderstandings, conflicts or protests amongst the public on proposed measures to improve air quality. These will especially apply to initiatives such as 'Green Lockdowns', curbs on certain types of public transport, school and office closures, and closures of

			be accessible by the identified vulnerable groups.	markets and public centers during poor air-quality events.
Results Area 2: Transport Sector Abatement Measures				
DLI 5: Expanding Cleaner Public Transport in Lahore Division (Electric Buses)	Positives Impacts: Expanded public transport using e-buses will reduce air pollution caused by buses with combustion engines. The expanded routes will also provide the public with safer, cleaner and more accessible public transport. Social Risks: • Occupational health and safety (OHS) risks for workers exist during construction activities of e-bus depots. • There may be OHS risks for workers related to driving and maintenance of e-buses, including long work hours and lack of protective equipment. • Construction activities carry a risk of child labor or forced labor by contractors. • Community Health and Safety (CHS) risks may be related to construction	Social Risks: • This DLI includes the construction of e-bus depots on public or acquired private lands. Four parcels of private lands have tentatively been identified for this purpose. Sizes and locations of the tentatively identified private land parcels, the largest of which is 10 acres, is provided in Annex 6. • Any construction activities on government lands including potentially e-bus depots carry a risk of forced evictions and removal of informal settlers/encroachers from these lands through anti-encroachment drives (AEDs)	Positive Impacts: Bus routes are planned through extensive studies to ensure they cover high demand areas such as hospitals, markets and educational institutes. These expanded bus routes will provide increased access to public transport for the public, including women and youth. E-buses that are being procured by the government have a ground clearance of 300mm and are equipped with wheelchair ramps. Social Risks: • Risk of exclusion: The design of the e-buses and bus stops may exclude persons with disabilities and elderly from accessing the buses, such as height of buses, misalignment of bus stops and buses, no dedicated space to park wheelchairs etc. • Risk of exclusion: Distance based fare policy may result in higher fares for individuals from low-income areas	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. The introduction of additional e-buses and routes will help improve the trust of citizens in the state to provide safe and reliable public transport options. Social Risks: Loss of income due to less demand for other means of public transport may result in protests and conflict between the government and private transport contractors operating on the proposed e-bus routes.

	activities of e-bus depots in public areas as well as road safety improvements. • Further CHS risks may be related to use of public transport including accidents, overloading of passengers and injuries from automatic doors, especially for elderly and children. • SEA/SH: There is a risk of SEA/SH for women and girls using public transport, and for women employed in the operations of e-buses and at e-bus depots.		located at a distance from commercial centers. This will exclude them from using the buses for daily commutes. • The introduction of new routes and e-buses may result in loss of income for operators of other means of transport including private passenger vans, taxis, rikshaws/3-wheelers, and ride-share services.	
DLI 6: Transition to Electric Two- and Three-Wheelers (new vehicles)	Positives Impacts: Use of electric two and three wheelers will directly benefit the citizens of Punjab by helping improve the air quality in the Greater Lahore region and improving the well-being of the public. Social Risks: • The use of electric 2 wheelers will have associated community health and safety risks. These risks will be greater for women and youth who are new users of 2-wheeler vehicles and may be	Social Risks: N/A	Positive Impacts: Financing facilities and incentives for women and youth will increase their access to the program benefits. Social Risks: • Risk of Exclusion: Women may be excluded from benefiting from the program benefits due to cultural constraints around women riding 2 wheelers and lack of opportunities to learn to ride. • Risk of exclusion of women, youth and low-income individuals from accessing the financing facility due to	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. Financing facilities and incentives for the adoption of electric 2 and 3 wheelers will improve citizen state trust. Social Risks: Cultural barriers around allowing women and girls to ride 2 wheelers may result in family and community conflicts.

	compounded by operating vehicles without a license; not using protective gear such as helmets for drivers and passengers; and overloading the vehicles. As electric vehicles do not have engine sounds, CHS risks due to increased probability of accidents may also arise for pedestrians. • SEA/SH: Women operating 2 wheelers may experience harassment, especially in crowded public areas.		excessive and stringent eligibility requirements by banks including providing collateral, having a credit history etc. • Risk of exclusion of low-income 3-wheeler operators from accessing the financing facility due to excessive and stringent eligibility requirements by banks including providing collateral, having a credit history etc.	
DLI 7: Expand vehicle inspections DLI 8: Buyback of aged polluting vehicles	Positives Impacts: Increased inspection and buyback of polluting vehicles will directly benefit the citizens of Punjab by helping improve the air quality in the Greater Lahore region and improving the well-being of the public. Social Risks: • OHS risks related to vehicle inspection, maintenance and de-commissioning of retired vehicles. • Recycling of buyback vehicles will need effective management under labor and occupational and	Social Risks: N/A	Social Risks: • There is a risk of exclusion of lower income individuals with polluting vehicles who are unable to retrofit or change their vehicles through the buyback scheme. • Vehicles inspections carry a risk of unequal penalties for poor or 'less influential' owners of non-compliant vehicles. • Retiring polluting heavy-duty vehicles may result in a loss of livelihoods for vehicle operators or financial burdens to retrofit.	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. Buyback schemes and financial incentives will improve citizen state trust. Social Risks: Expanded vehicle inspections, fines and retirement of polluting vehicles may result in protests and conflicts due to loss of income.

	health laws. Poor E&S management of recycling facilities will result in health and occupational hazards for the workers and communities.		Sudden bans resulting from the inspection process may also result in loss of livelihoods for individuals operating ridesharing or delivery services.	
Results Area 3: Agriculture Sector Abatement Measures				
DLI 9: Expanding mechanized rice straw management.	Positive Impacts: Use of super seeders and other straw collection and management machinery to small farmers as an alternative solution for rice crop residue burning will improve the air quality in Punjab, contributing to the well-being of farmers, their families and public at large. Social Risks: Super seeders and proposed rice straw management machinery have recently been introduced by the Punjab government, and their use is not common. Health and safety risks may exist for farmers using this machinery, farm labor working in fields, and children due to accidents resulting from lack of information on machinery	NA	Positive Impacts: The activity will provide incentives to small farmers for the use of super seeders and rice straw management machinery. Social Risks: - Risk of exclusion of small farmers who may not have the technical skills or financial capacity to invest in the super seeders and straw management machinery. - Risk of exclusion of women farmers from accessing the subsidy/incentive due to the ballot system and lower number of women farmers applying for the subsidy as compared to men.	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. Incentives to access new machinery and improve farming practices will improve citizen state trust. Social Risks: Possible protests and conflicts by farmers who do not receive program benefits due to the ballot system.

	operations and safety measures.			
Digital access to straw management machinery and crop residue management service.	Positive Impacts: Increased use of alternatives to crop burning will improve the air quality in Punjab, contributing to the well-being of farmers, their families and public at large. Social Risks: Use of rental machinery by farmers with little to no experience of using them carries a risk of accident and injury to farmers, labor and family members working in fields.	N.A.	Positive Impacts: Subsidies for rental of super seeders and straw management machinery will provide access for small and women farmers without the financial means to buy the machinery. Social Risks: Rentals through digital solutions carry a risk of exclusion due to lack of accessibility for women, illiterate, persons with disabilities, rural communities and those without access to digital mediums or reliable internet.	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. Incentives to access new machinery and improve farming practices will improve citizen state trust. Social Risks: None
Awareness raising and enforcement.	Positive Impacts: Improved public awareness on alternatives to crop burning and penalties will increase the uptake of these alternatives and improve air quality in the region. Social Risks: None	N.A.	Positive Impacts: Communications campaigns will provide critical information on crop burning alternates, penalties and digital mediums for rental to a large audience including women and small farmers. Social Risks: This activity carries a risk of social exclusion due to lack of accessibility of communications and awareness materials by women, illiterate, persons with disabilities, rural	Positive Impacts: There are no fragile or post conflict areas in the Lahore Division. Improved awareness on crop burning alternates and government support is likely to improve the air quality and enhance citizen-state trust. Social Risks: Lack of clarity in communications materials may result in misunderstandings, conflicts or protests amongst farmers on proposed measures to

			communities and those without access to digital mediums and reliable internet. • Risk of unequal enforcement of crop burning bans on small and less influential farmers.	reduce crop burning. These will especially apply to criteria for penalties, access to benefits and eligibility criteria. • Unequal enforcement of crop burning bans and penalties on small and less influential farmers may result in protests and conflicts of farmers with enforcement authorities.
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4. POLICY AND LEGAL FRAMEWORK

52. The Punjab province in Pakistan has established a comprehensive legal framework to manage environmental and social risks associated with the Punjab Clean Air Program (PCAP). Below is an overview of key legislations in Punjab. The detailed description of national legislations and other relevant policies/acts are discussed in Annex - 1:

4.1. Environmental Legislation:

- **Punjab Environmental Protection Act, 1997 (Amended up to 2017):** This foundational law provides for the protection, conservation, rehabilitation, and improvement of the environment, aiming to prevent and control pollution.
- **Punjab Environmental Protection (Amendment) Act, 2017:** This amendment refines the original 1997 Act, enhancing provisions related to the delegation of powers and environmental governance.
- **Punjab Hazardous Substances Rules, 2018:** These rules regulate the generation, handling, storage, and disposal of hazardous substances to prevent environmental contamination.
- **Environmental Impact Assessment (EIA) Regulations:** These regulations mandate that proposed projects undergo environmental assessments to identify and mitigate potential adverse impacts.
- **Punjab Environmental Protection (Smog Prevention and Control) Rules, 2023:** Implemented to mitigate smog and air pollution, these rules outline procedures and impose restrictions on activities contributing to air quality deterioration.
- **Provincial Motor Vehicle Ordinance, 1965 & Motor Vehicle Rules, 1969 (Amended by Provincial Motor Vehicles Amendment Act, 2023):** These laws govern vehicular emissions and standards, with recent amendments focusing on reducing vehicular pollution.
- **Punjab Local Government Act 2019, Cantonments Act, and Lahore Canal Heritage Act 2013** prohibiting tree cutting and establishing strict protocols where it is unavoidable.

53. In addition, relevant Punjab Environmental Quality Standards (PEQS), which aim to regulate air emissions and effluents of industry and other big polluters are also relevant especially related to Municipal and Liquid Industrial Effluents, Motor Vehicle Exhaust and Noise, Ambient Air and Noise.

4.2. Social Legislation:

- **Punjab Local Government Act, 2019:** This act defines the structure and functions of local governments, emphasizing community involvement in development and land use planning.
- **Punjab Occupational Safety and Health Act, 2019:** This legislation ensures workplace safety and health standards, protecting workers from occupational hazards.
- **Factories Act, 1934:** An established law regulating labor rights, working conditions, and safety measures in factories.

- **Punjab Restriction of Employment of Children Act, 2015:** This act prohibits child labor and outlines conditions for the employment of adolescents, aiming to protect minors from exploitation.
- **Land Acquisition Act, 1894 & Punjab Land Acquisition Rules, 1983:** These laws govern the process of land acquisition for public purposes, ensuring fair compensation and addressing resettlement issues.
- **Punjab Labour Policy, 2018:** This policy outlines the rights and obligations of employers and employees, promoting fair labor practices and dispute resolution mechanisms.
- **Punjab Antiquities Amendment Act, 2012:** This amendment focuses on the preservation of cultural heritage, regulating activities that may affect historical sites.
- **Guidelines for Public Consultation, 1997:** These guidelines emphasize the importance of stakeholder engagement and public participation in developmental projects.
- **Punjab Minimum Wages Act, 2019:** This act establishes minimum wage standards to ensure fair compensation for workers across various sectors.
- **Punjab Industrial Relations Act, 2010:** This legislation facilitates the formation of trade unions and promotes harmonious industrial relations.
- **Punjab Right to Public Services Act, 2019:** This act ensures timely delivery of public services to citizens, enhancing transparency and accountability.
- **Punjab Transparency and Right to Information Act, 2013:** This law grants citizens the right to access information held by public bodies, promoting openness and informed participation.
- **Punjab Protection of Women Against Violence Act, 2016:** This act provides a legal framework to protect women from various forms of violence and abuse.
- **Punjab Empowerment of Persons with Disabilities Act, 2022:** This legislation aims to empower and ensure the inclusion of persons with disabilities in all spheres of life.
- **Punjab Women Development Policy, 2018:** This policy outlines strategies for the socio-economic development and empowerment of women in the province.

54. The legal framework of Punjab shows commitment to addressing environmental and social challenges, ensuring sustainable development, and safeguarding the rights and well-being of its citizens. However, there are gaps in legislative framework and institutional capacities which are discussed in next section.

5. COMPARATIVE ANALYSIS OF BORROWER ENVIRONMENTAL AND SOCIAL MANAGEMENT SYSTEMS AND BANK POLICY CORE PRINCIPLES

5.1. Introduction

55. Program for Results financing requires the proposed Program to operate within an adequate environmental and social management system that can manage environmental and social effects (particularly adverse impacts and risks) identified during the ESSA process. This includes an adequate legal and regulatory framework and institutional setting to guide environmental and social impact assessment and the management of environmental and social effects.
56. The Section V assesses whether the Program's environmental and social management systems are consistent with the core principles and key planning elements contained in the PforR and whether the involved institutions have the requisite capacity to implement these systems' requirements. Both elements (e.g. Program systems and capacity) are necessary towards ensuring that the environmental and social effects identified in Section III are effectively managed. Through both analyses, the ESSA team has identified gaps in both areas, which are addressed in Section VIII: Recommendations and Proposed Actions.
57. A Program system is constituted by the rules and "arrangements within a Program for managing environmental and social effects," including "institutional, organizational, and procedural considerations that are relevant to environmental and social management" and that provide "authority" to those institutions involved in the program "to achieve environmental and social objectives against the range of environmental and social impacts that may be associated with the Program." This includes existing laws, policies, rules, regulations, procedures, implementing guidelines, etc. that are applicable to the program or the management of its environmental and social effects. It also includes inter-agency coordination arrangements if there are shared implementation responsibilities in practice.
58. *Program capacity* is the "organizational capacity" the institutions authorized to undertake environmental and social management actions to achieve effectively "environmental and social objectives against the range of environmental and social impacts that may be associated with the Program." This ESSA has examined the adequacy of such capacity by considering, among other things, the following factors:
- a. Adequacy of human resources (including in terms of training and experience), budget, and other implementation resources allocated to the institutions.
 - b. The adequacy of institutional organization and the division of labor among institutions.
 - c. Effectiveness of interagency coordination arrangements where multiple agencies or jurisdictions are involved; and
 - d. The degree to which the institutions can demonstrate prior experience in effectively managing environmental and social effects in the context in projects or programs of similar type and magnitude.

59. This ESSA examines and discusses only those aspects of the proposed Program's environmental and social management systems that the ESSA Team found to be relevant, considering its identified environmental and social effects. This section provides a *summary assessment* of the Program's systems as these relate to each of the core principles and key planning elements.
60. The details of the comparative analysis are included in the following sections and a detailed matrix of findings and recommendations, systems and capacity details are provided in the tables.

5.2. Environmental Systems Capacity Assessment

61. Key implementation agencies of the Program activities will be the Environmental Protection and Climate Change Department (EPCCD), Transport and Mass-transit Department (T&MD) (in collaboration with PMA and PTC) and Agriculture Department (and AD), whereas Planning and Development (P&D) Board will be overall responsible for the overall coordination among all the implementing agencies and monitoring the Program performance. The Industries Department will be involved in the implementation of solar power plants at small industries. Punjab Energy Efficiency and Conservation (PEECA) will be mainly involved in the studies conducted under the IPF component. EPCCD has developed its capacity to manage environmental and social aspects of the donor funded program under the World Bank PforR financed 'Punjab Green Development Program (PGDP)'. EPCCD has been executing PGDP for over seven years and thus has prior experience of the bank's environmental and social PforR requirements. However, they have limited experience in implementing the World Bank's ESF policy and staffing for environmental specialists under PGDP has been a challenge. Transport and Agriculture departments are also the implementing agencies under PGDP and have experience with the E&S requirements. The PGDP's experience of implementing E&S policy and regulatory requirements by the Implementing Agencies (IAs) was deficient. As the E&S responsibilities of each IA were not spelled out in the program, therefore most of the IAs transferred their responsibilities of implementing E&S requirements of their tasks to the EPCCD due to its overall environmental mandate in the province, which created conflict whilst affecting effective E&S policy implementation, monitoring and reporting.
62. The Program should clearly define E&S responsibilities of each IA. P&D has established P-CAP Coordination and Monitoring Unit (CMU). The E&S staff will be appointed in the Unit. Remaining IAs including EPCCD will designate E&S focal persons from its existing staff to ensure compliance with E&S safeguard requirements.
63. Institutional capacity of Transport and Agriculture departments for E&S management is assessed as low. The Transport department will be responsible for larger share of project implementation. Accordingly, a Program Implementation Unit (PIU) will be established in the Transport department. E&S specialists will be part of the PIU team. The scope of Agriculture department activities is relatively small. Agriculture department will designate focal E&S persons from its existing staff.

64. Overall, EPCCD has limited capacity to regulate waste batteries and e-waste management and recycling and enforce safety requirements for e-buses and e-2/3 wheelers and lithium battery storage and charging stations in the province. The EPCCD is lacking in infrastructure, capacity and resources (human, financial and technical) to enforce environmental regulations in the province.
65. The implementing departments also lack awareness regarding hazards associated with the improper recycling of waste batteries and other e-waste), environmentally safe recycling and disposal practices, and lithium batteries specific fire hazards. E-waste is treated along with other wastes and sold to waste contractors without any environmental consideration for its disposal.
66. The assessment of environmental system in comparison with core principles 1, 2 and 3 is given below and in the Table 5.

5.2.1. Core Principle 1 – Environmental and Social Management

67. Environmental and social management procedures and processes are designed to (a) avoid, minimize, or mitigate against adverse impacts; (b) promote environmental and social sustainability in program design; and (c) promote informed decision making relating to a program's environmental and social effects.

1. Program procedures operate within an adequate legal and regulatory framework to guide environmental and social impact assessments at the programmatic level.

68. Legal and regulatory framework exists at national and provincial level to assess the environmental and social impacts of the projects. The guidelines for the preparation and review of environmental reports include the assessment of environment, social, health, and economic and fiscal impacts under sections 3.7 to 3.9. The assessment mandates focus on the demographic, cultural and sociocultural impacts under the section related to social impacts. Other sections are related to health, economic, and fiscal-related impacts.
69. IEE/EIA studies (Punjab Environmental Protection Act requires IEE/EIA whereas the World Bank system requires ESIA with increased focus on social issues), which assess project environmental and social impacts, are prepared and submitted to the environmental agencies for acquiring NOC to construct and operate projects. However, these assessments are only required for those projects which are listed in Schedules I and II of the IEE/EIA regulations and those projects for which Punjab EPA deemed necessary for conducting IEE/EIA based on the scope and setting of the project. The proposed bus depots under the Program need IEE/EIA (under World Bank system ESIA) and they belong to latter category stated above.
70. Generally, the hazardous materials are regulated under PEPA 97, Punjab EPA 97, and Pakistan Penal Code but there are no specific provisions for e-waste and waste

batteries management in these regulations which generate hazardous material during recycling.

71. As the impact assessment is not required for e-buses and battery storage and charging areas, therefore, the fire hazards associated with the lithium batteries are not assessed and regulated.
72. Punjab Environmental Quality Standards (PEQS) are in place and therefore, the program activities will be required to ensure compliance for air, noise and effluent discharge.
73. The implementing departments do not have their own procedures of environmental and social impacts assessment and management. They follow the provincial regulations.

II. Program procedures incorporate recognized elements of environmental and social assessment good practices, including:

a) Early Screening of Potential Effects:

74. Under 'Review of IEE and EIA Regulations' projects are screened under Schedules I and II. Schedule I projects, whose environmental and social risks are lower, require to submission of Initial Environmental Examination (IEE) for NOC. Schedule II projects, whose environmental and social risks are higher, require submission of Environmental Impact Assessment (EIA) for NOC.
75. IEE or EIA are carried out for new projects prior to their construction and operation. The Program will prepare ESIA of the proposed bus depot/s. Vehicle inspection facilities are small facilities and do not require IEE/EIA. At the operational stage these do not result in sizeable environmental impacts. The Program needs to conduct E&S screening of vehicle inspection facilities and accordingly prepare Environmental and Social Management Plan (ESMP).
76. EIA and environmental compliance under Punjab Environmental Protection Act is required for the facilities where hazardous waste or material is handled including e-waste, battery waste and scrap recycling facilities.

b) Consideration of Strategic, Technical and Site Alternatives (including the 'no action' alternatives):

77. Site alternatives are considered during environmental impact assessments under Review of IEE and EIA Regulations. IEE/EIA to be conducted under the Program for bus depot/s will include site alternative analyses.
78. Waste batteries and e-waste recycling facilities are not regularly monitored by the EPA.

c) Explicit Assessment of Potential Induced, Cumulative and Trans-Boundary Impacts:

79. Most of IEE/EIA are conducted under spot analyses. Determination of cumulative/aggregated environmental impacts is not covered by the scope of IEE/EIA. International transboundary impacts are not relevant to the Program.

d) Identification of Measures to Mitigate Adverse Environmental or Social Impacts that Cannot be Otherwise Avoided or Minimized:

80. IEE/EIA studies identify measures to mitigate adverse environmental and social impacts of the project activities that cannot be otherwise avoided or minimized. Measures to mitigate adverse environmental and social impacts of bus depot/s will be included in the IEE/EIA.

81. E-waste, including waste batteries and recycling facilities are not regulated under any legal instrument, therefore their EIAs are not conducted under the law.

e) Clear Articulation of Institutional Responsibilities and Resources to Support Implementation of Plans:

82. Institutional responsibilities and resources for preparation, implementation, monitoring, and inspection are clearly spelled out by relevant regulations (PEPA 2012, Review of IEE/EIA Regulations, 2000). The mandate for monitoring the implementation of IEE/EIA lies with EPCCD.

83. Under the Program, the implementing departments are responsible for the management of environmental and social risks of the Program activities, but their environmental and social responsibilities are not defined and assigned.

f) Responsiveness And Accountability Through Stakeholder Consultation, Timely Dissemination of Program Information and Through Responsive Grievance Redress Measures:

84. Public consultation with relevant stakeholders, local authorities, and representatives of communities and organizations directly affected by projects is required (guidelines for public consultation). Aspects of public consultation are better in projects financed by banks, and other multilateral and bilateral institutions.

85. EIA reports are disseminated to the public for review and feedback. A complaint cell is active at EPCCD to address complaints by communities related to environmental and social aspects.

86. There are many government-operated and department specific GRMs such as GRM at PM and CM portals operational in the Punjab province. The public grievances related to environmental and social issues are effectively addressed and resolved by the specific department.

5.2.2. Core Principle 2 – Natural Habitats and Physical Cultural Resources

87. Environmental and social management procedures and processes are designed to avoid, minimize, and mitigate adverse impacts on natural habitats and physical cultural resources resulting from the program.

I. Includes appropriate measures for early identification and screening of potentially important biodiversity and cultural resource areas.

88. The scope of IEE/EIA covers the profiling of natural habitats, flora and fauna, national parks, important ecological areas, and impacts on downstream freshwater bodies. Impacts matrix of project components and activities during construction and operational phases of the projects is an essential part of IEE/EIA. Accordingly, EIA requires that the project should include mitigations to avoid such impacts.

89. The main legislation on conservation of archaeological heritage is the Pakistan Antiquities Act of 1975. This Act was adopted by Punjab in 1985. The Antiquities Act 1975 only focused on conservation of a monument whereas the Punjab Law also includes conservation of the area surrounding the monument. The site screening guidelines with respect to the historical sites are also provided in the sectoral guidelines for industrial estates issued by EPA as a part of EIA Procedures.

90. Under the law, chance finds are required to be reported to the Archaeology Department within seven days. If ruins are discovered, then all construction work must be stopped. In this case, the Archaeology Department will conduct a site assessment and look for signs of buildings or habitation.

II. Supports and promotes the conservation, maintenance, and rehabilitation of natural habitats, avoids the significant conversion or degradation of critical natural habitats, and if avoiding the significant conversion of natural habitats is not technically feasible, includes measures to mitigate or offset impacts or program activities.

91. This aspect is covered under IEE/EIA guidelines in line with the best international practices and protocols. In EIAs of most of the large projects in the Northern Areas of Pakistan, this aspect is taken care of. Under the Program, this aspect will be important for the sites where bus depots will be developed. Program's other activities will be carried out within cities built up areas, where the significance of this aspect will be low.

III. Considers potential adverse impacts on physical cultural property and as warranted, provides adequate measures to avoid, minimize, or mitigate such effects.

92. Pakistan Antiquities Act of 1975 is relevant here.

93. Chance Finds Procedures stated above are henceforth, also relevant here.

94. EIA guidelines, under social impacts, emphasize to consider cultural resource impacts

including changes in archaeological, historical and cultural artifacts and structures and environmental features with religious and ritual significance and measures to avoid, minimize or mitigate these impacts.

5.2.3. Core Principle 3 – Public and Worker Safety

95. *Environmental and social management procedures and processes are designed to protect public and worker safety against the potential risks associated with (a) construction and/or operations of facilities or other operational practices developed or promoted under the program; (b) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials; and (c) reconstruction or rehabilitation of infrastructure located in areas prone to natural hazards.*

I. Promotes community, individual, and worker safety through the safe design, construction, operation, and maintenance of physical infrastructure, or in carrying out activities that may be dependent on such infrastructure with safety measures, inspections, or remedial works incorporated as needed.

96. For the safety of the buildings and structures, with respect to fire and earthquake, Building Code of Pakistan-Fire Safety Provisions-2016 and Seismic Provisions-2007 are considered. Local government and development authorities building bylaws and safety standards are applied while extending approval of the infrastructures.

97. The Punjab Occupational Safety and Health Act, 2019 focuses on the safety of the construction workers. e-waste and waste batteries recycling facilities are not regulated, resulting in potential safety risks for the workers and community.

II. Promotes the use of recognized good practice in the production, management, storage, transport, and disposal of hazardous materials generated through program construction or operations; and promotes the use of integrated pest management practices to manage or reduce pests or disease vectors; and provides training for workers involved in the production, procurement, storage, transport, use, and disposal of hazardous chemicals in accordance with international guidelines and conventions.

98. Under the Program, e-waste in the form of waste solar panels and IT equipment will be produced. Spent batteries will also result from the e-buses and solarization activities. Their improper recycling will ultimately produce hazardous materials. Current regulations do not cover specific provisions on waste batteries and e-waste management. The Punjab Environmental Protection Act includes general provision on the handling of hazardous substances but not the e-waste. Hazardous Substance Rules do not mention e-waste as hazardous substance, so no rules are applied on it. There are no regulated e-waste and batteries recyclers or recycling facilities in the country. There are no SOPs for e-waste handling in the implementation departments.

III. Includes measures to avoid, minimize, or mitigate community, individual, and worker risks when program activities are located within areas prone to natural hazards such as floods, hurricanes, earthquakes, or other severe weather or climate events.

99. These aspects of the proposed projects are fully covered by the scope of IEE/EIA. In most of IEE/EIAs these aspects are covered under DMP which is part of the environmental management of IEE/EIA reports.

100. Details on environmental systems capacity assessment are discussed in Table 5.

Table 5: Environmental Systems Capacity Assessment – Core Principles 1, 2 and 3

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
Core Principle 1: Environmental and social management procedures and processes are designed to (a) promote environmental and social sustainability in program design; (b) avoid, minimize, or mitigate adverse impacts; and (c) promote informed decision-making relating to a program’s environmental and social effects.			
Program procedures operate within an adequate legal and regulatory framework to guide environmental and social impact assessment at the program level	Legal and regulatory framework exists at national and provincial level to assess the environmental and social impacts of the projects. The guidelines for the preparation and review of environmental reports include the assessment of environment, social, health, and economic and fiscal impacts under sections 3.7 to 3.9. The assessment mandates focus on the demographic, cultural and sociocultural impacts under the section related to social impacts. Other sections are related to health, economic, and fiscal-related impacts. IEE/EIA studies, which assess project environmental and social impacts, are prepared and submitted to the environmental agencies for acquiring NOC to construct and operate projects. However, these assessments are	The regulators do not have the capacity to regulate waste batteries, e-waste and enforce safety requirements for e-buses and lithium battery storage and charging stations. Overall, the regulators are lacking in infrastructure, capacity and resources (human, financial and technical) to enforce environmental regulations in the province. The implementing departments lack awareness regarding hazards associated with the improper recycling of waste batteries and other e-waste, its environmentally safe recycling and disposal practices and lithium batteries specific fire hazards. E-waste is treated along with other wastes and sold to waste contractor without any	- Prepare ESIA of the bus depots (which include battery storage and charging facilities) and implement its recommendations - Prepare and implement e-waste management SOPs and Plan for IAs - Mobilize EPCCD to devise Waste Battery Management Rules to regulate recycling facilities - Ensure procurement of e-buses equipped with lithium battery specific fire suppression system

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	required for those projects which are listed in Schedule-I and II of the IEE/EIA regulations and required by Punjab EPA due to scope and setting of the project. Generally, the hazardous materials are regulated in PEPA 97 and Punjab EPA 97, and Pakistan Penal Code but there are no specific provisions for e-waste and waste batteries management in these regulations As the impact assessment is not required for e-buses and battery storage and charging areas, therefore, the fire hazards associated with the lithium batteries are not assessed and regulated. The implementing departments do not have their own procedures for assessing and managing environmental and social impacts. Instead, they adhere to provincial regulations.	environmental consideration for its disposal.	
Program procedures incorporate recognized elements of environmental and social assessment good practices,	Under 'Review of IEE and EIA Regulations' the projects are screened under Schedule I and II. Schedule I projects, whose	Regulator EPA has adequate capacity for review and approval of IEE/EIA and monitoring its compliance requirements.	Screening of the Program activities will be carried out under ESSA and relevant environmental and social instruments will be prepared and

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
including (a) Early screening of potential effects	environmental and social risks are lower, require Initial Environmental Examination (IEE) for NOC. Schedule II projects, whose environmental and social risks are higher, require Environmental Impact Assessment (EIA) for NOC. IEE or EIA is carried out for new projects prior to their construction and operation. EIA is also required for those projects for which Punjab EPA considers it necessary based on scope and setting of the project and facilities where hazardous waste or material is handled. Requirement of EIA for bus depot is not mentioned in schedule I and II. EPA informed that conducting EIA of bus depot is in practice and it will be required for the bus depots to be constructed under the Program		implemented during the Program implementation.
(b) Consideration of strategic, technical, and site alternatives (including the 'no action' alternatives)	Site alternatives are considered during environmental impact assessment under Review of IEE and EIA Regulations. Site alternative analysis will be included in the IEE/EIA of bus depots.	As Above	ESIA of the bus depots will be conducted with site alternative analysis. The Program will mobilize EPCCD to devise legal framework to regulate Waste Battery recycling facilities.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
c) Explicit assessment of potential induced, and trans-boundary impacts	Most of IEE/EIA are conducted under spot analyses. International transboundary impacts are not relevant to the Program.	Data on ambient conditions such as status of air pollution and pollution levels of receiving bodies are collected under IEE/EIA and impacts are determined for immediate surroundings. However, integrated analyses for impacts from various simultaneous activities under air and water pollution dispersion modelling are not done.	The TORs for the ESIA to require that zoning and land-use planning and development initiatives be reviewed and mitigation measures proposed in the ESMP's against this broader context. This would integrate the assessment of planning decisions into the ESIA's and their recommendations
d) Identification of measures to mitigate adverse environmental or social impacts that cannot be otherwise avoided or minimized	IEE/EIA studies identify measures to mitigate adverse environmental and social impacts of the project activities that cannot be otherwise avoided or minimized. E-waste and waste batteries recycling facilities are not effectively regulated.	Regulator EPA has adequate capacity for review and approval of IEE/EIA and monitoring its compliance requirements.	ESIA of the bus depots will be carried out under the Program. Program will prepare e-waste management SOPs and implement. The Program will mobilize EPCCD to cover waste battery recycling facilities under legal framework and certify them.
e) Clear articulation of institutional responsibilities and resources to support implementation of plans	Institutional responsibilities and resources for preparation, implementation, monitoring, and inspection are clearly spelled out by relevant regulations (PEPA 2012, Review of IEE/EIA Regulations, 2000). The mandate for monitoring the implementation of IEE/EIA lies with EPCCD.	The regulators lack resources (human, finance and technical) to enforce and monitor the implementation of the plans. The implementing departments don't have focal persons for environmental and social impacts assessment and management.	Under the Program, one Coordination and Monitoring Unit (CMU) and 03 Program Implementation Units (PIUs) will be established with environmental and social focal persons to assess and manage environmental and social risks of the Program activities.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	Under the program, the implementation departments are responsible for the management of environmental and social risks of the Program activities, but their environmental and social responsibilities are not defined and assigned.		
f) Responsiveness and accountability through stakeholder consultation, timely dissemination of program information, and through responsive grievance redress measures	Public consultation with relevant stakeholders, local authorities, and representatives of communities and organizations directly affected by projects is required (guidelines for public consultation). Aspects of public consultation are better in projects financed by banks, and other multilateral and bilateral institutions. EIA reports are disseminated to the public for review and feedback. A complaint cell is active at EPCCD to address complaints by communities related to environmental and social aspects. There are many government operated and department specific GRMs such as GRM at PM and CM	Although the timings and techniques of consultations are clearly stated in the guidelines, generally public consultation is carried out at two stages, that is, during the socioeconomic and inventory surveys at the baseline data collection stage of the EIA and during public hearing during the processing of EIA reports. In consultations conducted during GoPb projects, the objectives of consultation are not met because consultations are not conducted in true spirit and procedures.	The Program will ensure that the stakeholder and public consultation processes are inclusive during each stage of investment planning and implementation, to ensure compliance with EPA guidelines for public consultation. GRM is also implemented at the Program level.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	portals operational in the Punjab province. The public grievances related to environmental and social issues are effectively addressed and resolved by the specific department.		
Core Principle 2: Environmental and social management procedures and processes are designed to avoid, minimize and mitigate against adverse effects on natural habitats and physical cultural resources resulting from program			
Includes appropriate measures for early identification and screening of potentially important biodiversity and cultural resource areas.	The scope of IEE/EIA covers the profiling of natural habitats, flora and fauna, national parks, important ecological areas, and impacts on downstream freshwater bodies. Impacts matrix of project components and activities during construction and operational phases of the projects is an essential part of IEE/EIA. Accordingly, EIA requires that the project should include mitigations to avoid such impacts. The main legislation on conservation of archaeological heritage is the Pakistan Antiquities Act of 1975. This Act was adopted by Punjab in 1985. The Antiquities Act 1975 only focused on conservation of a monument whereas the Punjab Law also	Most governmental agencies do not implement the Antiquities Act as per established procedures.	Under the Program, ESIA to be conducted for bus depots will include appropriate measures for early identification and screening of potentially important biodiversity and cultural resources of the sites selected for bus depots.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	includes conservation of the area surrounding the monument. The site screening guidelines with respect to the historical sites are also provided in the sectoral guidelines for industrial estates issued by EPA as a part of EIA Procedures. Under the law, chance finds are required to be reported to the Archaeology Department within seven days. If ruins are discovered, then all construction work must be stopped. In this case, the Archaeology Department will conduct a site assessment and look for signs of buildings or habitation.		
Supports and promotes the conservation, maintenance, and rehabilitation of natural habitats, avoids the significant conversion or degradation of critical natural habitats, and if avoiding the significant conversion of natural habitats is not technically feasible, includes measures to mitigate or offset impacts or program activities.	This aspect is covered under IEE/EIA guidelines in line with the best international practices and protocols. In EIAs of most of the large projects in the Northern Areas of Pakistan, this aspect is taken care of. Under the Program, this aspect will be important for the sites where bus depots will be	Regulator EPA has adequate capacity for review and approval of IEE/EIA and monitoring of its compliance requirements.	The Program will ensure that this aspect is taken care of according to PEPA 2012 and its guidelines, particularly during selection of sites for the development of bus depots.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	developed. The Program's other activities will be carried out within cities built up areas, where the significance of this aspect will be low.		
Considers potential adverse impacts on physical cultural property and as warranted, provides adequate measures to avoid, minimize, or mitigate such effects.	The main legislation on conservation of archaeological heritage is the Pakistan Antiquities Act of 1975. This act was adopted by Punjab in 1985. The Antiquities Act 1975 only focused on conservation of a monument whereas the Punjab law also includes conservation of the area surrounding the monument. Under the law, chance finds are required to be reported to the Archaeology Department within seven days. If ruins are discovered, then all construction works must be stopped. In this case, The Archaeology Department will conduct a site assessment and looks for signs of buildings or habitation. EIA guidelines, under social impacts, emphasize to consider cultural resource impacts including changes in archaeological, historical and	Application of antiquities laws during project planning need improvement as sometimes its provisions are ignored. Punjab EPA has adequate capacity for review and approval of IEE/EIA and monitoring of its compliance requirements.	During preparation of ESIA of the bus depots, the proposed sites will be screened for possible impact on sites of archaeological significance.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	cultural artifacts and structures and environmental features with religious and ritual significance and measures to avoid, minimize or mitigate these impacts.		
Core Principle 3: Environmental and social management procedures and processes are designed to protect public and worker safety against the potential risks associated with (a) construction and/or operations of facilities or other operational practices developed or promoted under the program; (b) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials; and (c) reconstruction or rehabilitation of infrastructure located in areas prone to natural hazards.			
Promotes community, individual, and worker safety through the safe design, construction, operation, and maintenance of physical infrastructure, or in carrying out activities that may be dependent on such infrastructure with safety measures, inspections, or remedial works incorporated as needed.	For the safety of the buildings and structures, with respect to fire and earthquake, Building Code of Pakistan-Fire Safety Provisions-2016 and Seismic Provisions-2007 are considered. Local government and development authorities building bylaws and safety standards are applied while extending approval of the infrastructures. The Punjab Occupational Safety and Health Act, 2019 focuses on the safety of construction workers. E-waste and waste batteries recycling facilities are not effectively regulated. Operations of these facilities are potential safety risks for the workers and	There is lack of enforcement of building and workers safety regulations in the province.	The Program will consider safety aspects while developing bus depots with respect to structure safety and fire safety. The Program will mobilize EPCCD to cover waste battery recycling facilities under legal framework and certify them for the safety of workers and the community.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	community.		
Promotes the use of recognized good practice in the production, management, storage, transport, and disposal of hazardous materials generated through program construction or operations; and promotes the use of integrated pest management practices to manage or reduce pests or disease vectors; and provides training for workers involved in the production, procurement, storage, transport, use, and disposal of hazardous chemicals in accordance with international guidelines and conventions.	Under the Program, e-waste and waste batteries will be produced whose improper recycling will ultimately produce hazardous materials. Current regulations do not have specific provisions on e-waste management. There are no regulated e-waste recyclers or recycling facilities in the country. There are no SOPs for e-waste management in the implementation departments.	The regulators do not have the capacity to regulate e-waste. The implementing departments don't have awareness regarding hazards associated with the improper recycling of waste batteries and other e-waste, its environmentally safe recycling and disposal practices. e-waste is treated along with other wastes and sold to waste contractors without any environmental consideration for its disposal).	- Prepare and implement e-waste SOPs for the IAs - Mobilize EPCCD to devise a legal framework to regulate waste battery recycling facilities
Includes measures to avoid, minimize, or mitigate community, individual, and worker risks when program activities are located within areas prone to natural hazards such as floods, hurricanes, earthquakes, or other severe weather or climate events.	These aspects of the proposed projects are fully covered by the scope of IEE/EIA. In most of IEE/EIAs these aspects are covered under DMP which is part of the environmental management of IEE/EIA reports.	Punjab EPA has adequate capacity for review and approval of IEE/EIA and monitoring of its compliance requirements.	ESIA of the bus depots will consider natural hazards and provide recommendations to build climate resilient infrastructure.

5.3. Recommendations to Strengthen Environmental Management Systems

101. Based on the above comparative analysis of the borrower's environmental and social management system and bank policy core principles, the following are the recommendations to strengthen the environmental and social management system for the program activities.
102. The proposed bus depot's tentative locations are in the well-developed areas of Lahore. Development of bus depots require an ESIA, as informed by The Punjab Environmental Protection (Amendment) Act 2017. Under the Program, ESIA of the e-bus depot/s, charging stations and operations will be carried out and its recommendations will be implemented during designing, construction and operational phases. The ESIA will include: i) Incorporation of environmentally friendly and climate resilient design elements as part of the design; ii) Site alternative analysis; iii) Environmental impact assessment due to the increase of traffic on the connecting roads of bus depots, load on utilities especially water supply and drainage, and land use of the area especially expansion of commercial area along the connecting roads of bus depots, incorporation of environmentally friendly and climate resilient design elements as part of the design, iv) Appropriate measures for early identification and screening of potentially important biodiversity and cultural resources at the subproject sites, v) Screening of the proposed site for possible impact on sites of archaeological significance, vi) Safety aspects while developing bus depot/s with respect to structure safety and fire safety and; vii) assessment of and measures to address risks to workers and community health and Safety from depot construction and e-bus operations. The ESIA will review the zoning and land-use planning and development initiatives in and around the proposed program area and propose mitigation measures in the ESMP against this broader context.
103. The e-waste and battery waste recycling facilities being hazardous in nature come under Hazardous Substances Rules, but these are not specifically mentioned in the rules. The e-waste management SOPs will be prepared and implemented under the Program. The Program will also mobilize EPCCD to formulate battery waste management rules under Hazardous Substance Rules to regulate waste battery recycling facilities for environmental protection and safety of workers and the community.
104. The safety of e-buses with respect to lithium battery specific fire is not considered under existing legal framework. The program will ensure procurement of e-buses equipped with lithium battery specific fire suppression system to avoid fire hazards and save the public and the government assets.

5.4. Social Management Systems Capacity Assessment

105. This section provides a summary of the assessment of the national and provincial systems and capacity of the implementing agencies in managing the social risks of the project. Details of the legal and policy frameworks as well as institutional capacity are provided in Annexures 1 and 2. The social management systems capacity has been assessed in comparison to Core Principles 1, 3, 4, 5 and 6. The detailed description on comparative capacity assessment is given below and summarized in Table 6.

5.4.1. Core Principle 1 – Environmental and Social Management

106. As assessed in the previous sections, social risks of the program are mainly related to transport related activities under RA2, including occupational and community health and safety due to e-bus infrastructure construction and e-bus operations, as well as land acquisition for the construction of e-bus depots and removal of encroachers/informal settlers from government lands identified for e-bus infrastructure. Risk of exclusion of vulnerable groups from project benefits exist across the results areas including exclusion of women, youth and low-income individuals from accessing subsidies for e-vehicles and farm machinery; women and persons with disabilities from using e-bus infrastructure; and lack of accessibility of communications and awareness materials by women, illiterate, persons with disabilities, rural communities and those without access to digital mediums. Retiring polluting vehicles and closure of polluting industries may result in loss of livelihoods and social conflicts. The magnitude of these risks may be further increased due to ineffective communications and exclusion of vulnerable groups from consultations during the IPF design and feasibility studies, as well as at the PforR activities preparation and implementation stages. Risks of SEA/SH exist for women employed in the operations of e-buses, women using public transport, and for women driving e-2 wheelers.

107. Key findings for the social and capacity assessment under Core Principle 1 show that:

- EPCCD has a Social Specialist on staff under PGDP. T&MD, AD, PMA, PTC and P&D do not have designated staff dedicated to the management of social impacts.
- Stakeholder consultations by law are only mandated for activities requiring an EIA. The requirements are weak, do not cover activities not requiring an EIA, and lack mandates for inclusion and transparency as per World Bank standards.
- Information dissemination is governed by the Punjab Transparency and Right to Information Act. Public Information Officers (PIOs) as mandated by the Act are present in the IAs, and the IAs have functioning websites with detailed information on services, structures and grievances.
- Regional Transport Authorities (RTAs) have a performance matrix which includes indicators for public engagement through advertisements and information sharing.
- Extensive GRMs at the provincial and departmental level are established and functioning. However, the functionality of the GRMs for vulnerable groups is not known. Except for EPCCD, there is no mechanism in the departments for tracking and compiling complaints received through various channels.

108. **Recommendations** based on the gaps identified under the social and capacity assessment for Core Principle 1 are as follows:

- Appoint/nominate social & gender specialists in PCAP-CMU and T&MD PIU.
- Nominate existing staff at EPCCD, AD, PTC, PMA and EPA as social & gender focal points for PCAP.
- Appoint/nominate staff in IAs for handling timely consultations, disclosure, dissemination of information and grievance redress measures.
- Expand the Stakeholder Engagement Plan (SEP) prepared under the IPF component of PCAP to a 'Program-Wide SEP' (P-SEP) to ensure consistent, transparent and inclusive consultations throughout the program life.

- Assess the accessibility of current GRM systems in the IAs for vulnerable groups and establish a mechanism in PCAP to compile and ensure resolution of program specific grievances received through various channels.
- Conduct annual Citizen Feedback Surveys on program performance.

5.4.2. Core Principle 3 – Workers and Public Safety

109. As indicated in table 5 below, and in greater detail in annexures, the assessment of Program systems under this principle determined that there are social risks related to workers health and safety, and community safety during construction of public transport infrastructure such as e-bus depots. There is also a risk of child labor and forced labor during construction activities. Risks related to SEA/SH exist for women employees of the IAs, women employed for e-bus operations, women using public transport, and women driving 2 wheelers. An increase in the number of accidents for users (especially inexperienced users) of 2 wheelers also exists, as well as risk of accidents for pedestrians. Under the agriculture component, there is a risk of community health and safety to farmers, farm labor and farmers families working in the fields due to accidents caused by inexperienced operators of super seeders and rice straw management machinery.

110. **Key findings** for the social and capacity assessment under Core Principle 3 show that:

- Worker's health and safety is governed by the Punjab Occupation Health and Safety Act 2019.
- Legislation regarding community health and safety is weak. SEA/SH risks for women at the workplace are governed by the Punjab Protection Against Harassment of Women at Workplace (Amendment) Act 2012, and use of child labor is restricted by the Punjab Restriction on Employment of Children Act, 2016.
- Community health and safety in IAs is most often considered for activities under donor financed projects requiring specific measures to be in place as per donor requirements.
- Overall implementing agencies are limited by human resources availability, capacity and infrastructure to target and respond to risks faced by communities during construction activities.

111. **Recommendation** based on the gaps identified under the social and capacity assessment of Core Principle 3 are as follows:

- Section on occupational health and safety of the workers and the community should be part of the EHSS Guidelines. Implementing departments' staff should be trained in these guidelines.
- Prepare Occupational Health and Safety Plan and Community Health and Safety Plan as part of the ESIAs.
- Prepare and implement Gender Inclusion and SEA/SH Action Plan
- Ensure all construction contracts include community health and safety, labor safety, child labor and SEA/SH prevention clauses.
- Train Program focal points on community health and safety, child labor and SEA/SH as master trainers.

5.4.3. Core Principle 4 – Land Acquisition

112. Given the scope of the proposed Program's activities, the ESSA concludes that the Program is unlikely to have any large-scale adverse impacts caused by the acquisition of land or the restriction of use or access to land or natural resources, including the loss of income caused by such actions. Land requirements under the program are expected under RA 2 for the establishment of e-bus depots. T&MD has in the past prioritized government lands for public transport activities, and will continue to do so under P-CAP. However, due to the specific requirements of the e-bus depots, T&MD has also identified four parcels of private lands to tentatively acquire for the construction of e-bus depots, with a total area of 33.85 acres. The size of each parcel ranges from 5.88 acres to 10 acres as provided in the table below. Satellite images of the identified land parcels are provided as Annexure 6 and indicate these parcels to be agricultural lands.

<i>Sr. No</i>	<i>Location</i>	<i>Area (Acre)</i>	<i>Coordinates</i>	
			<i>Latitude</i>	<i>Longitude</i>
1	Near Khaira Bridge BRB Canal	10	31°35'15.05"N	74°29'57.20"E
2	Near Gajjumata Interchange along Hudhara Drain	9	31°23'23.51"N	74°21'16.99"E
3	Near Muridwal Village M-2 Link	8.97	31°29'21.52"N	74°13'45.87"E
4	Near Barki Village	5.88	31°28'21.94"N	74°30'58.37"E

113. T&MD will use the Land Acquisition Act (LAA, 1894) and Punjab Land Acquisition Rules (PLAR 1983) with the help of the Revenue Department to acquire lands required for the construction of the e-bus depots. Details of the LAA 1894 and PLAR 1983 are provided in Annexure 2-Social Policy and Legal Framework. Specifically, the PLAR 1983 under its Rule 4 states "On receipt of the application under Rule 3 the collector of the district shall examine its feasibility taking into consideration the genuineness of the public purpose involved, the minimum requirements of the acquiring agency, and suitability of the area proposed for requisition keeping in view its alternative uses if any."

114. There are lapses in the implementation of the LAA, as well as gaps between this law and the provisions for land acquisition prescribed by the World Bank in the PforR core principles. Specifically, the LAA only allows payment of land and lost assets (structures, crops, trees) at market price to titleholders and has no similar provisions for non-titleholders. As such, application of the law means a loss of shelter and assets and increased vulnerability for the latter category of those affected. In certain cases, the law also enables 'emergency' acquisition of land (Section 17) without public consultation or payment of compensation before possession. This power leads to sudden dispossession of land and hence raises the risk of vulnerability of those affected.

115. The LAA also does not consider compensation for loss of livelihood due to displacement or dislocation from the land for both titleholders and non-titleholders and as such carries the risk of enhancing vulnerability. In addition, the lack of payment for land in line with representative prevalent market rates, as prescribed by the LAA, leads to long drawn-out legal cases and financial loss. Most often public lands are informally used by non-title holders, usually the landless, poorest segments of society. Possible use of public lands for construction of public transport infrastructure in crowded urban areas carries a potential

risk for non-titled users of the land (residents or those using them for commercial activities) to be forcibly removed as the LAA does not entertain compensation to informal settlers or users of the land, including encroachers, slum dwellers and street hawkers. This process of forced evictions is generally carried out by the district or local government authorities under Anti-Encroachment Drives (AEDs) without compensating the affected persons/households for resettlement and livelihood loss as required under the WB guidelines.

116. Given the issues identified with the existing land acquisition regulations, the Program's land acquisition activities could pose social equity and reputational risks due to private land acquisition during Program implementation; and if informal settlers exist on acquired public lands. Such risks have also been identified in ongoing World Bank PforRs in Punjab, and mitigation measures put in place. Particularly, standard operating procedures (SOPs) for Land Acquisition and Resettlement have been prepared by EPCCD under the Punjab Green Development Program (PGDP) to address the gaps in procedures for private land acquisition and compensation, as well as the weakness in proper implementation of the law. The SOPs have been assessed by the ESSA team and may be used by PCAP to tailor for land acquisition under PCAP, adjusted for the varying nature of the land acquisition under both programs, and to address gaps such as addressing areas affected by anti-encroachment drives (AEDs). The PGDP SOPs cover the following:

- Detailed analysis of PGDPs land requirements, identification of land parcels, status of identified lands and mechanisms for land acquisition.
- Legal framework including LAA 1894, PLAR 1983, Pakistan Antiquities Act 1975 and World Bank's PforR Core Principle 4; and gap analysis.
- Assessment of land acquisition risks and mitigation measures including:
 - Screening processes and screening tools.
 - Preparation of Resettlement Plans.
 - Community participation and consultations.
 - Categories of affected persons.
 - Entitlements matrix including compensations for non-title holders.
 - Implementing arrangements.
 - Chance finds procedures; and
 - Grievance redressal mechanism for land acquisition.

117. The risk of forced removal of encroachers has been assessed under previous World Bank PforR as well. The ESSA for World Bank funded Punjab Cities Program (PCP) and Punjab Affordable Housing Program (PAHP) report that compensation for loss of business to non-title holders in Punjab has been made through special provisions created for Metro Bus Projects in Lahore and Rawalpindi (not WB financed). Examples for management of risks related to forced evictions of encroachers in ongoing World Bank PforR are provided below:

- Under PCP, the Punjab Municipal Development Fund Company (PMDFC) has prepared an ESMF which covers the land acquisition process and is assessed to be in line with the requirements of this core principle. The ESMF states that sites where anti encroachment drives (by GoPb) have been conducted will be avoided as priority, and where avoidance is not possible, they will only be considered with implementation of the Entitlement Framework given in the ESMF.

- The PAHP ESSA recommends that the site selection eligibility criteria should include conditions of compensations to the illegal residents of the selected sites. Further, the land selection criteria veto factors include that the land should have clean title without encumbrances.
- Under PDGP, SOPs on land acquisition and resettlement have been prepared to address the gaps between Core Principle 4 and LAA 1894 and have been assessed by the ESSA team to be adequate. The Screening Checklists and Entitlements Matrix in these SOPs include procedures for identification of risks and compensation for non-title holders and encroachers.

118. **Recommendations** based on the gaps identified under the social and capacity assessment under Core Principle 4 are as follows:

119. The Program will apply two key exclusion criteria regarding land acquisition and resettlement. Sites identified for e-bus depots will exclude: (i) areas displacing 200 or more individuals per site; and (ii) areas which have legacy anti-encroachment drives (AED) starting from the approval date of the Program Concept (December 9, 2024).

120. The SOPs on land acquisition and resettlement from PGDP will be tailored for PCAP, addressing gaps in areas affected by anti-encroachment drives (AEDs). Key provisions will include:

- Land acquisition and resettlement screening checklists for each e-bus depot against the exclusion criteria and to identify resettlement impacts;
- Compensation for affected landowners, tenants, and non-titleholders, aligned with market rates; and
- Preparation of Abbreviated Resettlement Action Plans (ARAP) or Resettlement Action Plans (RAP) before commencing work (if required).

5.4.4. Core Principle 5 – Indigenous Peoples and Vulnerable Groups

121. Given the scope of the proposed Program's activities are limited to Punjab province and more specifically Lahore Division, the ESSA Team concludes that there are no recognized indigenous people in the program area. The only recognized Indigenous People in Pakistan are the Kalash residing in the Chitral District of Khyber Pakhtunkhwa.

122. As determined in the risk assessment in the previous sections, the Program activities are likely to carry risks of exclusion of groups vulnerable to hardship or disadvantage, including, the persons with disabilities, women and children, students, the elderly, illiterate, small farmers and people with limited income. These risks are related to access to public transport infrastructure, access to financing facilities for electric 2 and 3 wheelers, access to information and communications materials, and opportunities to benefit from agriculture subsidies for super seeders and machinery rentals.

123. Equitable access is governed by the Punjab government through the Punjab Women Development Policy 2018, Punjab Agriculture Policy 2018, Punjab Social Protection Policy 2022 and Empowerment of Persons with Disabilities Act 2022. The ESSA team concludes that while these legal provisions exist, the implementation of the requirements for inclusion and equity in the IAs is weak. Overall implementing agencies are limited by

human resources availability, capacity and infrastructure to target and respond to risks and challenges faced by the vulnerable groups such as encroachers, ethnic minorities and persons with disabilities. Projects financed by donors in the IAs have staff and activities mandated to work towards the welfare of vulnerable people as required by the donor agencies. However, these are mostly limited to specific project activities.

124. Key findings by the ESSA team related to this include:

- Low number of female staff in IAs except EPCCD.
- Newly procured e-buses by TMD are equipped with wheelchair ramps, however bus stops and pedestrian access to the bus stops are not always accessible. The only fully accessible public transport for persons with disabilities is the Metro Bus and Orange Line train.
- Buses operating under PTC and PMA have dedicated sections for women as well as double doors to provide women with a dedicated door to enter and exit the bus, limiting the risk of sexual harassment.
- E-bike subsidy scheme by the Punjab Government has prioritized the provision of subsidies to women who have submitted applications.
- Women farmers are a fraction of the number of male farmers. The general balloting system for super seeders does not account for this low number of women farmers, resulting in low probability of them being chosen to receive the subsidies.

125. **Recommendation** based on the gaps identified under the social and capacity assessment under Core Principle 5 are as follows:

- It is recommended to review the implementation of the Women Development Policy 2018, The Punjab Empowerment of Persons with Disabilities Act 2022, and Punjab Agriculture Policy 2018 in the IAs with respect to inclusion of vulnerable people.
- Prepare a Disability Inclusion Plan to ensure implementation of The Punjab Empowerment of Persons with Disabilities Act 2022 across RA 2 activities and to ensure that public transport infrastructure including buses, bus stops and pedestrian access to bus stops such as pavements and road shoulders are accessible as per international good practices and standards.
- Prepare a Gender Inclusion Action Plan for the Program activities related to ensuring inclusion of women in accessing program benefits.
- Conduct a social assessment to establish a baseline of access of women and persons with disabilities to public transport and e-vehicles.
- Prepare SOPs for ensuring accessibility of communications materials and awareness campaigns for vulnerable groups.

126. Review the accessibility of the existing provincial and department wise Grievance Redressal and Right to Information forums for vulnerable groups.

5.4.5. Core Principle 6– Social Conflict

127. As indicated below and to a greater extent in the Annexures, the ESSA team concludes that there are no fragile, post conflict or areas subject to territorial disputes in the Lahore Division. The Program activities carry a low risk of conflict between the public and authorities, and public protests. This risk may arise from increased enforcement of

pollution abatement measures such as lockdowns, retiring polluting vehicles, running e-buses on routes favoured by private contractors, and penalties for crop burning.

128. The formal grievance mechanisms at the provincial and department level are present for addressing public queries and complaints. Grievances can also be escalated by the public to the Provincial Ombudsman. However, local authorities have a history of dealing with protests and conflict situations with force.

129. Experience by the EPCCD shows that effective communication on Green Lockdowns has been helpful in managing public expectations. Therefore, ensuring accessible, effective and continued communications on the air pollution abatement measures will be critical in keeping the public engaged and informed, and will reduce the risk of conflicts and protests.

130. **Recommendation** based on the gaps identified under the social and capacity assessment under Core Principle 6 are as follows:

- Ensure IAs have an effective system to track and resolve complaints received from national, provincial and departmental GRMs.
- Ensure the communications campaigns are inclusive, understandable and inclusive for women, persons with disabilities, illiterate, students, small farmers, and those without access to digital mediums or internet.
- Conduct annual Citizen Feedback Surveys on program performance and publish results.

131. Table 6 provides the details of key planning elements, social system assessment, capacity assessment and recommendations in line with Core Principles 1, 3, 4, 5 and 6.

Table 6: Social Systems Capacity Assessment – Core Principles 1, 3, 4, 5 and 6

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
Core Principle 1: Environmental and social management procedures and processes are designed to (a) avoid, minimize, or mitigate adverse impacts; (b) promote environmental and social sustainability in Program design; and (c) promote informed decision-making relating to a Program’s environmental and social effects.			
1.2 (a) Responsiveness and accountability through stakeholder consultation, timely dissemination of Program information, and through responsive grievance redress measures	• All investment projects are duly approved by the national and provincial assemblies as a part of annual budget which are disclosed on web sites of respective ministries and departments. • The Punjab government does not have any formal stakeholders’ consultation process for the activities of the Program. However, all investment projects are duly approved by the provincial assembly as a part of annual budget which are disclosed on websites of respective IA(s). • Guidelines for Public Consultation, 1997 address possible approaches to public consultation and techniques for designing an effective Program of consultation that reaches all major stakeholders and ensures the incorporation of their concerns in any impact assessment study, specifically EIAs. • The Punjab Transparency and Right to Information Act 2013 ensures citizens have access to public information to make the	Adequate processes exist for information dissemination and disclosure across the IAs. Provincial Capacity: • Punjab Government hosts an online portal (www.punjab.gov.pk) with information on service delivery platforms, digital services, Annual Development Plan, published tenders, administrative departments, investment schemes, public benefit schemes, laws and citizen rights amongst other information. • The Punjab Portal lists helplines for various administrative departments and has a province level citizen relationship management feature called “Awaz e Khalq” that allows citizens to register complaints related to any government department. There is also a toll-free Chief Minister’s Complaint Cell Helpline Number (0800 02345) that can be used by the public to register complaints. • The Provincial Ombudsman of Punjab provides avenues for citizens to lodge complaints against any government department or service through their physical or online channels. Capacity of Implementing Agencies:	Appoint/nominate staff in IAs for handling timely consultations, disclosure, dissemination of information and grievance redress measures. Ensure consultations as per guidance available in SEP prepared for the TA component Assess the accessibility of current GRM systems in the IAs for vulnerable groups. Establish a mechanism in PCAP to compile and ensure resolution of program specific grievances received through various channels.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	government accountable to citizens as well as enforcing the fundamental right of access to all information in regard to all matters of public importance. The law places crucial duties and responsibilities on the designated Public Information Officer (PIOs), to liaise between the public and government institutions under the RTI Law.	• EPCCD has a functioning website with updates on activities, updated AQI, information regarding EPA requirements and an RTI tab with contact details of Chief Information Officer and Public Information Officer for citizens to get information or lodge complaints. The EPCCD also has a dedicated WhatsApp number, email a helpline number. A separate GRM under PGDP is also functional, with details provided on the PGDP website. • The Transport and Masstransit Department has a functioning website updated with details on services, routes, fares and contact details. The website has a section for complaints listing two complaints helplines for Route Permits and Fare Overcharging and can also be contacted through a dedicated WhatsApp number with queries and complaints. • The Punjab Masstransit Authority has an updated website, a helpline number 1762, and dedicated helpline numbers for metro-bus systems in Lahore, Multan and Rawalpindi. • Punjab Transport Company has an updated website and has established a dedicated commuter support centre with a dedicated UAN helpline, email address and online portal. • The Agriculture Department has a dedicated website with information on services, directorates and contact information. The	

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
		department has established the toll-free Punjab Agriculture Helpline for farmers and has also initiated a SMS Helpline Service to extend technical guidance to farmers using mobile phones. The extensive field network of the Department's Agriculture Extension Wing provides farmers with access to department staff for information, guidance, complaints and services. • The Planning & Development Board has a dedicated website with details on P&D programs, major development projects by the Punjab government, Annual Development Program and foreign aided projects. The website also has a resource center and an RTI tab with details of the Chief PIO. Public complaints related to P&D may be lodged through the PIO or through the provincial channels. Anonymous complaints can be made in writing through postal mail.	
Core Principle 3: Program procedures ensure adequate measures to protect public and worker safety against the potential risks associated with (a) construction and/or operations of facilities or other operational practices developed or promoted under the Program and (b) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials.			
3 (a) Promotes community, individual, and worker safety through the safe design, construction, and O&M of physical infrastructure, or in carrying out activities that may be	• The Punjab Protection Against Harassment of Women at Workplace (Amendment) Act 2012 is present to address risks for female employees of IAs.	• Overall implementing agencies are limited by human resources availability, capacity and infrastructure to target and respond to risks faced by communities during construction activities. • OHS measures required by donors or financing entities are ensured by the IAs by	• Section on occupational health and safety of the workers and the community should be part of the EHSS Guidelines. IAs staff should be trained in these guidelines.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
dependent on such infrastructure with safety measures, inspections, or remedial works incorporated as needed.	- Harassment of women on public transport infrastructure is not governed by any provincial law. - The Women Development Department provides details for women on their rights, services for women, opportunities for women inclusion and referral directories for GBV. - Employment of children in construction activities is governed by the Punjab Restriction on Employment of Children Act, 2016. - Workers' health and safety are governed by the Punjab Occupational Safety and Health Act, 2019 - The performance of implementing departments is not satisfactory to ensure the safety of the construction workers and the community.	- adding them to procurement contracts for contractors. - There are Departmental Committees constituted in most agencies for preventing harassment of women at workplace, mostly chaired by the respective Secretaries (heads of the departments). There is also a Punjab Ombudsperson for Protection Against Harassment of Women at the Workplace. - A helpline number 1043 is operational at the Provincial level for women to register complaints related to gender-based discrimination, harassment, or violence.	- Prepare Occupational Health and Safety Plan - Prepare Community Health and Safety Plan - Prepare Gender and SEA/SH Action Plan - Ensure all construction contracts include community health and safety, labor safety, child labor and SEA/SH prevention clauses. - Train Program focal points on community health and safety, child labor and SEA/SH as master trainers
Core Principle 4: Land acquisition and loss of access to natural resources are managed in a way that avoids or minimizes displacement, and affected people are assisted in improving, or at least restoring, their livelihoods and living standards.			
4 (a) Avoids or minimizes land acquisition and related adverse impacts	The main legal tool used for land acquisition in Punjab is the Land Acquisition Act (LAA) 1894. The Law takes into consideration the genuineness of the public purpose involved, the minimum requirements of the acquiring agency, and suitability of the area	- The Transport & Mass Transit Department Has tentatively identified private lands for acquisition for the e-bus depots. - T&MD uses the LAA 1894 and PLAR 1983 for land acquisition. - Cases where land acquisition is minimized due to adverse social impacts are minimal.	The Program will apply two key exclusion criteria regarding land acquisition and resettlement. Sites identified for e-bus depots will exclude: (i) areas displacing 200 or more individuals per site; and (ii) areas which have legacy anti-

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	proposed for requisition, keeping in view its alternative uses if any.	Measuring social impacts is usually not on the radar of the implementing agencies.	encroachment drives (AED) starting from the approval date of the Program Concept (December 9, 2024). Tailor and adopt PGDP SOPs for Land Acquisition and Resettlement. These SOPs review LAA 1894 to respond to system and capacity gaps in line with the principles of the ESSA. The tailored SOPs for PCAP will include the process for assessing land prices in line with market rates, compensation for assets and livelihoods for landowners, tenants, and non-titleholders, and processes for removal of informal settlers.
4 (b) Identifies and addresses economic and social impacts caused by land acquisition or loss of access to natural resources, including those affecting people who may lack full legal rights to assets or resources they use or occupy	LAA 1894 has clear provisions and procedures for compensation in this regard when it comes to land titleholders under its Sections 23 and 24. The use of Section 17 under LAA leads to further issues due to emergency acquisition of land without prior public consultation. The Punjab Local Government Act 2022 allows removal of moveable and immovable encroachments from local government lands and application of fines.	The Law does not cover non-titleholders and encroachers for any type of compensation. There is a strong resistance to this as most public officials consider encroachments as organized crime where influential individuals occupy public land and lease/rent it out to the landless.	
4 (c) Provides compensation sufficient to purchase replacement assets of equivalent value and to meet any necessary transitional expenses, paid before taking of land or restricting access	According to LAA 1894, compensation is paid for all assets at market value rather than replacement cost. The process for the determination of market price relies on recent land sale transactions of a similar nature in the nearby area. The market rates are also notified by the relevant deputy commissioner on an annual basis.	Replacement costs for assets not considered. One of the reasons cited has been the antiquity of the Law. Such concepts were not present in the colonial rule of 1894.	
4 (d) Provides supplemental livelihood improvement or restoration measures if taking of land causes loss of income-generating	Livelihood improvement or restoration measures are not a consideration under the Law.	A serious shortcoming in the Law. The law is silent on such compensation for even the titleholders. Economic losses due to the removal of informal businesses established on	

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
opportunity (for example, loss of crop production or employment)		encroached land, including temporary establishments, are not covered by law.	
4 (e) Restores or replaces public infrastructure and community services that may be adversely affected.	This is not covered under the Law.	Relevant department for whom the public land has been acquired, or whose impact is on public infrastructure and community services, restores and replaces it.	
Core Principle 5: Due consideration is given to cultural appropriateness of, and equitable access to, Program benefits, giving special attention to the rights and interests of indigenous peoples and the needs or concerns of vulnerable groups.			
5. (c) Gives attention to groups vulnerable to hardship or disadvantage, including, as relevant, the poor, the persons with disabilities, women and children, the elderly, or marginalized ethnic groups. If necessary, special measures are taken to promote equitable access to Program benefits.	- Punjab has enacted a Women Development Policy 2018 aiming at improving overall status of women in Punjab through their socio-economic, political and legal empowerment. Salient features of the policy include access to public spaces, provision of quality health care, reducing gender gap in education, creation of employment and income generation opportunities to encourage their equal participation in all political processes and to ensure elimination of all forms of discrimination and violence against women and girls through legislation and its effective implementation. - The provincial government has also passed the Punjab Protection Against Harassment of Women at the Workplace Act 2012.	Policies for the inclusion of women and persons with disabilitiespersons with disabilities exist for public transport and agriculture. However, IAs do not have dedicated services or for catering to the inclusion of vulnerable groups, including women and persons with disabilities. Therefore, the capacity to mitigate the mentioned risks in preceding sections is limited in the implementing agencies. - Overall implementing agencies are limited by human resources availability, capacity and infrastructure to target and respond to risks and challenges faced by the vulnerable groups such as encroachers, ethnic minorities and persons with disabilities. - Projects financed by donors in the IAs have staff and activities mandated to work towards the welfare of vulnerable people as required by the donor agencies. However, these are mostly limited to the project activities. - Regarding the use of digitized AQM systems, these have been in use by the EPCCD staff	- It is recommended to review the implementation of the Women Development Policy, The Punjab Empowerment of Persons with Disabilities Act, and Punjab Agriculture Policy in the IAs with respect to inclusion of vulnerable people. - Prepare a Gender Action Plan for the Program activities related to ensuring inclusion of women in accessing program benefits. - Conduct a social assessment to establish a baseline of access of women and persons with disabilities to public transport and e-vehicles. - Prepare SOPs for ensuring accessibility of communications materials and awareness campaigns for vulnerable groups.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	• Punjab Commission on the Status of Women Act 2014 has also been enacted. The Commission has been constituted and given an autonomous status for effective and improved performance, efficiency and responsiveness, to provide effective services for promoting women rights and to eliminate all forms of discrimination against women and for matters connected therewith. The • The Punjab Empowerment of Persons with Disabilities Act 2022 provides for the establishment of disability-friendly infrastructure in public places, including public transport. • Access for elderly and senior citizens to public places is not governed by law. However, the Senior Citizens Welfare Bill 2024 is under consideration by the Punjab Government. • The Punjab Transport Company provides Free Transport Card for senior citizens and persons with disabilities to travel free on PTC buses within Lahore City. • Social Protection Policy 2022 in Punjab is designed to provide a comprehensive, multi-faceted	along with other digitization measures in the department.	• Review the accessibility of the existing provincial and department wise Grievance Redressal and Right to Information forums for vulnerable groups.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	approach to supporting the welfare of its citizens, especially those facing economic hardship and social exclusion. • Punjab Right to Public Services Act 2012 is a significant piece of legislation aimed at improving governance and ensuring timely delivery of essential public services to citizens. The law aims to hold public officials accountable and make service delivery more efficient, transparent, and responsive to the needs of the people. • Punjab Agriculture Policy 2018 includes several key provisions to support small farmers and improve their livelihoods.		
Core Principle 6: Avoid exacerbating social conflict, especially in fragile states, post-conflict areas, or areas subject to territorial disputes.			
Considers conflict risks, including distributional equity and cultural sensitivities	There are no fragile, post conflicts or areas subject to territorial disputes in the Lahore Division. Possible protests and conflicts between public, private contractors and government may arise due to pollution abatement measures, enforcement of bans and penalties, and introduction of new public transport systems, as discussed in the earlier sections. These	There are multiple GRM platforms at the provincial level such as Punjab Portal, individual GRM of service delivery departments, and Provincial Ombudsmen office, which serve the citizens at large. The Complaint Cell at the EPCCD maintains a record and tracks complaints received through different national and provincial channels. The other IAs do not have such an integrated system and can benefit from the same.	• Ensuring accessible, effective and continued communications on air pollution abatement measures will keep the public engaged and reduce the risk of conflicts and protests.

Key Planning Elements	System Assessment	Capacity Assessment	Recommendations
	may be mitigated by effective communications and GRM systems. The Punjab government has put in place GRM systems at the provincial level, which are online and can be accessible by any individual. The Punjab Portal allows filing complaints from any aggrieved individual and is housed under the Chief Minister's office. In addition, each IA has its own GRM in place focusing on their own performance. The Provincial Ombudsman Office is mandated to receive and resolve complaints against any public institution and has a user-friendly system in place.	Effective communications by the EPCCD on Green Lockdowns have been helpful in managing public expectations.	

6. DISCLOSURE AND CONSULTATION

6.1. Disclosure

132. This draft ESSA will be disclosed in-country and on the World Bank's external website, prior to formal appraisal of the relevant PforR, to serve as the basis for discussion and receipt of formal comments. Following incorporation of the feedback received from different stakeholders, the revised ESSA will be disclosed in-country and on the World Bank's external website.

6.2. Stakeholder Consultations

133. Consultations with relevant institutions, program affected peoples, experts, and beneficiaries are essential in the proper planning and preparation of development projects and programs, to ensure effective identification and assessment of environmental and social effects, and to recommend measures to improve environmental and social management capacity. Stakeholder consultations were an integral part of the ESSA process and were carried out consistent with applicable World Bank principles.

134. In addition to the formal Bank organized multi-stakeholder workshop that was held in Lahore on January 30, 2025, the Bank has and will continue to engage in other formal and informal consultations as needed prior to finalization of this ESSA and the PforR. Multiple meetings, interviews and checklists were used to gather information from the relevant implementing agencies to identify and validate the risks mentioned in earlier sections of this ESSA, and to agree upon the recommendations.

135. This sub-section highlights important consultations conducted during the process, and the following sub-sections summarize major findings and recommendations received. Consultations and meetings with key relevant stakeholders at the provincial level, particularly with those involved in environmental and social assessment and management as well as planning, implementation and monitoring of the Program were conducted as follows:

136. Institutional consultations to identify risks and assess institutional environmental and social management capacity and associated environmental and social management systems of the implementing agencies were held virtually on 24th December 2024, followed by in person detailed meetings with P&D, TMD, EPCCD and AD in Lahore from 15th to 17th January 2025.

137. The World Bank subsequently held a multi-stakeholder workshop on the proposed Program in Lahore on January 30, 2025.

6.3. Summary of the Multi-Stakeholder Workshop

138. The Bank-organized multi-stakeholder workshop in Lahore on January 30, 2025 was attended by 35 stakeholders from government, academia, the private sector, and civil society representing citizens, women and persons with disabilities as shown below. A detailed list of participants is provided in Annexure 5:

- **Government Departments:** Planning & Development Board, Transport & Masstransit Department, Agriculture Department, Environmental Protection and Climate Change Department, Energy Department, Industries Department, Punjab Energy Efficiency and Conservation Agency, Special Education Department, Social Welfare & Bait-ul-Mall Department, Lahore Development Authority, Punjab Small Industries Corporation.
- **E&S Specialists from World Bank Financed Programs and Projects:** PIUs staff from the Punjab Green Development Program (PGDP), Punjab Cities Program (PCP), Punjab Affordable Housing Program (PAHP), Punjab Resilient and Inclusive Agriculture Transformation Project (PRIAT)
- **Civil Society & NGOs:** WWF Pakistan, Pakistan Air Quality Initiative (PAQI).
- **Private Sector:** Pakistan Agricultural Machinery and Implements Manufacturers (PAMIMA), Dawood Agro Company (progressive rice farmer)
- **Academia:** Lahore University of Management Sciences (LUMS)

139. Major points raised by the participants can be summarized as follows:

- **Feedback on Program Design:** CSOs and NGO representatives highlighted that the expansion of air quality monitoring stations must be scientifically planned to ensure comprehensive coverage and real-time public access. Multiple stakeholders including government, academia, NGOs and private sector are conducting AQM activities, but are working in silos. Bringing them to one platform is essential. Institutional strengthening of the EPCCD and technical capacity building is essential for effective AQM and monitoring. Private sector representatives in agriculture emphasized that super seeders should be provided in targeted areas / hot spots for crop burning such as Gujranwala Division and Lahore Division which grow 50% of the rice crop of Punjab. This will ensure maximum positive impact to air quality. To encourage production of agricultural machinery to combat crop burning, the program may also consider providing incentives to machinery manufacturers.
- **Stakeholder Engagement:** Feedback received from CSOs, and government largely focused on ensuring transparency and accountability of the Program through citizens' engagement and monitoring. Ensuring that air quality data is real-time accessible and actionable for both policymakers and the public is crucial for behavioral change.
- **Inclusion of Vulnerable Groups:** Government stakeholders from EPCCD and the Social Welfare Department stressed on the importance of ensuring the public transport infrastructure is accessible for persons with disabilities and the relevant government legislations on ensuring access of public facilities to vulnerable groups are implemented under PCAP activities.

- **Feedback on ESSA Recommendations:** Participants expressed that the ESSA recommendations should be institutionalized across government departments so that they can have an impact beyond the Program.
- **Site Selection for Bus Depots:** Representative from the Punjab Green Development Program (PGDP) emphasized that the site selection for bus depots under the program would be critical. The eco-sensitive areas should have been defined and avoided for the bus depots. The sites should have been carefully screened environmentally and then selected. E&S team ensured that the 'Environmental and Social Assessment (ESIA)' of all the bus depots would be conducted under the program to assess environmental and social effects of the development of the bus depot at the specific site prior to the finalization of the site and its construction. The EIA of the bus depots would also be conducted under the law including site screening and site alternative analysis. The ESIA would include, inter alia, i) site alternative analysis, ii) environmental impact assessment, iii) appropriate measures for early identification and screening of potentially important biodiversity and cultural resources of the sites selected for bus depots, iv) screening of the proposed site for possible impact on sites of archaeological significance, v) safety aspects while developing bus depots with respect to structure safety and fire safety, and vi) consider natural hazards and provide recommendations to build climate resilient infrastructure.
- **Baseline Health Study of the Punjab:** The representative of PGDP suggested that the program should have conducted baseline health study of the Punjab province so that the performance of the program could be evaluated against the baseline health situation in the province after the completion of the program. It was appreciated and recognized it as a good suggestion. It was confirmed that this suggestion would be discussed with the program management.
- **Facilities for Persons with disabilities in the e-buses:** Participants were in the viewpoint that the e-buses should have provided facilities for the persons with disabilities such as wheelchair ramps etc. for their boarding and alighting the bus. One participant also commented that there was no participation of the disables at this workshop and without the representation of this vulnerable class, their specific concerns could not be understood and addressed. It was explained that the concerns regarding the persons with disabilities boarding and alighting the bus had been addressed in the ESSA and recommended to consider this aspect during procurement and provide such facilities in the buses. E&S team admitted that they would consider including disable persons in the next consultation to listen their concerns about the program activities.
- **Third Party Validation of the Program Performance:** Participants emphasized that there should have been the third-party validation of the program performance. The World Bank team responded that the Program performance evaluation at different stages of the program was always inbuilt in Program design.
- **Clear Allocation of the E&S Responsibilities of the Implementation Agencies:** Representative from the PGDP emphasized that the E&S responsibilities of all the IAs under P-CAP should have been clearly defined/allocated to avoid any confusion among IAs with respect to allocation of E&S compliance requirements in future. The context of this recommendation was that the PGDP experience of

implementing safeguard requirements by the Implementing Agencies (IAs) was not good. The environmental and social safeguard responsibilities of each IA were not spelled out in the PGDP, therefore most of the IAs transferred their responsibilities of implementing environmental and social safeguard requirements of their tasks to the EPCCD due to its overall environmental mandate in the province which created conflict among the IAs.

- **Inclusion of Transplanters in the Program:** Representative of Dawood Agri Company gave his opinion that the program should have also provided the transplanters to the farmers along with super seeders which would improve the farmers' efficiency. It was replied that program funds were only allocated for the super seeders, transplanters addition was not possible in the Program. Representative from Agriculture Department also added that transplanters would not create any impact on achieving the Program objectives.
- **Focus on Rice Crop Hot Spots:** Representative of Dawood Agri Company also emphasized that the program should have focused on rice crop hot spots of the Punjab such as Gujranwala and other large rice crop areas. It was explained that these hot spots would be identified under the Program and covered. However, most of the hot spots were already covered under the Program.
- **Promotion of e-bike Manufacturing Industries:** Participants were in the view that the program should have considered to promote the manufacturing of the e-bikes in the country, though there were few manufacturers already operational and producing e-bikes in the country. It was explained that the Program would promote e-bikes in the province by providing easy access to the financing through banks, which would ultimately promote e-bikes manufacturing.
- **Solar System Installation by the Agriculture Machinery Manufacturers and Reduction in Sales Tax/Income Tax on Super Seeders:** Representative from Pakistan Agriculture Machinery Association requested the Program that the responsibility of solar system installation under the Program should have been granted to the agriculture machinery manufacturer. He also requested that the Program should have facilitated the farmers by reducing sales tax and income tax on Super Seeders which would ultimately reduce 23% of the cost of the Super Seeders. It was responded that these aspects were not under the mandate of the Program. Limited number of high air polluting industries will be provided technical and financial support by the Program.
- **Solid Waste Legislation:** Representative from WWF Pakistan enquired that as there was no legislation on solid waste (including e-waste and waste batteries) in the province, would program work on to formulate legislation on solid waste, including e-waste. It was explained that the Program will prepare and adopt SOPs for e-waste management. Furthermore, , the Program will support the Punjab EPA to formulate waste batteries management and recycling rules under Hazardous Waste Management Act.
- **Stakeholders on Air Quality, Working in Silos:** Representative from WWF Pakistan added that there were many stakeholders in the Punjab working on air quality but all of them were working in silos. These stakeholders do not collaborate with each other and share their data. There is an urgent need to engage all the stakeholders and put combined effort to solve the air quality issue

of the province. It was agreed and explained that all the stakeholders of the program would be engaged and involved in the program activities.

- **Special Infrastructure for the e-buses:** Someone from the participants asked about the e-buses infrastructure and whether the same infrastructure would be utilized, or new infrastructure be built. It was explained that the same infrastructure would be utilized except for the bus depots.

7. RECOMMENDATIONS AND PROPOSED ACTIONS

7.1. Introduction

140. This section summarizes the E&S risks and impacts of the program activities, along with the recommendation measures which are included in the Program Action Plan (PAP). These actions will be executed during Program implementation to address important gaps and key elements of the Program systems identified with reference to the PforR core principles.

141. As noted in Section III, the ESSA identified the following E&S issues:

- a. **Lithium battery specific fire hazard in the e-buses and at battery storage/charging areas pose Life and Fire Safety (LFS) risks and specific legal provisions are not available to address this risk.** Lithium battery used in e-buses poses LFS risks, owing to the higher fire risk. When the battery is damaged, it presents a significant fire hazard that cannot be controlled by a traditional fire extinguishing system. Battery charging stations and battery storage areas will also be potential fire prone areas.
- b. **Improper recycling of waste lithium batteries and other e-wastes is a potential risk to the environment, workers and the community. Specific legal provisions are not available to address and manage waste batteries and e-waste in the province.** After a few years, lithium batteries used in the e-buses and other e-vehicles (2/3 wheelers), will expire and accordingly will require proper disposal. Batteries used in solar systems will also expire needing disposal. Similarly, the Program activities, under expansion of air quality monitoring regulatory grade stations across Punjab, with real-time data reporting, will require procurement and installation of IT equipment. Waste IT equipment and IoTs will also be produced by the end of their life. The waste batteries and e-waste recycling facilities in Punjab are in the informal sector where recycling practices are not environmentally friendly, resulting in potential environmental and health and safety impacts to the associated workers and the community due to discharge of heavy metals, dust and hazardous material.
- c. **Bus washing activities will consume substantial quantity of water and generate wastewater and sludge.** Bus washing activities at bus depots will consume substantial amount of water, generate wastewater and sludge. As Punjab is recently experiencing extensive water shortage, therefore, bus washing activities with increased water consumption will put stress on the water resources. Improper disposal of wastewater sludges will be another environmental concern at the bus depots.
- d. **Development and construction of bus depots is expected to result in air and noise pollution from construction activities,** potential soil and water contamination, disruption of local ecosystems, and traffic congestion impacts. Additionally, failure to integrate climate and environmentally friendly design elements at an early stage in bus depot construction can lead to increased energy consumption and greenhouse gas emissions due to reliance on inefficient lighting and cooling systems. Poor stormwater management may cause flooding and soil

erosion, while extensive use of impermeable surfaces can exacerbate runoff issues particularly in monsoon. The absence of shading, green buffers, or reflective materials may contribute to the urban heat island effect, increasing cooling costs and worker discomfort. Inefficient waste and water management can lead to higher operational costs and environmental pollution, particularly from hazardous waste such as oils and batteries. Additionally, lack of climate-resilient materials may result in faster infrastructure degradation due to extreme weather events.

- e. **Implementing Agencies lack systems and capacity to ensure equitable access for vulnerable groups and ensure community health and safety.** As explained in detail in the earlier sections, implementing agencies lack capacity for ensuring equitable access to Program benefits for vulnerable groups such as women, persons with disabilities, people with low incomes and elderly. Similarly, IAs lack the capacity to identify and address risks to community health and safety including risk of SEA/SH, child labor and forced labor during construction activities.
- f. **Legal provisions on land acquisition do not cover compensations for encroachers.** Physical or economic displacement of encroachers or non-title holders due to construction of public transport infrastructure is not covered by the Land Acquisition Act 1894.

142. The recommendations and proposed actions are described in detail below:

7.2. Recommendations to be Included in the Program Action Plan

- 143. **Recommendation 1:** Program will ensure lithium battery fire safety during procurement by installing battery specific fire suppression systems in e-buses. E-Bus drivers and staff in fire extinguishing system to prevent thermal runaway (when battery damages or crashed) and enable early evacuation of passengers. Program should also install fire suppression systems at battery charging/storage areas/depots. Ensure fire-prone areas are separated and implement wastewater recycling for bus washing to conserve water and manage sludge disposal.
- 144. **Recommendation 2:** Program will develop SOPs e-waste management at the departmental level (for project IAs). The program will also mobilize EPCCD to develop and enforce waste batteries recycling management rules and certify recycling facilities.
- 145. **Recommendation 3:** Program should conduct ESIA for all bus depots, integrating site alternative analysis, environment and social impacts, measures for identification of biodiversity and cultural resources of sites, sites screening for archaeologically important places, site safety and fire aspects, occupational and public health and safety aspects, and climate resilience into design, construction, and operations. The scope of the ESIA will include bus depots, charging stations and the bus operations. As required under the prevailing laws, it will contain OHS plan and public safety measures.
- 146. **Recommendation 4:** Program should develop and implement strategy for safe disposal of buy-back or retired vehicles.

147. **Recommendation 5:** Program will build the capacity of institutions on updating/amending the existing rules/ordinance on Motor Vehicles emissions.
148. **Recommendation 6:** The fuel testing laboratories establishing under the Program will require to prepare and follow the ESMP and get ISO/IEC 17025 certification.
149. **Recommendation 7:** The institutional arrangement for E&S, as described in the previous section (PIU and E&S focal persons) to be put in place from the outset of the program. IAs to develop internal institutional procedures, appoint staff, and build their capacity on institutional procedures for environmental and social risk management. PCAP-PMU to clearly describe E&S roles and responsibilities of all relevant departments under the program; develop and oversee monitoring and reporting procedures.
150. **Recommendation 8:** Ensure E&S Considerations in Procurement & Program Operations. Conduct E&S assessments for super seeders & agricultural machinery and develop SOPs for deployment of super seeders including awareness raising for its safe E&S performance. Develop buses procurement, and operator manuals with E&S provision.
151. **Recommendation 9: Workers and Community Health and Safety:**
- Prepare an Occupational Health and Safety Plan, and Community Health and Safety Plan for the Program activities related to construction and operation of e-bus infrastructure and e-bus operations as part of the ESIA. These plans are to be prepared as part of the ESIA by the Transport & Mass Transit Department and implemented by PMA, PTC and construction contractors. Transport & Mass Transit Department to ensure all construction contracts include community health and safety, labor safety child labor and SEA/SH prevention clauses.
 - Prepare a Gender Inclusion and SEA/SH Action Plan for the Program activities related to ensuring inclusion of women in accessing program benefits, and SEA/SH prevention in program activities, especially for women employed in e-bus operations and women using public transport.
152. **Recommendation 10: Fair Land Pricing and Displacement Mitigation:** The Program will apply two key exclusion criteria regarding land acquisition and resettlement. Sites identified for e-bus depots will exclude: (i) areas displacing 200 or more individuals per site; and (ii) areas which have legacy anti-encroachment drives (AED) starting from the approval date of the Program Concept (December 9, 2024). The SOPs on land acquisition and resettlement from PGDP will be tailored for PCAP, addressing gaps in areas affected by anti-encroachment drives (AEDs). Key provisions will include:
- Land acquisition and resettlement screening checklists for each e-bus depot against the exclusion criteria and to identify resettlement impacts;
 - Compensation for affected landowners, tenants, and non-titleholders, aligned with market rates; and
 - Preparation of Abbreviated Resettlement Action Plans (ARAP) or Resettlement Action Plans (RAP) before commencing work (if required).

153. Recommendation 11: Stakeholder Engagement, Accessibility of Information and Inclusion of Vulnerable Groups

- Prepare a Program Wide Stakeholder Engagement Plan (P-SEP) to ensure consistent, transparent and inclusive consultations throughout the program life. This will build upon the SEP prepared for the TA component, will identify all relevant program stakeholders, engagement activities, methods of engagement and timelines. As part of the SEP, conduct annual Citizen Feedback Surveys on program activities and publish results.
- Conduct a social assessment to establish a baseline of access of women and persons with disabilities to public transport and e-vehicles. The social assessment will focus on establishing a baseline of the use of public transport by women and persons with disabilities, and cultural acceptance of women driving e-2 wheelers. This will allow the Program to monitor and establish a midline and an endline on the same indicators.
- Prepare Disability Inclusion Plan for implementation of The Punjab Empowerment of Persons with Disabilities Act 2022 across RA 2 activities to ensure that public transport infrastructure including buses, bus stops and pedestrian access to bus stops such as pavements and road shoulders are accessible.
- Prepare SOPs for ensuring accessibility of communications materials and awareness campaigns for vulnerable groups. These SOPs should ensure that all communications and awareness campaigns conducted through the Program are inclusive and accessible to vulnerable groups including women, illiterate, persons with disabilities, rural communities and those without access to digital mediums.
- Prepare SOPs for inclusive communication, ensure awareness campaigns and materials are accessible to women, illiterate, persons with disabilities, and rural populations, including those without digital access.

154. Recommendation 12: Strengthening Grievance Mechanisms

- Review the accessibility of the existing provincial and department wise Grievance Redressal and Right to Information forums for vulnerable groups. Punjab government has existing forums on grievance redressal and RTI, whereas all IAs also have forums for receiving and responding to grievances. However, the accessibility of these forums for vulnerable groups including women, gender minorities, the persons with disabilities, illiterate and rural communities is lesser known. By reviewing their accessibility, the Program can ensure grievance redressal and information sharing is more inclusive and responsive towards vulnerable groups.
- Prepare SOPs for IAs to track, collate, respond to and report on program related grievances received through different national and provincial channels.

155. The recommended inputs for PAP are given in Table 7.

Table 7: RECOMMENDED INPUTS FOR PROGRAM ACTION PLAN

No	Action Description	Source	Responsibility	Timing	Completion Measurement
1	Appoint one Environmental Specialist, and one Social/Gender	Environmental and Social Systems	Specialists – PCAP-	Within 90 days of Program effectiveness	Listed specialists and focal points appointed and maintained

	Specialist at the PCAP-CMU and T&MD PIU, and one Environmental/Social Focal Point within each IA		CMU and T&MD PIU E&S Focal Points - AD, EPCCD, Industries Dept, PSIC, PMA, PTC and other agencies as relevant		throughout Program implementation
2	Prepare & operationalize E&S technical manual	Environmental and Social Systems	PCAP-CMU & T&MD PIU	Within 90 days of Program effectiveness	E&S technical manual acceptable to WB and approved (based on PSC minutes)
3	Tailor and adopt PGDP land acquisition and resettlement standard operating procedures.	Environmental and Social Systems	Developed and implemented by T&MD Overseen by PCAP-CMU	Within 90 days of Program effectiveness and before civil works start.	Tailored standard operating procedures acceptable to WB and approved (based on PSC minutes)
4	Develop and implement i) Gender Inclusion and SEA/SH Action Plan, ii) P-SEP including GM, iii) Disability Inclusion Plan	Environmental and Social Systems	i) & ii) PCAP-CMU, to develop and all IAs to implement; iii) T&MD to develop, and PMA and PTC	i) , ii) and iii) within 12 months of Program commencement and prior to the start of any Program activities	Gender Inclusion and SEA/SH Action Plan, Disability Inclusion Plan, P-SEP including GM prepared and submitted to Bank, disseminated to stakeholders and operationalized

			to implement; iv) PCAP-CMU in coordination with IAs		
5	Environmental and Social Impact Assessment (E SIA) prepared for bus infra and operations, incl. OHS and CHS plans.	Environmental and Social Systems	T&MD to develop PMA and PTC to implement EPCCD to monitor	Prior to the start of civil works	ESIA acceptable to WB and approved by EPA.
6	Tailor and adopt SOP for e-waste management	Environmental and Social Systems	Tailored and adopted by PCAP-CMU All IAs to implement	Within 6 months of Program effectiveness	SOPs acceptable to WB and approved (based on PSC minutes).
7	E&S checklists for deployment of super seeders	Environmental and Social Systems	Developed and implemented by Agriculture Department. Overseen by PCAP-CMU	Prior to the deployment of super-seeders	E&S checklists acceptable to WB and approved (based on PSC minutes).
8	Develop and implement trainings for IAs on E&S management, including CHS, child labor, SEA/SH, e-waste and lithium battery fire safety	Environmental and Social Systems	Developed by PCAP-CMU Implemented by IAs	Within 90 days of Program effectiveness and before civil works start	E&S training module developed and acceptable to the WB. E&S training completed

9	Conduct annual citizen feedback surveys and publish results	Environmental and Social Systems	PCAP-CMU	Annual	Citizen feedback surveys implemented and results disseminated
10	Exclude from consideration as a Pro ram area any areas subject to AED on or after December 9, 2024, or that may involve the displacement of 200 or more individuals	Environmental and Social Systems	PCAP-CMU & IAs	Continuous	Regular reporting by PCAP-CMU

ANNEX 1: POLICY AND LEGAL FRAMEWORK FOR MANAGING ENVIRONMENTAL AND SOCIAL IMPACTS OF THE PROGRAM

156. This section describes policy and legal framework at national and provincial levels for managing potential environmental and social impacts of the Program activities mentioned in Section III by the implementing agencies.

Policy Framework

157. In 2010, the 18th Constitutional Amendment was approved by the National Assembly of Pakistan. Under the amendment, environmental functions in the territory of the federal capital were delegated to Pakistan Environmental Protection Agency (Pak-EPA), and Provincial Environmental Protection Agencies were delegated the environmental management functions of provinces. In 2012, the federal government converted the Ministry of Environment to the Ministry of Climate Change (MCC). The MCC is mainly responsible for managing common national-level environment issues and climate change impacts and implementation of international conventions signed by the Government of Pakistan.

Environmental Policies and Legal Framework

158. The following section describes environmental policies at national and provincial level which are relevant with the Program activities:

National Environmental Policy, 2005

159. The National Environmental Policy provides an overarching framework for addressing the environmental issues facing Pakistan, particularly pollution of freshwater bodies and coastal waters, air pollution, lack of proper waste management, deforestation, loss of biodiversity, desertification, natural disasters and climate change. It also gives directions for addressing cross-sectoral issues as well as the underlying causes of environmental degradation and meeting international obligations.

160. Policy measures recommends to i) enact the National Clean Air Act, ii) ensure reduction and control of harmful emissions through regulatory programs, iii) promote cleaner production technologies, iv) introduce discharge licensing systems for industry, v) establish cleaner production centers and promote cleaner production techniques and practices, vi) encourage reduction, recycling and reuse of municipal and industrial solid and liquid wastes, and vii) provide financial and other incentives (reduction/elimination of tariffs, low interest loans, appreciation certificates and awards) for technology up-gradation, adoption of cleaner technology, implementation of pollution control measures and compliance with environmental standards.

Punjab Environmental Policy, 2015

161. The Punjab Environment Policy 2015 provides an overarching framework for addressing the environmental issues facing Punjab, particularly pollution of freshwater bodies, air pollution, lack of proper waste management, deforestation, loss of biodiversity, desertification, water logging, natural disasters and climate change. It also gives directions for addressing the cross-sectoral issues, underlying causes of environmental degradation to meet national and international obligations. The theme of the policy is sustainable development in the sense of enhancement of human wellbeing. Policy provides broad guidelines to Provincial Governments and Local Governments for addressing environmental concerns related to them and ensuring effective management, restoration, enhancement of environmental resources and ensuring their sustainable use. Policy is a guide for actions in regulatory reform, programs and projects for environmental conservation; and will review and ensure the enactment of legislation, by Provincial and Local Governments. The policy has been approved by the Punjab Environmental Protection Council headed by the Chief Minister Punjab under Section 4 (b) of the Punjab Environmental Protection Act 1997 (amendment Act 2012) and is enforced with immediate effect.
162. The policy addresses effluent and sewage treatment plants, wastewater, air and soil pollution control, groundwater management, energy efficiency and renewable energy, climate change and ozone depletion, and industrialization and environment.
163. This policy also focuses on hazardous waste and recommends that the hazardous and air polluting industries be relocated from residential areas to outside the cities. The viable models of public-private partnerships will be developed and implemented for setting up and operating secure landfills, incinerators, and other appropriate techniques for the treatment and disposal of toxic and hazardous waste, both industrial and biomedical, on payment by users. The recycling technology and establishing system for segregation, recycling, and reuse of municipal solid wastes will be promoted. The policy recommends the establishment of cleaner production centers and promotion of cleaner production techniques.

National Climate Change Policy (Updated October 2021)

164. The goal of the updated National Climate Change Policy (NCCP) 2021 is: To ensure that climate change is mainstreamed in the economically and socially vulnerable sectors of the economy and to steer Pakistan towards climate compatible development.
165. Mitigating and adapting actions are the two ways of combating climate change. The more immediate and pressing task for the country is to prepare itself for adaptation to climate change. The updated NCCP 2021 highlights the government's flagship "Ten Billion Tree Tsunami Programme, in addition to the Prime Minister's 'Urban Forest Project', 'Clean Green Pakistan Movement', 'Protected Areas and National Park Initiatives'. The objective of the last two initiatives is to expand protected areas to at least 15% of Pakistan's area by 2023. Further, the policy's major emphasis is on 'Eco System Restoration Initiative (ESRI) for facilitating transition towards environmentally resilient Pakistan by mainstreaming adaptation and mitigation through ecologically

targeted initiatives. In the policy, the vulnerabilities of various sectors to climate change have been highlighted and appropriate adaptation measures spelled out. These cover policy measures to address issues in various sectors such as water, agriculture, forestry, coastal areas, biodiversity and other vulnerable ecosystems. Furthermore, the updated policy document has been designed in accordance with the requirements of the Paris Agreement on climate change, Sustainable Development Goals and Sendai Framework for Disaster Risk Reduction. Hence, appropriate measures relating to disaster preparedness, capacity building, institutional strengthening; technology transfer and international cooperation have also been incorporated as important components of the Policy.

166. The main objectives of Pakistan's National Climate Change Policy (NCCP) include: 1. To pursue sustained economic growth by appropriately addressing the challenges of climate change; 2. To integrate climate change policy with other inter-related national policies; 3. To focus on pro-poor gender sensitive adaptation while also promoting mitigation to the extent possible in a cost-effective manner; 4. To build climate-resilient infrastructure; 5. To track the impact of climate change on water, food and energy security of the country, and to implement remedial plans to support water, energy and food policies; 6. To minimize the risks arising from the potential increase in frequency and intensity of extreme weather events such as floods, droughts and tropical storms; 7. To develop climate-resilient agriculture and food systems for all agro-ecological zones in the country; 8. To promote the country's transition to cleaner, lower emission and less carbon intensive development; 9. To accelerate the policy coherence and integration to achieve the United Nations' Sustainable Development Goals (SDGs) in the light of its Sustainable Development Report 2020 (SDR2020) and Nationally Determined Contributions; 10. To strengthen inter-ministerial and inter-provincial decision making and coordination mechanisms on climate change; 11. To facilitate effective use of the opportunities, particularly financial, available both nationally and internationally; 12. To foster the development of appropriate economic incentives to encourage public and private sector investment in adaptation and mitigation measures; 13. To enhance the awareness, skill and institutional capacity of relevant stakeholders; 14. To promote tree plantation, conservation of natural resources, nature-based solutions and long term sustainability.
167. The main policy measures proposed in the NCCP 2021 include: Develop appropriate digital simulation models for assessment of climate change impacts on physical, chemical, biological and financial aspects of agricultural production systems in various agro-ecological zones; Develop new varieties of crops which are high yielding, resistant to heat stress, drought tolerant, less vulnerable to heavy spells of rains and less prone to attack by insects and pests; Improve crop productivity per unit of land and per unit of water by increasing the efficiency of various agricultural inputs, in particular irrigation water; Promote energy efficient farm mechanization to increase yields and laborsaving; Promote horizontal expansion of cultivated lands through development of wastelands, and rainwater harvesting through community based approaches to development; Promote feed conservation techniques and fodder banks in arable areas; Develop a proper risk management system including crop insurance to safeguard against crop failures due to extreme events (such as floods and

droughts); Protect ground water through management and technical measures such as regulatory frameworks, water licensing, slow action dams, artificial recharge especially for threatened aquifers, and adoption of integrated water resource management concepts; Ensure rational ground water exploitation by avoiding excessive pumping; Ensure recycling of wastewater through proper treatment and its reuse, for example in agriculture, artificial wetlands and ground water recharge; Aggressively pursue afforestation and reforestation programs with plantation suited to the effects of climate change; Restore degraded mangrove forests in the deltaic region and prevent their further degradation by allowing minimum necessary environmental flows downstream of Kotri; etc.

National Sustainable Development Strategy, 2012

168. The goal of the National Sustainable Development Strategy (NSDS) is 'vibrant and equitable economic growth' that delivers benefits to all, particularly the poor and the vulnerable, in a way which does not lead to undue exploitation or degradation of natural resources. The need for the NSDS is stark since the country faces a series of significant challenges: Growth rates are currently largely coupled with resource and natural material use, leading to continued environmental degradation. This has contributed to increasing water scarcity, a significant problem given that storage capacity and water efficiency delivery is low. In the energy sector, thermal efficiency is low, distribution losses are high, and power cuts are reported to be a common frustration for domestic and business users. In addition, strategic energy reserves are low. These problems are set against a backdrop of high poverty and illiteracy rates in addition to increased impacts of natural hazards, notably the impact of disastrous flooding. The three core program areas addressed are:

- Economic: Sustainable trade, cleaner production, and sustainable consumption
- Environment: Natural capital and biodiversity
- Social: Social protection, poverty alleviation and equal opportunity/human development.

169. The strategic goals of the program are to:

- Promote green investment and green jobs
- Improve eco-efficiency by changing production and consumption systems
- Internalization of environmental costs into pricing
- Develop sustainable infrastructure focusing on transport and communication
- Develop demand for sustainable consumption among consumers through raising awareness
- Account for depletion of natural resources in national accounts
- Promote efficient use of energy and water, including through improved watershed management and reforestation
- Improve biodiversity management and increase forest cover; prepare lists of endangered species
- Deliver basic services of acceptably high quality to all citizens. These would cover 10 years of schooling, healthcare, food, water, shelter and energy

- Ensure preparedness for natural and human-made calamities and emergencies through mitigation and integration of disaster contingencies in broader development strategies.

National Clean Air Policy, March 2, 2023

170. The policy interventions of National Clean Air Policy under transport, agriculture, industry and clean cooking are mentioned below:

Transport
Implement fuel quality standards in transport to comply with Euro-5 or Euro-6 leading to complete shift to minimum Euro-5 by 2025, or Euro-6 by 2030
Improved measures to ensure inspection of vehicles for compliance with NEQS/PEQS, through enforcement. Such measures include capacity building of Motor Vehicle Examiners (MVEs). Tools like digital maintenance of emission data need to be introduced for compliance/non-compliance of vehicles to regulate movement in identified areas
Introduce mechanisms for regulation on non-compliant vehicles, such as emission taxation or penalties
Introduce plan for gradual phasing out of obsolete technology in engines
Further development and improvement of mass transit systems (multi-modal transport model) to at least 10 major cities by 2030, to reduce reliance on personal vehicles and develop plan for expanding such systems to other cities.
Expedite the market penetration of EV's in Pakistan through legal mandates, subsidies, and investment in related infrastructure. Phased modernization of public fleet towards EV technology
Promote low carbon, fuel efficient infrastructure and technology within railways, maritime, aviation and development of clear road map that promote the use of freight services
Introduce mechanism for zoning on the basis of AQI to regulate traffic movement
Strengthen regulatory control for traffic management (focusing on big cities), through measures like strengthening of parking facilities, introduction of mechanism for imposing taxation to discourage motorized transport
Promote non-motorized mobility, through development and management of infrastructure like removal of encroachments, development of micro infrastructure, while ensuring gender considerations
Implement measures to reduce dispersion of particulate matter/dust, specially from construction sites/roadsides
Agriculture
Impose immediate and complete ban on burning of crop residues, and introduction of low-cost alternative options for agriculture waste management. Promotion of circular economy in agriculture waste management
Enhanced public-private partnerships and adopt and upscale available solutions for agriculture systems, like utilizing engineering controls to control particulate matter generation from tilling and threshing during crop harvesting
Introduce and upscale measures to reduce emissions from livestock sector
Industry
Revision of existing industrial emission standards, where necessary, and the establishment of sectoral industrial emission standards to address and mitigate emissions of air pollutants specific to each industry type
Strengthen regulatory measures to ensure compliance with NEQS/PEQS, with priority focus on high impact industry and conduction of third-party audits
Restriction on establishment of new industries within cities by establishing industrial sites outside

cities
Gradual upgradation of industries towards low-emission technologies, like upgradation of all brick kilns to zigzag by 2025
Impose immediate and complete ban on high polluting industries like pyrolysis of tires, burning of coal in traditional furnaces, etc., and practices like elimination of Hydro-Floro Carbon (HFC) emissions from contained and emissive application and elimination of gas flaring in oil and gas sector
Develop standards for diesel power generation are set below 800 Kilo Watt (kW) category. For already operational diesel generation sets, ensure usage of either of the two options:(a) use of retrofitted emission control equipment having a minimum specified PM capturing efficiency of at least 70%, type approved by one of the accredited labs; or (b) shifting to gas-based generators by employing new gas-based generators or retrofitting the existing diesel generation sets for partial gas usage
Introduce de-carbonization framework for power sector towards carbon neutrality targets
Clean Cooking
Introduce and promote measures to reduce indoor air pollution like improved cookstoves, provision of alternate domestic fuels like Liquid Petroleum Gas (LPG) and renewable energy sources to reduce exposure to pollutants

Punjab Clean Air Policy, March 2023

The policy interventions of Punjab Clean Air Policy under transport, agriculture, industry and clean cooking are mentioned below:

Transport
Expansion of public transport system and induction of electric buses in mass transport system (2023-26).
Implementing a comprehensive, old and used vehicles phasing out plan, through a consortium of vehicles registering, inspecting and regulating agencies, considering options of buy-back/subsidies, early retirement, and conversion to electric mode (2023-30).
Promoting electric and hybrid vehicles particularly to replace Qing-Qi/auto-rikshaws by creating market-based linkages (2023-24).
Registration of electric vehicles and charging stations with considerable proportion of power supply from alternate energy sources like solar energy etc. (2023-26).
i) Promote non-motorized means of transportation within urban territories. ii) With the facilitation of the Federal Government, ensuring supply of Euro-V and above compliant fuel at retail points in metropolitans to steadily increase its market share (2023-30).
i) Enforcement of VICS certification for ensuring emissions compliance with PEQS in all districts for passenger vehicles and private vehicles. ii) Append the payment of token tax with valid VICS certificate. iii) Banning the entry of vehicles older than 30 years in Lahore during the months of October to December. iv) Removal of encroachments along main roads and commercial markets (TEPA, Traffic Police and MCs) and introduce hourly parking rate to ensure efficient parking turnover (2023-24).
i) For curtailing emission contributions from the vehicular sector, revise the Punjab Environmental Quality Standards for new and used vehicles. ii) Employing an Intelligent Transportation System (ITS) and enforcing driving licenses in major cities to improve traffic management.

iii) Upgradation of the system to smart stickers – digitally readable at main entry points.
iv) Provision of drop lanes in school premises for preventing road choking, after feasibility study (2023-26).
i) The VICS certificate would be issued to vehicles having appropriate catalytic converters/Diesel Particulate Filters and compliance of PEQS in consonance with the New European Driving Cycle (NEDC). Certificate to specifically mention the above.
ii) Devising a mechanism to avoid roadside congestion and subsequent air pollution with concepts like School Districts and Walkable Commercial Areas, after feasibility study (2023-30).
Agriculture
i) Zero-tolerance policy on burning of stubble/crop residues and agricultural waste.
ii) Ban on crop residue burning.
iii) Fixation of rate for provision/sharing of happy seeders on rental basis (2023-24).
Sorting out the areas based on soil type and cropping pattern etc., for promotion of zero-tillage technology (2023-26).
Introducing climate smart practices for reducing contribution of Green House Gases (GHGs) from agriculture and livestock sectors (2023-30).
Environment friendly agricultural machinery for management of crop residues like happy seeder/Pak Seeders preferably on 60% subsidy with due consideration of repair and maintenance system (2023-24).
Promoting straw mix technique as an alternative to prevent stubble burning and improving nitrogen efficiency (2023-26).
Piloting projects for demonstration of biofuel, biochar, and raw material for paper and board industry for environment friendly use of agriculture waste (2023-30).
Industry
i) Strict compliance of Punjab Environmental Quality Standards for emissions by raising Environment Enforcement Squads.
ii) Ban on use of substandard fuel including but not limited to waste plastics, rubber, and cloths/rugs etc.
iii) Conversion of conventional brick kilns to Zig-Zag Technology (2023-24).
Creation of special industrial zones like Tannery zone for shifting industry of similar nature from residential area (2023-30).
i) Promotion of green investment and cleaner technologies among industrial sectors.
ii) Incentivizing the industry through loaning/financing at lowest possible interest.
iii) Greening industry through encouragement of use of renewable energy resources and providing tax exemptions or other incentives to compliant industry (2023-26)
i) Setting up industry specific environmental quality standards.
ii) Restricting industrial establishments outside the designated industrial areas/estates.
iii) Absolute restriction on illegal tyres pyrolysis plants, use of carbon powder and used tyres as a fuel.
iv) Restrict departments from issuance of NOCs and electric connection for establishment of an industrial unit outside the designated industrial areas (2023-24).
i) Maintaining the emission load inventory by each industry to keep check over their respective contribution and providing access to Environmental Monitoring Center.
ii) Registration and implementation under the Self-Monitoring and Reporting Tool (SMART) Rules (2023-26).
Regulating the industries for production of energy-efficient appliances (2023-30).
Mandatory installation of online stack emission analyzer in major plants/units with their online access to EPA (2023-24).
i) Adoption of cleaner production practices for pollutants quantity control through latest technologies.

ii) Promotion of solar energy in existing commercial markets/ plazas/ areas.
iii) Mandatory provision of solar systems in all new commercial and industrial establishments (2023-26).
Replacing coal fired boilers with Combined Heat and Power units in chemical, paper, printing, dyeing, leather, pharmaceutical and other industries (2023-30).
Provision of catalytic converters upon power generators (2023-26).
Clean Cooking
Use of Improved Cookstoves to Reduce Indoor Air Pollution: Option to utilize improved cooking stoves to reduce the effect of indoor air pollution from biomass burning (2023-24).
Option to utilize clean and affordable energy options such as liquified petroleum gas (LPG) for indoor space heating and cooking to reduce biomass burning (2023-24).

Policy on Controlling Smog, 2017

171. The policy interventions of Policy on Controlling Smog, 2017 under transport, agriculture, and industry are mentioned below:

Transport
<u>Introduction of low sulfur fuels</u> Keeping in view the significant contribution of fuel Sulphur to particulate matter (PM) and Sulphur Dioxide (SO ₂) emission, maximum allowable content of Sulphur for all fuels used in the country to be reduced from 10,000 to 500 ppm by end of 2017. A liaison is to be made with the Federal Government in this regard.
<u>Adoption of Euro II Standards for reduction of vehicular emissions</u> Fuel efficient Euro II standard to be adopted and backed by a strict vehicle inspection regime, both for new and second-hand imported vehicles. Particulate matter control technologies, such as diesel oxidation catalysts (DOCs) and diesel particulate filters (DPFs) to be installed in the in-use vehicles.
<u>Installation of catalytic converters in vehicles for reduction of harmful emissions</u> For reduction of harmful vehicular emissions such as NO _x , Sox and CO to simpler non harmful levels, catalytic converters to be installed in motor vehicles. The Federal Government to be approached to issue directions to all vehicles manufacturers and importers for the purpose with a clear cut-off date where after no vehicles to be allowed to be manufactured or imported without a suitable converter. At enforcement level, a cut-off date, where no vehicle to be allowed to play in the Punjab unless it has a catalytic converter installed on it, to be announced.
Better traffic management for reduction of vehicular emissions
Agriculture
<u>Ban on burning of crop residue</u> Ban to be imposed on the burning of crop residue under section 144 of Code of Criminal Procedure. Strict enforcement of ban to be ensured.
Industry
<u>Shutting down of major smoke emitting industries</u> Shutting down of large smoke emitting industries including foundries, steel re-rolling mills especially in the vicinity of major cities for reduction of emission of pollutants.
<u>Greening of industrial processes</u> Industrial processes to be made green by mandating new industries to use such plants, machinery and processes which are environmentally friendly. To start with, such a plan to be implemented on a limited set of industries which are major contributors to air pollution such as cement manufacturing plants, steel mills, thermal power plants etc.

National Electric Vehicle Policy, 2019

172. The Policy was adopted in November 2020 with an aim to tackle transport emission. It provides targets and incentives aimed at achieving electric vehicles capture 30% of all passenger vehicle and heavy-duty truck sales by 2030, and 90% by 2040. The Policy provides a number of incentives: manufacturers, assemblers, and suppliers in the EV and related infrastructure industries will benefit from lower taxes – 1% GST for EVs as opposed to 17% for regular vehicles; import duty for charging equipment is being slashed to 1%; the government will lower the unit rate of electricity for charging station operators; the government will install at least one DC fast-charging station every 10 km² in all major cities and every 15–30 km on all motorways. Policy encourages registration of electric vehicles and establishment of national center for electric vehicles.

Environmental Legislations

173. The following section describes environmental laws and rules at national and provincial level which are relevant with the Program activities.

Pakistan Environmental Protection Act 1997

174. PEPA 1997 is the basic legislative tool empowering the Government to frame regulations for the protection of the environment. It is comprehensive legislation and provides the basic legal framework for protection, conservation, rehabilitation, and improvement of the environment. The act is applicable to a wide range of issues and extends to air, water, soil, marine, and noise pollution, and to the handling of hazardous wastes.
175. Environmental pollution control associated with hazardous waste is addressed in this act under Section 13 and 14. Under Section 13 'Prohibition of Import of Hazardous Waste', no person shall import hazardous waste into Pakistan and its territorial waters, Exclusive Economic Zone and historic waters. Under Section 14 'Handling Hazardous Substances', no person shall generate, collect, consign, transport, treat, dispose of, store, handle or import any hazardous waste except under a license issued by the Federal Agency and in such manner as may be prescribed or in accordance with the provision of any other law or of any international treaty, convention, protocol, code, standard, agreement or other instruments to which Pakistan is a party.

Pakistan Climate Change Act 2016

176. The Prime Minister established Pakistan Climate Change Council which coordinates and supervises the enforcement of the provisions of the Act, monitor implementation of the international agreements relating to climate change, approve and monitor implementation of comprehensive adaptation and mitigation policies, strategies, plans, programs, projects and other measures formulated by the authority to meet Pakistan's international obligations, monitor the implementation of National Adaptation Plan and its constituent provincial and local adaptation action plans,

approve guidelines for the protection and conservation of renewable and non-renewable resources, species, habitats and biodiversity adversely affected or threatened by climate change.

177. The Minister In-charge of the Federal Government shall establish the Pakistan Climate Change Authority to exercise the powers and perform the functions under the Act. The functions of the authority shall be to formulate comprehensive adaptation and mitigation policies, plans, programs, projects and measures designed to address the effects of climate change, establish institutional and policy mechanism for implementation of Federal and provincial adaptation and mitigation policies, plans, programs, projects and measures, prepare suitable adaptation and mitigation projects for submission to international and local institutions for funding, including Clean Development Mechanism (CDM), Global Environmental Facility (GEF), Green Climate Fund and Adaptation Fund, prepare National Adaptation Plan and its constituents provincial and local adaptation plans, carry out Technology Need Assessment and prepare Climate Change Technology Action Plan in accordance with international best practices, prepare projects for funding under the Reducing Emissions from Deforestation and Forest Degradation (REDD) Mechanism, prepare guidelines for the protection and conservation of renewable and non-renewable resources, species, habitats and biodiversity which are adversely affected or threatened by climate change, advise Government regarding appropriate legislative, policy and implementation measures and actions relating to disaster preparedness, capacity building, institutional strengthening and awareness raising in relevant sectors affected by climate change, advise the Government regarding implementation of international conventions, design, establish and maintain a national registry and database on greenhouse gas emissions etc.

178. Pakistan Climate Change Fund shall be established under the Act. The monies received in the form of donations, endowments, grants and gifts, and raised by the authority for the execution of programs and projects of the authority shall be paid into the fund. This fund shall be managed by the authority.

Punjab Environmental Protection Act 2012

179. Punjab Environmental Protection Act 2012 addresses prohibition of discharges and emissions of wastewater and air emissions, noise control, requirements of initial environmental examination and environmental impact assessment for the newly established development projects, and hazardous waste and substances.

180. Under this act, no person is allowed to discharge or emit any effluent or waste or air pollutant or noise in an amount, concentration or level which is more than the Punjab Environmental Quality Standards. No person shall be allowed to import hazardous waste into the Punjab, generate, collect, consign, transport, treat, dispose of, store, handle or import any hazardous substance except he/she gets license from the authority.

181. Section 12 is for the environmental impact assessment of the new projects which

states that no project construction or production will be allowed to commence without conducting and submitting Initial Environmental Examination (IEE) or Environmental Impact Assessment (EIA) study to the provincial agency and getting no objection certificate.

182. Under Section 16, Provincial Agency can issue Environmental Protection Order where agency is satisfied that the discharge or emission of any effluent, waste, air pollutant or noise, or the disposal of waste, or handling of hazardous substance, or any other act or omission is likely to occur, or is occurring, or has occurred, in violation of any provision of this act, rules or regulations or of the condition of license, or is likely to cause, or is causing, or has caused an adverse environmental effect. The Provincial Agency may, after giving the person responsible for such discharge, emission, disposal, handling, act or omission an opportunity of being heard, by order, direct such person to such measures as the Provincial Agency may consider necessary with such a period as may be specified in the order.

Pakistan Penal Code

183. The Pakistan Penal Code discusses offences where public or private properties and/or human lives are affected due to intentional or accidental misconduct of an individual or body of people. The Code defines the penalties for violations concerning pollution of air, water bodies and land.

Hazardous Substance Rules 2003

184. Under the Hazardous Substance Rules 2003, made under PEPA 1997, a license will be required for the import and transportation of hazardous substances from the Federal or Provincial agency. The application for the grant of license for the industrial activity involving generation, collection, consignment, transport, treatment, disposal, storage, handling or import of hazardous substances, will also be accompanied with EIA report and safety plan. The validity of the license will be for three years from the date of issue. The licensee will notify any major accident occurring at licensed facility to provincial and federal agencies. There will be packing and labelling requirements, the safety precautions for the premises and workers which will have to be followed. The licensed facility may be inspected by the provincial or federal staff.

Punjab Hazardous Substances Rules 2018

185. These rules are made under Punjab Environmental Protection Act 2012 to manage hazardous substances in the province for their collection, generation, handling, consignment, transport, treat, dispose of, manufacturing and storage. The names of the hazardous substance and their threshold quantities are listed in Schedule 1, 2, 3, and 4 which are regulated under these rules.
186. The concerned authorities are required to inspect the subject industrial activity once a year and submit the report on the compliance of the rules by the occupiers to the

EPA annually. The occupier is required to notify the concerned authority in case of the major accident within the premises or outside the premises of the licensee within 48 hours during manufacturing, loading or unloading, supply, storage, marketing, and transportation of hazardous substances. The notified officer will take appropriate actions to prevent accidents from recurring.

187. The occupier of the subject industrial activity will require to acquire license from concerned authority i.e. EPA to operate the facility after submitting Hazardous Substance Report. The occupier is also required to submit a safety report to the concerned authority 90 days before commencing industrial activity. This safety report will be updated annually. The occupier is also required to prepare and keep up to date an onsite emergency plan. It shall be the duty of Rescue 1122 to prepare and keep up to date an adequate off-site emergency plan with details that how emergencies relating to a possible major accident on that site will be dealt with.

188. The rules provide guidelines to the occupier regarding packaging and labelling of the hazardous material, conditions to be maintained for the premises where hazardous substance is generated, collected, consigned, treated, disposed of, stored or handled, general and specific safety precautions to be taken at the facility and for the workers, and requirements of the safety plan and waste management plan.

Punjab Occupational Safety and Health 2019

189. Under the act, the employer would be responsible to ensure the health and safety of the workers at workplaces (construction sites are also considered as workplaces under the act). This act mentions health and safety requirements which need to be ensured to be complied with by the employer/site in-charge and the workers. The Chief Inspector and the inspectors appointed under the act shall be responsible for enforcing health and safety requirements prescribed by the act. Penalties shall be imposed in case of noncompliance of the requirements.

Social Policy and Legal Framework

The Constitution of Pakistan

Article 38 (c) and the Employees' Old-Age Benefits Act, 1976

190. Government of Pakistan had initially promulgated the Employees' Old-age Pensions Ordinance in 1972. Subsequently, in 1976, this Ordinance was substituted with an act of Parliament, namely Employees' Old-Age Benefits Act, 1976. This social insurance system was started to achieve the objective of article 38 (c) of the Constitution, which is stated as under:

191. The State shall: *Provide for all persons employed in the service of Pakistan or otherwise, social security by compulsory social insurance or other means;*

192. The Government has created special social security systems for public-sector employees (where Civil Pension Rules are applicable) that include members of the armed forces, police officers, employees of statutory bodies, local authorities, and railways.

193. Other than these, the Government is also managing social assistance programs for the welfare of destitute and needy citizens. Under the Zakat and Ushr Ordinance, 1980, benefits are provided to the poor Muslim citizens of Pakistan while under the Pakistan Baitul Mal Act, 1992 and Benazir Income Support Program Ordinance, 2010 (Program was started in 2008), assistance is being provided to all the citizens of Pakistan irrespective of their religion or location.

194. The Employees' Old-Age Benefits Act is applicable to all firms (industrial or commercial, including banks) where 5 or more workers, whether contractual or regular, are employed or were employed during past 12 months. The laws remain applicable even if the number of persons employed is subsequently reduced to less than five.

195. As for the benefits, it provides the following four types of benefits to insured persons or their survivors:

- Old-Age Pension (or Reduced Pension)
- Survivors' Pension
- Invalidity Pension
- Old-Age Grant (if an employee is not eligible for pension)

196. The Act specifically provides for Survivors' Pension (which is equal to minimum pension) after the death of an insured person in insurable employment for a period of at least three years. If the deceased had become eligible for old age or invalidity pension, the spouse will receive that pension of deceased for lifetime. In case of death of spouse who was receiving the insured person's pension, this pension will be divided among children in equal share, until they attain the age of 18 years (or in the case of girls, before she marries or attains the age of 18 years, whichever comes earlier; however, since July 2010, daughters can receive this pension until their marriage). In case an insured person has not left a spouse or children behind, his parents would be entitled to the survivors' pension for a period of five years.

197. Old-Age Grant is paid to those employees who are not eligible for old-age pension, as they don't meet the requirement of 15 years of contributions. However, if these employees have completed at least 2 years of insurable employment, they are entitled to a lump sum payment of one month of earnings, for each year of insured employment.

Women Rights

198. The constitution of Pakistan provides supportive platform to achieve women rights and women empowerment goals as it does not discriminate between men and women. The Constitution of Pakistan has significant human rights content, while

human rights concepts can be found from the preamble onwards; there is also a separate chapter on fundamental rights of citizens.

199. A brief overview of relevant constitutional provisions is given below:

- Article 3 calls upon the State to eliminate all forms of exploitation
- Article 4 provides for the right of individuals to enjoy the protection of the law and to be treated in accordance with the law. This applies to the citizens as well as “to every other person for the time being within Pakistan”. This article also clearly states that certain rights cannot be suspended. Under 8 any existing law or practice, inconsistent with or in derogation of the fundamental rights, shall be void. It further prohibits the State from enacting any law or policy, in conflict with Fundamental Rights except “any law relating to members of the Armed Force, or of the police ... charged with maintenance of public order ... for the purpose of ensuring the proper discharge of their duties ...”.
- Article 25 ensures equality before the law and equal protection of the law and states that there shall be no discrimination based on sex alone.
- Articles 26 and 27 provide for equal access to public places and equality of employment in the public and private sectors.
- Articles 11 and 37 (g) prohibit trafficking in human beings as well as prostitution.
- Article 32 makes special provisions for the representation of women in the Local Government.
- Article 34 directs the State to take appropriate measures to enable women to participate in all spheres of national life and community activities. In addition, Articles 25(3) & 26(2) allow the state to make special provisions for the protection of women and children.
- Article 35 asks the State to protect the marriage, the family, the mother and the child.
- Article 37 (e) directs the State to make provisions for securing just and humane conditions of work and ensuring that children and women are not employed in vocations unsuited to their age or sex, and for ensuring maternity benefits for women in employment. (The wording of the provision is archaic compared to present thinking on women’s rights, but the maternity benefit mentioned in the provision is available to and exercised by women as a matter of right. The issue of women not being employed in “vocations unsuited to their age or sex” was inserted as a guarantee against exploitation and was appropriate for the time and context in which the Constitution was drafted. However, it has not been used to hinder the entry of women in nontraditional areas/ fields of work as is evidenced by the increasing numbers of women entering the uniformed services – the police, army, air force etc
- Articles 51 and 106 provide for the reservation of seats for women in the legislatures⁴.

Labor Rights

200. The constitution of Pakistan contains a range of provisions regarding labor rights found in Part II: Fundamental Rights and Principles of Policy:

⁴ http://wdd.punjab.gov.pk/women_rights

- Article 11 of the constitution prohibits all forms of slavery, forced labor, and child labor.
- Article 17 provides for a fundamental right to exercise the freedom of association and the right to form unions.
- Article 18 prescribes the right of its citizens to enter into any lawful profession or occupation and to conduct any lawful trade or business.
- Article 25 lays down the right to equality before the law and prohibition of discrimination on the grounds of sex alone.
- Article 37(e) makes provision for securing just and humane conditions of work, ensuring that children and women are not employed in vocations unsuited to their age or sex, and for maternity benefits for women in employment.

201. Pakistan's labor laws trace their origins to legislation inherited at the time of partition. The laws have evolved in response to socioeconomic conditions, shifts in governance, state of industrial development, population and labor force expansion, growth of trade unions, level of literacy, and the government's commitment to development and social welfare.

202. Under the constitution, labor is regarded as a 'concurrent subject,' which means that it is the responsibility of both the federal and provincial governments. However, for the sake of uniformity, laws are enacted by the federal government, stipulating that provincial governments may make rules and regulations of their own according to the conditions prevailing in or for the specific requirements of the provinces.

203. The labor laws are a comprehensive set of laws in Pakistan dealing with the following aspects: contract of employment; termination of contract; working time and rest time; working hours, paid leave, maternity leave and maternity protection, and other leave entitlements; minimum age and protection of young workers; equality; pay issues; workers' representation in the enterprise; trade union and employers association regulation; and other laws.

Land Acquisition Act, 1894

204. LAA came into force on March 1, 1894. The act aimed to amend the then existing law for the acquisition of land for public purposes and for companies. LAA 1894, as amended from time to time, is the primary law for the purpose of land acquisition in the Province of Khyber Pakhtunkhwa. The Act provides for the acquisition of land for public purposes and for companies.

205. While the act is federal legislation, according to Article 142(c) of the constitution, a Provincial Assembly is competent to amend it. The act (as amended) is applicable to the Province of KP by virtue of West Pakistan (Amendment) Ordinance, 1969. This ordinance continues to be in force in the four provinces in view of the Adaptation of Laws Order, 1975.

206. The act is broadly grouped into 8 parts comprising 55 sections. The relevant sections are shown in Figure 1 and a brief overview is presented below:

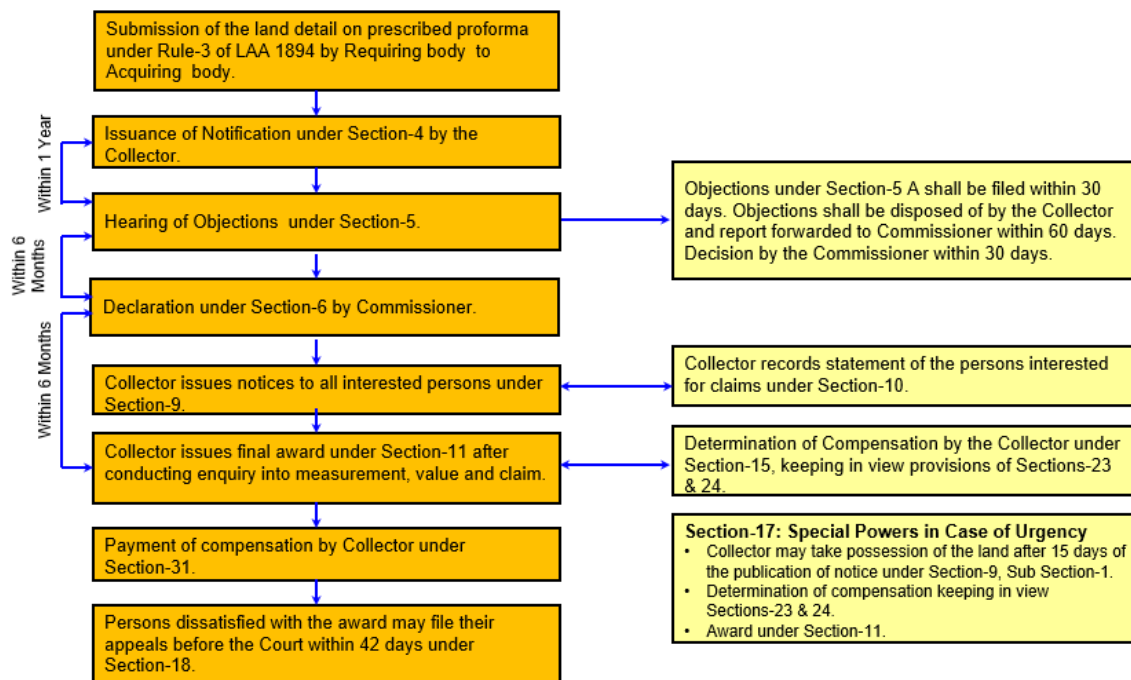


Figure 1. Land Acquisition Process under Land Acquisition Act, 1849

207. In accordance with the act, the legal process is initiated by an application from the government agency that requires the land as it is the requiring body. As land is a provincial issue according to the constitution, the next step is for the provincial government to deem it necessary to acquire the land, after which it takes the actions listed below and in Figure 1.

208. Under Section 4, it causes the publication of preliminary notification that the land is needed for a public purpose. This permits entry, survey, and investigation of the land in question by an authorized government servant so that the authorities can determine whether the land intended to be acquired is suitable for the purpose for which it is needed. Compensation shall be paid for any damage caused by such entry. The process of acquisition must start with a notification under Section 4. It is a condition precedent to the exercise of any further powers under the act.

209. Under Section 5, a formal notification is issued that the particular land is needed for a public purpose. This notification is published in the official gazette and the collector is required to issue a public notice of the substance of the notification. Issuance of Section 5 has to take place not later than one year after notification of Section 4.

210. Any person interested in any land which has been notified under Section 5 may, within thirty days after the issuance of the notification, object to the acquisition of the land under Section 5A. The collector shall hear the objection, make necessary enquiries, and submit a report within 90 days to the appropriate government authority. This authority must announce its decision, which shall be final, within 90 days, otherwise the objection shall be deemed to have been admitted, and the acquisition proceedings will come to an end.

211. When the provincial government is satisfied, after considering the report, if any, made under Section 5A that any land is required for public purpose, a declaration to that effect shall be made by an authorized officer of the provincial government under Section 6. This should follow within six months of the publication of the Section 5 notification.
212. After the declaration under Section 6, the commissioner of land acquisition collection shall “direct the collector to take order for the acquisition of land” under Section 7. The collector then notifies under Section 8 to cause the land to be marked out, measured and planned (if this was not done after Section 4).
213. Under Section 9, the collector gives notice to all interested people that the government intends to take possession of the land and if they have any claims for compensation that they be made to him at an appointed time. Section 10 delegates powers to the collector to record statements of persons possessing any interest in the land or any part thereof as co-proprietor, sub-proprietor, mortgage and tenant, or otherwise. Section 11 enables the collector to make enquiry into “measurements, value and claim and issue the final award.” Included in the award is the land’s true area, the collector’s view of what compensation is warranted, and the apportionment of that compensation to all interested people.
214. Though this section is the one that contains the final award, there are two other sections, that is, Section 23 and Section 24, which actually take place before Section 11. This is because these two sections pertain to compensation and the criteria to be followed (Section 23) or not to be followed (Section 24) in arriving at appropriate compensation.
215. Under Section 23 are included such items as the market value of the land at the time of notification of Section 6, and various damages that have been sustained at the time possession was taken.
216. Matters not considered in awarding compensation, that is, Section 24, include such items as the degree of urgency, which led to the acquisition, any disinclination of the person interested in the land to part with it, any expected increase in value of the land from its future use, and so on.
217. When the collector has made an award under Section 11, he will then take possession under Section 16 and the land shall thereupon vest absolutely with the government, free from all encumbrances.
218. Section 18 pertains to persons still dissatisfied with the award, who may request the collector to refer the case to the court for determination and decision. This does not affect the taking possession of the land.
219. In cases of emergency, where the Board of Revenue considers it expedient to take possession of any land at any time before an award under Section 11 has been made, it shall notify this act in writing to the collector intimating in addition the date by which

the land is required. Under Section 17, the collector can, after causing a notice to this effect to be served on the person or persons interested in the land, take possession of the land subject to the liability to pay any amount, which may be incurred on account of acquisition. This specific section will not be applicable in the proposed Program, as the acquisition of land is being done through normal procedures.

Punjab Land Acquisition Rules, 1983

220. The Punjab Land Acquisition Rules, 1983, describe the land acquisition procedure for public purposes or for a company. The Punjab Land Acquisition Rules comprises 16 rules pertaining to area notification and surveys, inquiry and award, compensation and apportionment, awards and dispute resolution, and exceptions. Important rules for acquiring land are described below:
221. Rule 4. This rule states that the acquiring agency shall submit an application under Rule 3 to the collector of the district concerned on a prescribed form for the acquisition of land. Then, the collector of the district shall examine the feasibility of the land acquisition, taking into consideration the genuineness of the public purpose involved.
222. Rule 5. According to Rule 5, the collector of the district, after the examination of feasibility, issues a notification under Section 4 stating clearly the details of the area to be acquired.
223. Rule 6. In accordance with Rule 6, after the issuance of the notification under Section 4, the collector shall take immediate steps to have the land surveyed and submit a report to the commissioner not later than sixty days from the date of the publication.
224. Rule 7. The rule describes the procedure for the issuance of notification under Section 5 by the commissioner, where the land is to be acquired for a public purpose, and the reporting procedure to the BoR in case any delay occurs in the issuance of notification.
225. Rule 8. This rule states that where the land is to be acquired for a company, the commissioner, on receipt of the survey report of the collector under Rule 4, forwards the same to the BoR for issuance of notification under Section 5.
226. Rule 9. According to Rule 9, objections received under Section 5A by the collector shall be disposed of with least possible delay and along with the recommendations shall be forwarded to the commissioner within a prescribed time limit for decision.
227. Rules 10 (1), (2), and (3). This rule states that the commissioner of the division, when issuing a notification under Sections 5 or 17, shall ensure required details for the acquisition of land.
228. The rules also describe the procedure for land acquisition where a company makes an urgent request for invoking the provision of Section 17 of the act. It is also mentioned that no land lying near a town, meant for fodder cultivation or for orchards or otherwise cultivable shall be notified for acquisition for the establishment of any industry except if it is declared by the agriculture, health, and industry departments that no risk would be involved due to such land acquisition.

- 229. Rule 11. The rule states that the collector, while forwarding draft notifications under Sections 5 and 17 to the commissioner, shall ensure that the certificates, documents, and information mentioned in the act are appended.
- 230. Rule 12. The rule states that the estimated cost of the land worked out under Rule 10(2) is approved by the collector/commissioner/BoR.
- 231. Rule 13. According to Rule 13, the collector or the commissioner may not deliver possession of land sought to be acquired by the department or agency concerned unless sufficient funds for the payment of compensation are placed at the disposal of the collector in advance.
- 232. Rule 14. The rule states that if any land acquired by the department or local authority for public purpose is not used as proposed, that land should be handed over to the collector for disposal by the order of the government.
- 233. Rule 15. Where land has been acquired by a company it shall not be sold or disposed of before the approval of the government.
- 234. Rule 16. This rule deals with penalties and states that where land is used for a purpose other than that for which it was acquired by the acquiring agency, it shall be repossessed immediately, and the acquiring agency shall be liable to penalty.

Employment of Child Act, 1991

- 235. Article 11(3) of the constitution of Pakistan prohibits employment of children below the age of 14 years in any factory, mine, or any other hazardous employment. In accordance with this article, the ECA 1991 disallows such child labor in the country. The ECA defines a child to mean a person who has not completed his/her fourteenth year of age. The ECA states that no child shall be employed or permitted to work in any of the occupations set forth in the ECA (such as transport sector, railways, construction, and ports) or in any workshop wherein any of the processes defined in the act is carried out.
- 236. Presently the government of Punjab has drafted a new act titled 'Punjab Restriction of Employment of Children Act 2015' dealing with child labor in the commercial and industrial sectors of Punjab. The act aims to banning children up to 14 years from all types of employment. The approval of the act is ongoing and expected to be approved by the Punjab Assembly in a short span of time.

Punjab Prohibition of Child Labor at Brick Kilns Act, 2016

- 237. Adding onto the Employment of Child Act, 1991, Under Section 13, any Parent or Guardian who employs or permits a child (Person under the age of 14 years) to work at a brick kiln shall be liable to 6 months' imprisonment and a fine of 500,000 Rupees. This act directly addresses the widespread use of child labor in Brick Kilns and ensures improvement in social sustainability.

Punjab Restriction on Employment of Children Act, 2016

238. Under section 11(a) of the Act, an occupier who employs or permits a child (person under the age of 15 years) to work in an establishment shall be liable to punishment with imprisonment for a term which may extend to 6 months, but which shall not be less than 7 days, and a mandatory fine between 10,000 and 50,000 rupees.

The Punjab Occupational Safety and Health Act, 2019

239. The Occupational Safety and Health Act ensures a safe work environment for the citizens of Punjab, whether its farming, industrial work, office work or working in Brick kilns. The Act ensures no Psychological or Physiological harm is brought onto employees.

240. The duties of the Employer under the Act are as follows:

- take all possible and reasonably practicable measures to ensure safety and health of the employees at the workplace.
- ensure systematic and effective identification of the existing and new hazards at the workplace on a regular basis.
- arrange for training of safety and health representatives every year to attend health and safety training as approved by the Government and shall bear all expenses thereof.
- get, at his own expense, the employees vaccinated and inoculated against occupational related deceases at such intervals as may be prescribed.
- ensure provision and application of processes, systems of work and tasks to be safe and without risks of harm to health.
- ensure provision and maintenance of tools, machinery, equipment and appliances which are safe and without risks of harm.
- decide to control and prevent physical, chemical, biological, radiological, ergonomics and psycho-social or any other hazards to the employees and other persons at the workplace.
- provide such information, instructions, training and supervision, as is necessary or required under the Act and the rules;
- keep the workplace in safe, clean orderly and harm-free condition, and provide and maintain safe means of access to and egress from the workplace;
- inform the employees in an understandable manner and in accessible written form, before any work commences, the hazards associated with their work, risks involved and the preventative and protective measures that need to be taken;
- provide where necessary to the employees, at his own expense, if hazards cannot otherwise be eliminated or controlled, adequate protective clothing and protective equipment as may be approved by the Government, to prevent every risks of harm and of adverse effects on health;
- maintain in the general register particulars of all accidents including dangerous occurrences, commuting accidents and suspected cases of occupational diseases at the workplace and submit the extracts thereof to the Government;
- provide where necessary measures including adequate first aid arrangements to deal with emergencies, dangerous occurrences, accidents and industrial disasters; and

- take all practical measures for the prevention of fires or explosions and for the provision of safety measures in the event of fire or explosions as well.

241. Similarly, the duties of the Employee under this Act are as follows:

- reasonable care for his own safety and that of other persons who may be affected by his acts or omissions at the workplace without willfully doing anything to endanger himself or other persons.
- proper use of any protective clothing and protective equipment.
- no willful interference with or misuse of any means, appliances, equipment or any other thing;
- his/her escape from the place of danger and report forthwith to the immediate supervisor of any situation.
- reporting the matter to the Council in the prescribed manner: Provided that where the threat cannot be controlled in view of the Council, it may direct the employees who could be affected, to cease work until remedial action is taken to the satisfaction of the Council: Provided further that where work ceases, the employer may allow the employees to undertake temporary alternative work on the same terms and conditions and continuance of his employment shall not be affected until the dispute is resolved by the Council.

Pakistan Labor Policy, 2010

242. Pakistan's Labor Policy aims at attaining its objectives in a manner best suited to the resources of the country and the present state of economy. There is an urgent need to revitalize the economy to increase the level of productivity, promotion of investment, and maximization of employment. There is an equally genuine requirement to create, among workers and employers, a better awareness of their obligations to the national objectives stated above. At the same time, the government recognizes that workers and employers must enjoy reasonable benefits. With these priorities in view, the government considers that a balanced labor policy should be based on the following objectives:

- Workers' rights to form unions and unions should be protected, and an institutional framework be made available to foster close cooperation between workers and employers at the establishment level.
- Equitable adjustment of rights between workers and employers should be ensured in an atmosphere of harmony, mutually beneficial to workers and the management.
- Consultations between workers and employers on matters of interest to the establishment and welfare of workers should be made more effective.
- Adequate security of jobs should be available to workers and there should be expeditious redressal of their grievances.
- Conditions should be created such that workers and employers are committed to enhancing productivity.
- Promotion to higher jobs should be ensured at all levels based on suitability and merit and for this purpose, arrangements should be made for in-service training facilities.

- Facilities for proper matching of job opportunities and job seekers should be strengthened and standard procedures streamlined.
- Social insurance schemes should be further strengthened.
- Just and humane conditions of work should be guaranteed to all workers.
- Forced labor in all its forms is to be eliminated.
- Provisions relating to the employment of children are to be strictly adhered to and enforced.

243. To address these concerns, the Labor Policy is divided into four parts: the Legal Framework; Advocacy: Rights of Workers and Employers; Skill Development and Employment; and Manpower Export.

Compliance Requirements under Generalized Systems of Preferences (GSP+)

244. Pakistan acquired the status of EU's GSP+, which allows increased access of some categories of Pakistani products by allowing duty-free exports to EU markets from January 1, 2014. GSP+ requires ratification and compliance of twenty-seven International Conventions including eight core labor standards of the ILO. These eight core labor-related conventions include the following:

- Convention concerning Forced or Compulsory Labor, No.29 (1930)
- Convention concerning the Abolition of Forced Labor, No.105 (1957)
- Convention concerning the Application of the Freedom of Association and Right to Organize, No.87
- Convention concerning the Application of the Principles of the Right to Organize and to Bargain Collectively, No.98 (1949)
- Convention concerning Equal Remuneration of Men and Women Workers for Work of Equal Value, No.100 (1951)
- Convention regarding discrimination in Respect of Employment and Occupation, No.111 (1958)
- Convention concerning Minimum Age for Admission to Employment, No.138 (1973)
- Convention concerning the Prohibition and Immediate Action for the Elimination for the Worst Forms of Child Labor, No.182 (1999)

245. The following labor laws (out of 24) relate directly to the implementation of the eight Labor Conventions:

- Bonded Labor System (Abolition) Act, 1992
- Employment of Child Act, 1991
- Minimum Wages Ordinance, 1961
- Industrial Relations Act, 2010
- West Pakistan Minimum Wages for Unskilled Workers' Ordinance, 1969

246. Two of the important labor-related aspects, which are required to be strengthened include the equal remuneration of men and women workers for work of equal value and removal of discrimination with regard to employment and occupation. Special

institutional arrangements have been proposed in the above five laws for ensuring implementation, as follows:

- Setting up of Vigilance Committees in the Bonded Labor (Abolition) Act
- Setting up of a cadre of inspectors for enforcement of the ECA Act.
- Establishment of Minimum Wage Boards in the Minimum Wages Ordinance
- Appointment of registrars of trade unions and establishment of labor courts under the Industrial Relations Act.

The Protection Against Harassment of Women at the Workplace Act, 2010

247. The act protects women against sexual harassment at the workplace. The act is composed of 13 sections elaborating definitions, composition of the inquiry committee, procedure for holding inquiry, penalties (minor and major), powers of inquiry committee, role of the employer, the process for appeal against minor and major penalties, ombudsmen and powers of the ombudsmen.

The Punjab Protection against Harassment of Women at the Workplace (Amendment) Act, 2012

248. The Punjab Protection against Harassment of Women at the Workplace Act (the Act) was passed in 2012 to prevent harassment of women at the workplace and encourage reporting of cases of harassment, through the Ombudsperson's office and Inquiry Committees.

- Under the Act, all Provincial and District Government bodies and authorities, including educational institutes and medical facilities, must constitute Inquiry Committees for harassment at the workplace and display the Code of Conduct for Protection of Women from Harassment at the Workplace in their office premises.
- A Complainant has the option to submit a complaint to the Inquiry Committee, or the Ombudsperson.
- The Inquiry Committee will inform the accused person that he has been accused of harassment within 3 days of receiving a complaint. If the accused fails to submit his defence within 7 days, the Committee will decide the matter without the accused person.
- Under the Act, the Ombudsperson has the power of a Civil Court, and can summon the accused, examine him on oath, receive evidence on Affidavits, and enter office premises for the purpose of inspection or investigation.
- The Inquiry Committee or the Ombudsperson can decide the case and penalize the accused for harassment, as outlined in section 4 of the Act.

Punjab Women Development Policy 2018

249. The Punjab Women Development Policy seeks to ensure that benefits of growth, industrialization, and development and livelihood improvements are targeted towards all sections of society particularly for vulnerable groups such as women and girls.

250. The Policy Objectives include to ensure gender balance and equality of access and opportunities for women across all economic empowerment initiatives of the provincial government as well as in informal sectors and to reduce labour market distortions and discriminations against women to enhance Female Labour Force Participation.

Punjab Empowerment of Persons with Disabilities Act 2022

251. The Act guarantees the rights of persons with disabilities to full equality and non-discrimination in the enjoyment of all civil, political, economic, social and cultural rights and to the full and effective participation and inclusion in society. The Act provides that persons with disabilities have access to public transport, public places, hospitals and streets.

Punjab Social Protection Policy 2022

252. The Social Protection Policy 2022 of Punjab aims to strengthen social safety nets and improve the wellbeing of vulnerable populations in the province. The policy focuses on Targeted Cash Transfers, Health and Education Support, Food Security and Nutrition, Livelihoods and Employment, Social Insurance and Pensions, Climate Change and Environmental Adaptation, Social Protection Delivery Systems, and Gender Equality and Inclusivity.
253. Overall, the Social Protection Policy 2022 in Punjab is designed to provide a comprehensive, multi-faceted approach to supporting the welfare of its citizens, especially those facing economic hardship and social exclusion.

Punjab Agriculture Policy 2018

254. Represents a comprehensive approach to transforming the province's agricultural landscape. It seeks to boost productivity, ensure sustainable practices, and improve the livelihood of farmers, while also tackling emerging challenges like water scarcity and climate change. The policy aims to create a more diversified and resilient agricultural sector, focusing not only on crop production but also on livestock, agri-processing, and market access.
255. Punjab Agriculture Policy 2018, a significant focus is placed on small farmers, recognizing them as the backbone of the agricultural sector in the province. The policy aims to address the challenges faced by small farmers, who often struggle with low productivity, limited access to resources, and vulnerability to climate change impacts. The policy includes several key provisions to support small farmers and improve their livelihoods including – access to credit a financial support; agricultural inputs and subsidies; training and extension services; improved market access; land ownership and tenure security; sustainable farming practices; gender inclusion; farming enterprises; and technology and innovation.

Pakistan Antiquities Act 1975 and Punjab Antiquities Amendment Act 2012

256. The current Antiquities Act 1975 (amended in 1990) redefined as ‘ancient’ any object that is at least 75 years old. It requires that all accidental discoveries are reported to the federal Department of Archeology. It also makes the federal government the owner of all buried antiquities discovered from any site, whether protected or otherwise. It bans all new construction within a distance of 200 feet from protected antiquities. The cultural heritage laws of Pakistan are uniformly applicable to all categories of sites regardless of their state of preservation and classification as monuments of national or world heritage. The Antiquities Act guarantees that no changes or repairs can be made to a protected monument even if it is owned privately without approval of the official agencies concerned with it. The Punjab Antiquities Amendment Act 2012 adopts the Act of 1975 with a few minor changes.

Guidelines For Public Consultation, 1997

257. The Guidelines for Public Consultation are part of the Environment Protection Act, 1997. These guidelines address possible approaches to public consultation and techniques for designing an effective program of consultation that reaches all major stakeholders and ensures the incorporation of their concerns in any impact assessment study. The guidelines cover consultation, involvement and participation of stakeholders; effective public consultation (planning, stages of an EIA where consultation is appropriate); and facilitation of involvement (including the poor, women, and NGOs). The guidelines are intended to ensure that the EIA process incorporated community participation and consider environmental issues.

The Punjab Transparency and Right to Information Act 2013

258. The Punjab Transparency and Right to Information Act was passed by the Provincial legislative to ensure citizens have access to public information to make the government accountable to citizens as well as enforcing the fundamental right of access to all information in regard to all matters of public importance. This Act is an improvement of the Punjab Freedom of Information Act 2010, which lacked transparency between the people and the State and the response to grievances was not time bound.

259. The Punjab assembly was the second provincial body, after Khyber Pakhtunkhwa, to pass this law, under the 18th amendment of the constitution, which states that is it mandatory for all provinces to pass and implement the RTI Act. This Act itself is a cornerstone to the democratic nature of our government, giving the power to the people to hold their government officials accountable.

260. The law itself puts responsibility onto Public Information Officers (PIOs) to serve as the “middlemen” between Government officials and citizens, to make sure any requests made by citizens to access public information is granted with utmost efficiency and transparency.

261. The public representatives and the government have to play a very important role but now it's for the citizens to come forward, to understand and own the Law and actively participate in its implementation to enjoy this fruit of democracy. Awareness about their rights under this Law will change the tide in favor of the common man, ushering in an era of transparency, freedom, peace and prosperity in Punjab.
262. The Punjab Transparency and Right to Information Act is overseen by the Punjab Information Commission, which consists of the Chief Commissioner who acts as the executive authority of the commission and two information commissioners who come from diverse backgrounds such as Law, media, Civil society of government service. Their key responsibilities as an independent statutory body are to ensure that all public bodies comply to the act, conducting certain inquiries or complaints filed by the citizens, issuing directives to ensure transparency, providing training to Public Information Officers (POIs) and running several awareness campaigns about the citizens right to information. Overall, they play a pivotal role in safeguarding citizens' right to public information in Punjab.

Right to Public Services Act 2019:

263. Right to Public Services commission (RTPS), a statutory body, was established under an Act of Provincial Assembly Punjab Right to Public Services Act in 2019. This legal framework ensures a time bound delivery of services to the citizens of Punjab in an efficient and transparent manner. Through this Act, bureaucratic delays are reduced, and accountability of public servants increases.
264. The general objective of this act is to guarantee timely delivery of specified public services to citizens and to establish a grievance redress mechanism for addressing service-related complaints. Each service is given a specific time within which it must be delivered and if public officials fail to meet said timelines, they tend to face penalties.
265. The commission is headed by the Chief commissioner who is often assisted by two commissioners. For departmental level oversight there are Designated officers that are often appointed within certain government departments or agencies, and lastly the district level oversight is done by District monitoring officers (DMOs) that are responsible for implementing the law on a district level. Some of the public services' that come under the act are issuance of birth, death and domicile certificates, property registration and land records, and health, education or municipal services.

Punjab Local Government Act 2022

266. The Punjab Local Government Act 2022 Section 132 and 134 read with 5th Schedule of PLGA 2013 deals with the municipal offences including encroachment and the cognizance of these offences' rests with Enforcement Inspector and Enforcement officers appointed by the Authority. The Act states that no person shall make an encroachment moveable or immovable on an open space or land vested or managed, maintained or controlled by a Local Government, or on, over or under a street.

Pakistan Citizen's Portal

267. The **Pakistan Citizen's Portal** (<https://web.citizenportal.gov.pk/>) is a government-operated mobile application designed to enhance public service delivery and facilitate efficient interaction between citizens and various government departments. Launched in **2018** under the vision of the **Prime Minister's Performance Delivery Unit (PMDU)**, the portal is a part of Pakistan's e-governance initiative to ensure transparency, accountability, and citizen participation in governance.

268. Through this application, citizens can lodge complaints about various issues, ranging from municipal services to governance inefficiencies, directly with the relevant authority. Categories include Environment and Climate change, Industries and Commerce, Transport and Vehicle Management, and Agriculture.

Government of Punjab Online Portal:

269. The Government of Punjab has a state-of-the-art online portal (www.punjab.gov.pk) with information on service delivery platforms, digital services, Annual Development Plan, tenders published by GoPb, administrative departments, investment schemes, public benefit schemes, laws and citizen rights amongst other information. The portal also lists helplines for various administrative departments. The Punjab Portal has a feature for citizen relationship management called the "Awaz e Khalq" that allows citizens to register specific complaints that may be associated with several departments such as DC office, Punjab police, Labor and Human resource, Environment protection and Healthcare to name a few. Figure 1 shows a screenshot of the Punjab Portal Homepage (www.punjab.gov.pk)

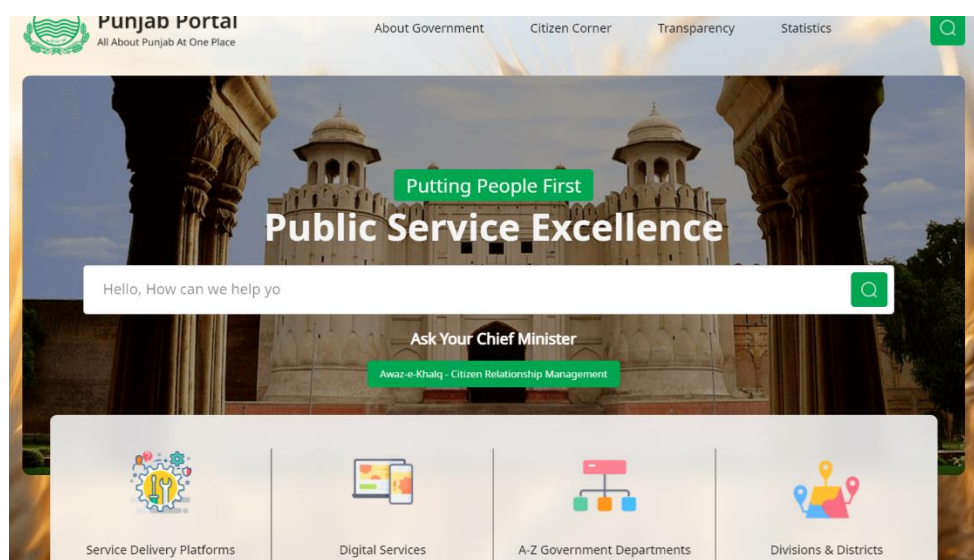


Figure 2: Punjab Portal Homepage

270. In order to register a complaint, its mandatory to set up an account using your CNIC, district, home address and your personal phone number and email ID. Once an account is set up, a complaint form can be filled either in English or Urdu.

271. The “Citizen Corner” feature on the website (as shown in figure 3 <https://punjab.gov.pk/helplines>) allows easy access to all forms of centers, platforms, mobile apps and even helplines that can cater to several citizen complaints or queries in an efficient manner. The options that are shown take into consideration people belonging to all socio-economic classes. The citizens who do not have access to android phones can easily lodge their complaints through several different helpline numbers listed in accordance with their departments such as the Chief Ministers complaint center number which is 0800-02345 and the Punjab Health Line 0800-99000.
272. On October 15th, 2022, the Chief Minister inaugurated the “Go Punjab” app, which is one of the many mobile applications that exist on the Punjab portal. The Go Punjab app provides citizens with easy access to a plethora of public services. A few of these services are Issuance of domicile certificates; birth, death, and marriage certificates; payment of utility bills; and health services. Another interesting feature it has is that it directly takes you to a Whatsapp chat with the Department of Transport and Mass Transit, assisting in any complaints or queries that may exist towards the department. The app also has a user-friendly interface, and it also contains a grievance redress mechanism allowing citizens to lodge complaints efficiently and with ease. The application itself is a significant step towards e-governance and improving the quality of public administration in the province. Other than the “Go Punjab” application, the portal has links to 44 different mobile applications that can be used for various departments that cater to citizen services, education, women safety, etc.
273. To ensure that there is complete transparency, the portal also gives access to the Annual budget, Punjab Laws, PC-1’s, Tenders and the Punjab Growth strategy 2023. Along with that, several statistics, be it population or economic statistics are also present on the Portal.

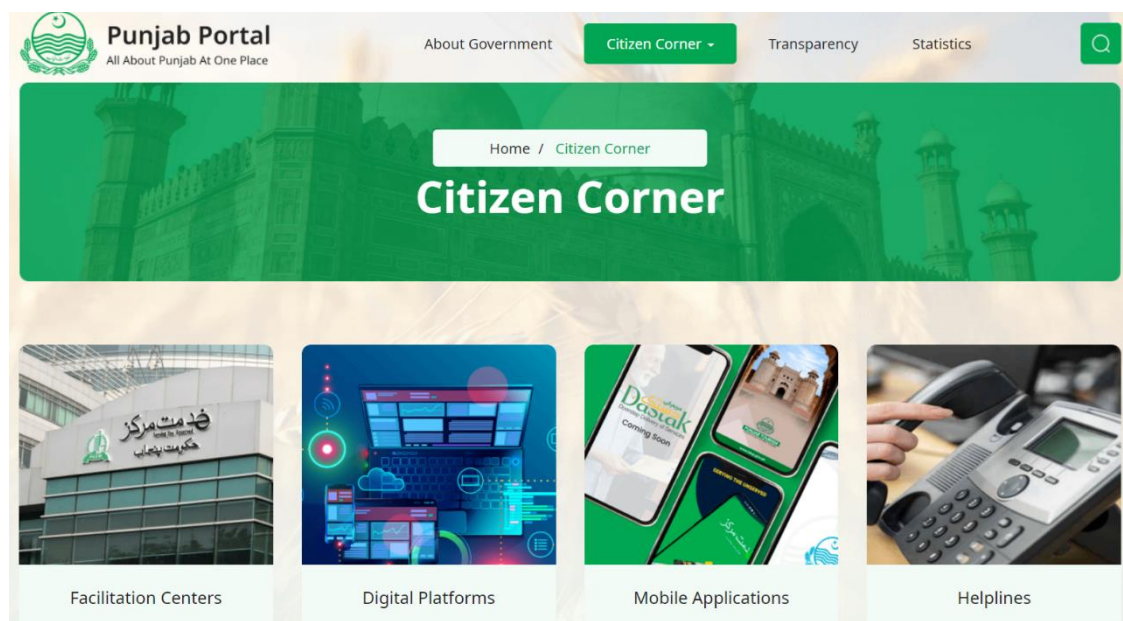


Figure 3: Citizen Corner page (<https://punjab.gov.pk/citizen-corner>)

Green Punjab App

274. The Punjab Government has launched a Green Punjab smartphone application to resolve smog-related concerns from citizens. Citizens can now file complaints against individuals who burn hospital waste and garbage, operate smoke-emitting kilns and vehicles, and factories that run without smoke control devices. Complaints submitted through the app will be swiftly reported to the appropriate authorities, and urgent action will be taken to resolve them. Complainants will also be kept up to date on the status of their complaints. Among the institutions linked to the app are PDMA, metropolitan corporation, traffic police, Parks and Horticulture Authority, and departments of agriculture, transportation.

275. The app also contains the SMOG complaint helpline, 1373 along with AQI reports generated in real time and the AQI wind Direction. Along with that it has a RASTA feature that helps in route planning and traffic advisory (<https://punjab.gov.pk/node/6322>).

Provincial Ombudsman of Punjab

276. The Ombudsman institution was first set up at the federal level in the year 1983 which later got established in Punjab first on a provincial level in the year 1997, making it a front runner at protecting citizens' rights and ensuring administrative accountability and good governance. The law essentially empowers the Ombudsman Punjab to consider complaints against any department, commission or even statutory co-operations that are established by the Provincial government, without including the High court. The Ombudsman has similar powers as are vested in a civil court under the Code of Civil procedure for summoning and enforcing the attendance of any person, compelling the production of documents, receiving evidence on affidavits, and issuing commission for examination of witnesses.

277. Any person who is aggrieved of mal-administration of any provincial government agency excluding courts and provincial assembly, can make a complaint on plain paper or online through their E-complaint portal (<https://ombudsmanpunjab.gov.pk/register-complaint-residential-pakistanis>) either in Urdu or English and can lodge the same in the Ombudsman Office either in person or through other means of communication. Figure 4 shows a detailed flow chart on how the complaint management process takes place. Every complaint must be accompanied by a copy of computerized National Identity Card (CNIC) of the complainant and supported by an affidavit to the effect that:

- The allegations are true and correct to the best of his/her knowledge and belief.
- Previously no complaint about the subject matter has been lodged.
- No suit, appeal, petition or any other judicial proceedings in connection with the subject matter of the complaint are pending before any court or judicial Tribunal.
- The complaint must bear the signature or thumb impression of the complainant and complete mailing address with Telephone or cell phone number if any. No anonymous or pseudonymous complaint is admissible.

- Alongside this, no complaint which is late than six months from the day on which the aggrieved person had the knowledge of the matter alleged in the complaint will be entertained unless special circumstances exist necessitating investigation in a complaint which is time barred.
- Findings in every complaint will be communicated to the complainant and the Agency on conclusion of the investigation and any person or agency aggrieved by an order of the Ombudsman, may make representation to the Governor within thirty days of the order.

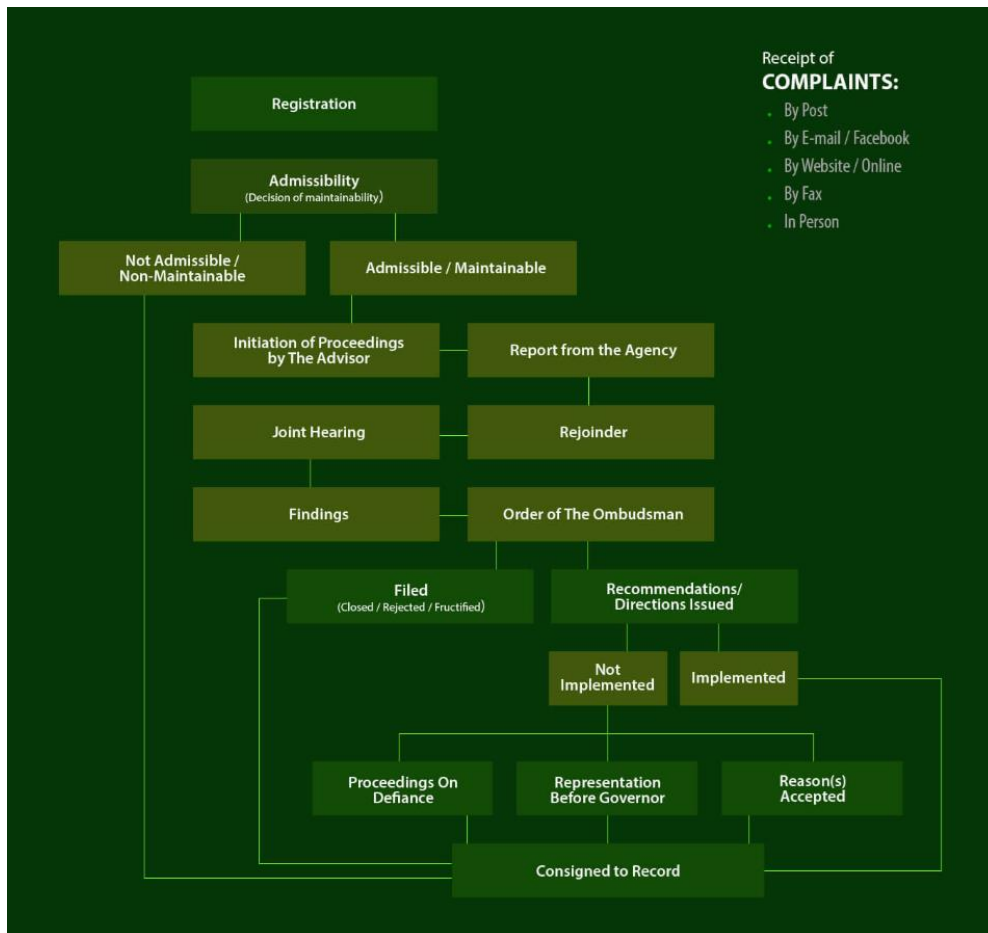


Figure 4: Complaint management process

ANNEX-2: E-WASTE MANAGEMENT IN PAKISTAN AND THE GLOBAL E-WASTE MANAGEMENT OPTIONS

E-Waste Issues in Pakistan

In Pakistan, e-waste is generated from three key sources: domestic manufacturing, domestic consumption, and import. After disposal, e-waste is collected by scrapers and vendors, who sometimes dismantle the waste in several parts which is, in turn, sold to extractors and dismantlers. Extractors and dismantlers, by using environmentally unsound technologies extract precious and valuable materials and discarded waste is then often disposed of either in dumping grounds or water bodies. Sometimes scrapers and dismantlers reassemble different parts of old equipment for resale. Laborers may not be fully aware of the potentially harmful consequences of recycling or dismantling electronics, thus exposing themselves to high toxicity. Crude recycling techniques are adopted like physical dismantling, open burning, acid bath, and use of blow torches to extract valuable metals. The informal recycling is growing day by day in Pakistan and is expanded all over the country as small and medium enterprises. Up till now, there is no formal e-waste recycling facility available in Pakistan; all the e-waste is recycled through illegal and/or informal means.⁵

Electronic and electrical waste is imported and labelled as 'second-hand equipment'. A very small amount of the imported material is reusable. After the removal of usable items, the bulk of the electrical waste is sent to recycling industry. Major recycling waste enterprises are located at Karachi. Lahore, Faisalabad, Peshawar, Gujranwala and Islamabad/Rawalpindi are also involved in the recycling and dismantling of the e-waste, but at a small scale compared to Karachi. Karachi, being a seaport, receives containers of e-waste from all around the world. After clearance from the port, this waste is sent to warehouses from where scrapers buy the items by weight. The e-waste is dismantled, burned or dumped depending on its composition. Hundreds of workers, including teenage children, earn their livelihood by dismantling and extraction of valuable items from the e-waste.⁶

The scrapers have no protective gears. They handle extremely toxic waste with their bare hands. Workers involved with the breaking of the computers are at the risk of inhalation of dust that may contain lead, barium oxide and phosphorus. Lead may cause neurotoxicity, high blood pressure, and muscle pains, and learning disabilities among children. Barium oxide can cause severe skin irritation and ingestion is harmful, and chronic exposure may lead to damage of Central Nervous System (CNS), spleen, liver, kidney or bone marrow. The cables, wires and motherboards that have Polyvinyl Chloride (PVC) and Brominated Flame Retardants (BFRs) release dioxins and furans when these are burnt. These are known carcinogens, which accumulate in the human body and may cause changes in the immune system, glucose metabolism and reproductive problems. Gold is extracted by mercury amalgamation and acid bath extraction processes which result in release of pollutants in the environment due to burning the contents and dumping of acid in the nearby drains.⁷

⁵ Iqbal, M., et al., Emerging issue of e-waste in Pakistan: A review of the status, research needs and data gaps, *Environmental Pollution* (2015), <http://dx.doi.org/10.1016/j.envpol.2015.09.002>

⁶ Ibid

⁷ Informal Electronic Waste Recycling in Pakistan:

https://www.researchgate.net/publication/308272840_Informal_Electronic_Waste_Recycling_in_Pakistan

E-Waste Management

The e-waste management in Pakistan needs backing of legislation and its enforcement. The e-waste should be included in the hazardous substance list so that its generation and processing could be regulated through license. The e-waste management guidelines are required to be developed on international best practices such as World Bank's EHS guidelines and Guidelines on Environmentally Sound Material Recovery and Recycling of End-of-Life Computing Equipment, prepared by the Partnership for Action on Computing Equipment (PACE) Working Group under Basel Convention. Training programs for the recyclers need to be developed and executed environmentally and economically sound material recovery and recycling from the e-waste. Enforcement of the guidelines at the recycling facilities, their licensing and monitoring will be a challenging task for the regulators.

Environmentally sound material recovery and recycling of e-waste requires setting up an effective recycling chain, comprising the steps of robust collection of e-waste, evaluation, testing/refurbishment/reuse if appropriate, preparing/dismantling of non-reusable parts, separation into material streams, final recovery of marketable raw materials, and disposal of non-recyclable fractions and processing residues. Some hazardous fractions will have to be sent to facilities for the destruction of the hazardous substances in order to ensure these are taken out of use.

Environmentally sound material recovery and recycling of e-waste is not simple and can cause exposures to hazardous substances if not done correctly. It should be well understood, managed and performed consistently with good practices to protect workers and communities. All steps should be taken to ensure that unsound e-waste material recovery and recycling practices are avoided.

A material recovery facility should first collect used/dead computing equipment for which these facilities are prepared, qualified and licensed to accept and process. Next, they should carefully remove and separate the most problematic constituents - those that contain hazardous substances that may contaminate other materials – such as mercury, which usually need additional processing and/or environmentally sound final disposal. Informal recovery operations, such as acid leaching, on circuit boards and other precious metal-bearing materials are inefficient, and expose workers, communities and the environment to cyanides, strong acids, toxic gasses and other hazards. Plastics may be recycled if these are separated by type, are mostly free of metals and other contaminants, and do not contain certain hazardous brominated flame retardants (BFRs). Plastics can be used in smelting operations as fuel and as reducing agents, if the smelter emissions are well controlled, especially for dioxins and furans. Residues from processing and pollution control systems that cannot be efficiently recovered are likely to contain metals and other substances of concern, which should be carefully managed, often as hazardous waste. These include bag house filters and dust, sweepings, glass fines, phosphors, plastics and slags. Because these waste residues are likely to contain metals, plastics and halogens, disposal in an incinerator that does not have efficient pollution control systems is not suitable. Similarly, because process residues may leach hazardous constituents, disposal in an uncontrolled landfill is also not suitable.⁸

⁸ Basel Convention, Partnership for Action on Computing Equipment (PACE), Revised Guideline on Environmentally Sound material Recovery and Recycling of End of Life Computing Equipment, July 26, 2013

ANNEX-3 CAPACITY ASSESSMENT FOR MANAGING ENVIRONMENTAL AND SOCIAL EFFECTS

This section assesses whether the involved institutions have the requisite capacity to implement the Program's environmental and social management systems. *Program capacity* is the "organizational capacity" the institutions authorized to undertake environmental and social management actions to achieve effectively "environmental and social objectives against the range of environmental and social impacts that may be associated with the Program." This ESSA has examined the adequacy of such capacity by considering, among other things, the following factors:

- (a) Adequacy of human resources (including in terms of training and experience), budget, and other implementation resources allocated to the institutions;
- (b) The adequacy of institutional organization and the division of labor among institutions;
- (c) Effectiveness of interagency coordination arrangements where multiple agencies or jurisdictions are involved; and
- (d) The degree to which the institutions can demonstrate prior experience in effectively managing environmental and social effects in the context of projects or programs of similar type and magnitude.

This ESSA examines and discusses only those aspects of the proposed Program's environmental and social management systems and related capacity that the ESSA Team found to be relevant, considering its identified environmental and social effects. This section provides a *summary assessment* of the Program's capacity.

The capacity of the implementing institutions was assessed through one-to-one meetings by raising specific environmental and social capacity questions. These meetings were held jointly by environmental and social team members of the ESSA Team.

This section describes the functions of various agencies involved in the implementation of the project and their capacity assessment to manage environmental and social impacts of the Program activities.

Planning and Development Board⁹

Overview: The Planning and Development Board, Government of Punjab, is the principal planning organization at the provincial level. It coordinates and monitors development programs and activities of various departments of the provincial government. The department also prepares an overall Medium Term Development Framework (MTDF) of developmental activities in the province. The Medium-Term Development Framework lays down the developmental activities to be carried out in the province in various sectors of the provincial economy. In this manner, the Planning & Development Board is one of the main actors in the growth of the economic potential of the province.

The mandate of the Planning and Development Board includes provision of technical support and coordination to various Government departments in their planning activities. The

⁹ <https://pnd.punjab.gov.pk/>

department is also the main government agency working with foreign donors in the province.

Objectives: The main objectives of the department are as follows:

- Assessment of the material and human resources of the province
- Formulation of long- and short-term plans
- Recommendations concerning prevailing economic conditions, economic policies or measures
- Examination of such economic problems as may be referred to it for advice
- Coordination of all economic activities in the provincial government

Functions: The functions of the department are listed in the Rules of Business of the Government of Punjab. These include:

- Formulation of provincial government vision, policies and strategies for economic planning and development in consultation with all stakeholders in the light of NEC's guidelines
- Annual Development Programme (ADP)/Medium Term Development Framework (MTDF):
 - Preparation in co-ordination with all departments of the government
 - Monitoring implementation
 - Evaluation of development Projects and Programs
- Economic Issues:
 - Conducting research/surveys
 - Reviewing/analysis of socioeconomic data
- Public Sector Development Programs (PSDP):
 - Preparing short terms and long terms provincial development plans
 - Coordination with Federal government
- Policy for the approval of development schemes
- Catalyst for different departments/sectors to improve the pace and quality of economic development
- Resource allocation, re-appropriation of development funds, appropriations from block allocations and disbursement of supplementary grants
- Secretariat for the Provincial Development Working Party (PDWP) and clearing house for development within the competence of CDWP and ECNEC
- Foreign Assistance:
 - Determination of key areas for foreign assistance and preparation of sector-wise portfolio for foreign assistance
 - Loan negotiations and securing federal financial guarantees, wherever required
 - Review of foreign aided projects
- Coordination of nominations for foreign training, seminars, conferences and workshops for all officials serving with the provincial government
- Capacity building of government departments, agencies and functionaries for good governance
- Focusing accelerated development of rain fed (barani) and less developed areas
- Framing guidelines for procurement of consultancy services. Policy formulation with respect to private sector development and promotion and public private partnership (PPP)
- Implementation, development and administration in respect to foreign assisted/ funded and mega ADP projects

- Matters relating to attached departments, autonomous bodies and special institution of P&D Department
- Information Technology:
 - I.T. Policy
 - Electronic Data Management
 - Control of and liaison with district I.T. Departments
 - e-Governance and e-Service Delivery
 - Web Content Management
 - Pre-qualification of firms to provide I.T. consultancy, software development and I.T. products to the government
 - Coordination with both public sector departments and private sector agencies in the field of I.T
 - Service matters of I.T. cadre both at provincial and district level
- Budget, accounts and audit matters
- Purchase of stores and capital goods for the department
- Service matters except those entrusted to service and General Administration Department
- Administration of the following laws and the rules framed there-under:
 - The Cholistan Development Authority Act 1976
 - The Punjab Economic Research Institute Ordinance, 1980
 - The Punjab Public-Private Partnership for Infrastructure Act, 2010
- Matters incidental and ancillary to the above subjects

Attached Organizations

Attached Departments

- Bureau of Statistics Punjab
- Agency for Barani Areas Development (ABAD)
- Directorate General Monitoring & Evaluation

Companies

- The Urban Unit
- Engineering Consultancy Services Punjab
- Punjab Skill Development Fund (PSDF)

Autonomous Bodies

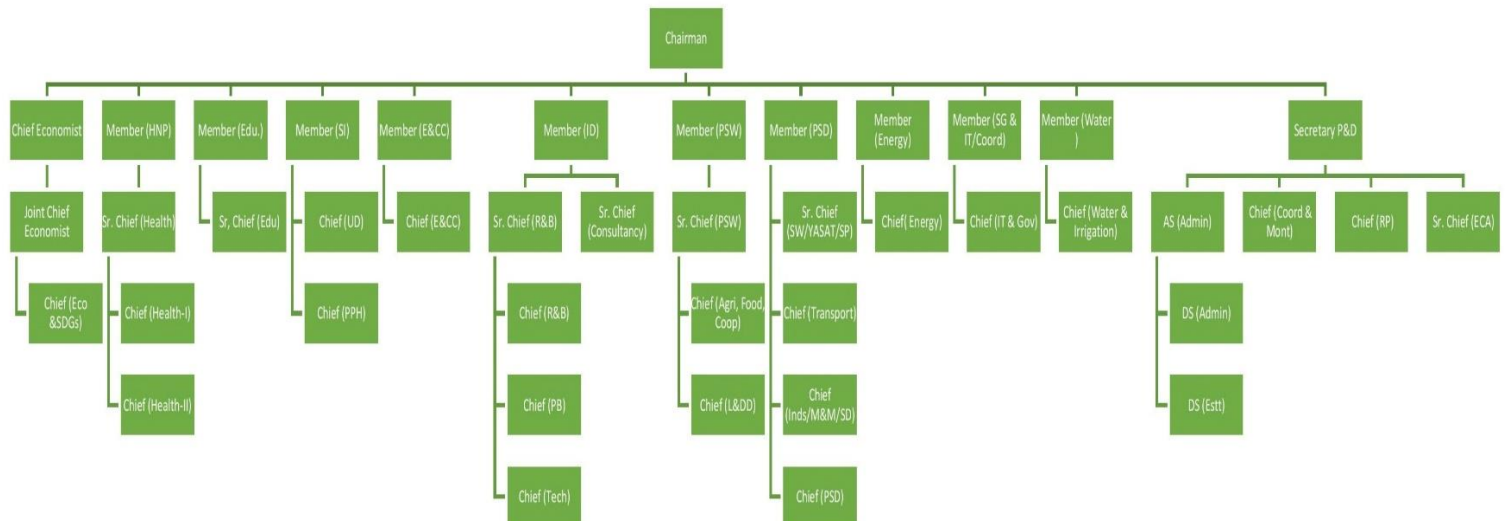
- Punjab Information Technology Board (PITB)
- Punjab Economic Research Institute (PERI)
- Punjab Public Private Partnership Authority
- Punjab Social Protection Authority (PSPA)
- Cholistan Development Authority

Project Management Units

- Punjab Resource Management Program (PRMP)

- Southern Punjab Poverty Alleviation Program
- Punjab Human Capital Investment Project (PHCIP)
- Koh-E-Suleman Improvement Project (KSIP)
- Punjab Tourism for Economic Growth Project (PTEGP)
- Punjab Public Private Partnership Cell
- Program Implementation Unit (PIU)

Organogram: The organogram of the organization is as under:



coordination & Monitoring Unit (CMU) is hereby established in Planning & Development Board for Punjab Clean Air Program (P-CAP) with the following composition:

- Program Coordinator
- Deputy Program Coordinator
- Program Manager

The Terms of the Reference (ToRs) of the interim CMU is as under:

- To facilitate initial groundwork, including preparation of concept paper and PC-I, and securing necessary approvals as per agreed timelines
- Liaison with all the implementing Departments / Agencies for their necessary input in preparation and approvals of concept paper and PC-t.
- To function as interim Coordination & Monitoring Unit (CMU) enabling the program to be launched promptly from day one till the establishment of regular Coordination & Monitoring Unit

Punjab Resource Management & Policy Unit (PRMPU), P&D Board will provide secretarial support to the interim CMU

Complaint Management: The website of P&D has an RTI tab with details for the Chief Public Information Officer who can be contacted for information or complaints. P&D projects financed by multi-lateral organizations have their own functioning GRMs. Complaints related to P&D may also be lodged through the provincial GRM channels.

Environmental Protection and Climate Change Department

The mandate of Environmental Protection and Climate Change Department (EPCCD) according to the PEPA 2012 is summarized as follows:

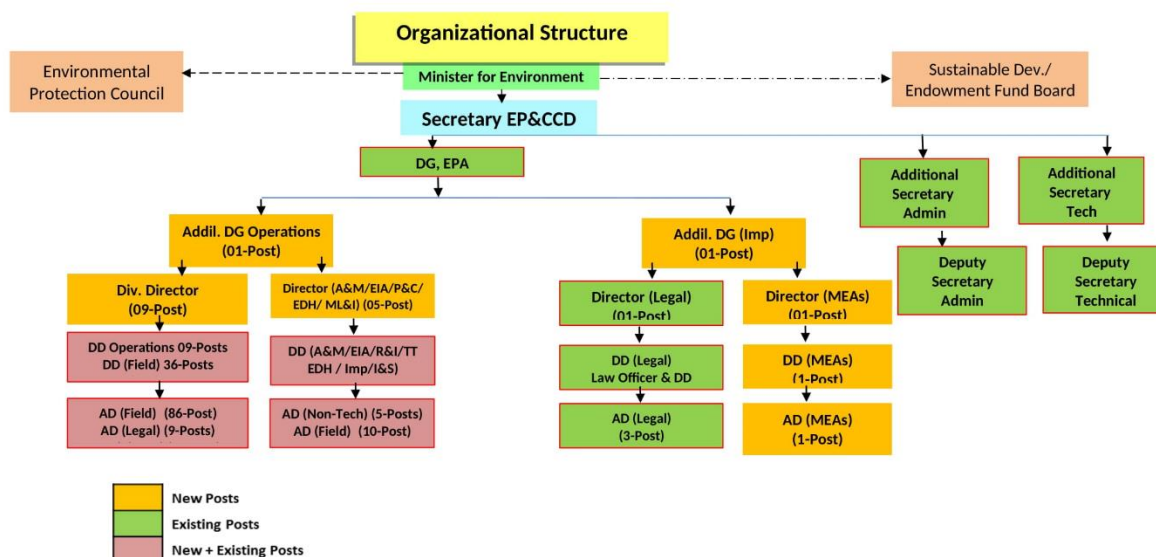
Implement rules and regulations prepared under PEPA 2012 and prepare additional legislation according to the needs of the province, prepare and implement provincial environmental standards, develop provincial systems for the implementation of pollution charges, conduct research and development for most viable environmental technologies, certify laboratories, engage LGs in the implementation of PEPA 2012, raise environmental awareness and incorporate environmental issues in educational curriculums, prepare provincial-level Environmental Disaster Management Plans (DMPs), collaborate and coordinate with stakeholders for the effective implementation of environmental policies and PEPA 2012, entertain inquiries and complaints raised by stakeholders, mobilize national and international financial resources for environmental projects, develop provincial-level fiscal programs and financial incentives for environmental compliance, fix pollution charges, conduct investigations against polluters, assist courts by generating field-level environmental data about polluters, establish environmental laboratories, implementation of IEE/EIA Rules and Regulations and Guidelines, manage hazardous waste under the Hazardous Substance Rules, and monitor vehicles for controlling air pollution.

Functions: The major functions of the EPCCD include:

- Administer and implement the provisions of Environment Protection Act and the rules and regulations made there under
- Prepare and establish the Punjab Environmental Quality Standards with the approval of the Council and Enforcement
- Take measures to promote research and development of science and technology which may contribute to the prevention of pollution, protection of the environment and sustainable development
- Identify the needs for, and initiate legislation in various sectors of the environment
- Provide information and guidance to the public on environmental matters
- Specify safeguards for the prevention of accidents and disasters which may cause pollution
- Encourage the formation and working of non-government, community, and village organizations to prevent and control pollution and promote sustainable development
- Take all necessary measures for protection, conservation, rehabilitation, and improvement of the environment, and for prevention and control of pollution
- Promotion of sustainable development

Organogram: The organogram of EPCCD is presented in below:

Environmental Protection & Climate Change Department, Punjab



Complaint Management at EPCCD: The EPCCD has established a website that gives access to air quality reports that are generated daily and are open to citizens to get an idea of the current Air Quality that exists in their district. This allows transparency and access to important environmental challenges that exist, to which the citizens should remain aware of at all times to safeguard Public Health (<https://epd.punjab.gov.pk/aqi>).

EPCCD has a Right to Information tab on their website which provides details for complaint management ([https://epd.punjab.gov.pk/right to information](https://epd.punjab.gov.pk/right_to_information)). EPA Punjab considers that for the enlightenment of citizens to hold public officers / officials accountable is their legal and ethical right. To maintain transparency, EPA Punjab has provided contact details of the Chief Information Officer which is 042-99231886 and the Public Information Officer 042-99231818. Citizens can also submit complaints directly to the office of Deputy Director Information and services EPA Punjab Lahore, by emailing on (ddisepa@punjab.gov.pk) or through the dedicated WhatsApp number 0335-9232227. The EPCCD also has a helpline number 1373 for public queries and complaints.

The EPCCD has also established a Complaint Cell under the administration section for the timely redressal of grievances received through various national and provincial complaint portals.

The Punjab Green Development Program (PGDP) also has a grievance redressal mechanism that resolves public grievances through the filing of complaints on environmental and social issues only related to the Punjab Green Development Program. To register a complaint, citizens would need to have their CNIC number and a mobile number. The GRM details are provided at their website <http://idm.pgdp.pk:8081/>.

Transport and Mass Transit Department¹⁰

Overview: Transport Department, Government of the Punjab was established in 1987. Earlier, it was a cell in the Services and General Administration Department (S&GAD) under the supervision of the Additional Chief Secretary, Government of Punjab. The establishment of the Transport Department is headed by Secretary with two Additional Secretaries, three Deputy Secretaries, and six Section Officers. In 2024, the department was renamed from "Transport Department" to "Transport & Mass Transit Department."

Vision: Sustainable transport system to support economic growth and improve mobility

Mission: Safe, affordable, and environment-friendly transport for the people of Punjab

Functions: Transport & Mass Transit Department under Punjab Government Rules of Business, 2011 performs the following functions:

- Legislation, transport policy, and planning
- Plan, design, operate, regulate and establish Mass Transit System in the province
- Route permits, fare / freights, matters relating to traffic speeds, loading, parking, and halting places, exemption cases of vehicles under Motor Vehicles Ordinance and Rules Grouping of stage carriages
- Inspection and checking of public service vehicles
- Payment of compensation in accident cases of private / public sectors, and allied matters
- Policy regarding student's concession and nationalization of road transport
- Inspection and certification of roads worthy vehicle
- Maintenance / management of public bus stands throughout the province
- Administration of motor vehicles examiners
- Budget, accounts and audit matters
- Purchase of stores and capital goods for the department
- Service matters except those entrusted to the Services and General Administration Department
- Administration of laws
- Matters incidental and ancillary to the above subjects

Attached Departments

Transport & Mass Transit Department has the following Attached Departments / Authorities along with a company and 36 subordinate offices at district level i.e. District Regional Transport Authorities.

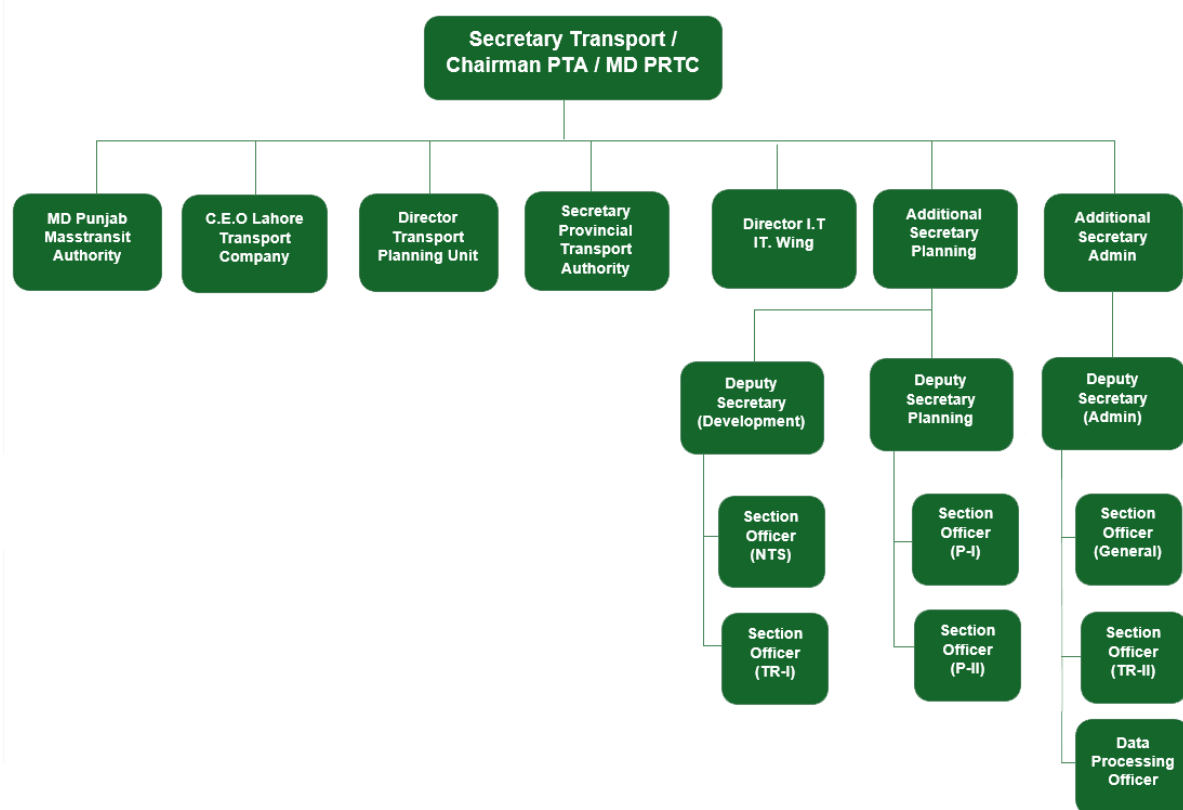
- Punjab Provincial Transport Authority (PTA)
- Punjab Mass Transit Authority (PMA)
- Transport Planning Unit (TPU)
- Punjab Transport Company (PTC)
- Punjab Road Safety Authority
- Punjab Road Transport Corporation (PRTC, Defiant)

¹⁰ <https://transport.punjab.gov.pk/>

Organogram: The organogram of the organization is as under:

Complaint Management:

- **Transport & Masstransit Department:** For any complaints regarding the Department, the Punjab Transport and Mass Transit website (<https://transport.punjab.gov.pk/complaints>) has a section for complaints listing two complaints helplines regarding Route Permits and Fare Overcharging: 042-99200484 and 042-99200329. The department can also be contacted through Whatsapp: 03210980980.
- **Punjab Masstransit Authority:** Helpline numbers for the metro bus systems in Lahore, Multan and Rawalpindi are provided at their website <https://pma.punjab.gov.pk/>. The helpline number for Lahore is (042-111-222-627) and may be used to lodge complaints. A helpline number 1762 is also available for quick dialing and is displayed in public transport operated by PMA.
- **Punjab Transport Company:** PTC has established a dedicated commuter support center for the assistance of commuters regarding public transport operations, transport cards and other queries related to fares and routes in Lahore. Any complaints / suggestions / general inquiries regarding public transport operations and transport cards can be addressed through their helpline (**111-582-111**). Public feedback is also accepted through email on complaints@ptc.gop.pk and through their public feedback page (https://ptc.punjab.gov.pk/commuter_support_center)



Agriculture Department¹¹

Functions: The main functions of Agriculture Department are as follows:

- Legislation, policy formulation and sectoral planning regarding:
 - Agriculture education, training and research including Agricultural University, Faisalabad and Pir Mehr Ali Shah University of Arid Agriculture Rawalpindi and pre-service/in-service training at Agriculture Training Institutes
 - Adaptive research and research farms
 - Improvement of agricultural and water management methods
 - Protection against insects, pests, prevention of plants diseases and quality control of pesticides
 - Soil fertility and soil conservation
 - Mechanization, reclamation of land, use of agricultural machinery ploughing, tube wells installation, and agricultural engineering research (agricultural machinery and implements), Water Management Training and Research Institute, Lahore
 - Agricultural information and publications / training
 - Agricultural statistics
 - Preparation and review of agricultural production strategy in coordination with district agriculture extension
- Arboricultural operations
- Monitoring of agriculture inputs like fertilizers, pesticides and irrigation through field extension staff
- Promotion of modern agriculture technologies and other extension activities through method / result demonstration, farmers gatherings, print and electronic media, etc.
- Training and research on Floriculture Seed Farms / Green Belts
- Market information and intelligence system and matters common to all Market Committees
- Agricultural loans / subsidies
- Water management operations, planning, research and coordination
- Production, multiplication and marketing of certified seed through Punjab Seed Corporation
- Coordination and strengthening of research activities in Agriculture, Livestock, Irrigation, Water Management, Forest and Fisheries sectors through Punjab Agricultural Research Board
- Economic planning and policy-making with respect to agriculture in the province
- Plant Protection:
 - Standardization of local and imported pesticides
 - Plant quarantine
- Economic studies for framing agricultural policy
- Farm management research for planning project formulating and evaluation in the province
- Grading of agricultural commodities other than food grains, for exports
- Agricultural commodity research (marketing research and laboratory research for laying

¹¹ <https://www.agripunjab.gov.pk/>

- down district, regional and provincial grades)
- Soil survey, comprehensive inventory of soil resources of province and their proper utilization
- Standardization of fertilizers for meeting provincial requirements
- Under-development areas:
 - Identification of under-development areas
 - Identification of the fields in which an area is under-developed
 - Measures necessary to remove the causes of under-development in different areas
- Socioeconomic studies for framing agricultural research policies
- Research for the introduction of improved germplasm, relating to agriculture
- Collection of statistics on agricultural research
- High level manpower training for agricultural research and on farm management
- Pest scouting, pest survey, pest warning, quality control of pesticides, research on plant protection, training of pesticides dealers, farmers and extension workers in plant protection
- Budget, accounts and audit matters
- Purchase of stores and capital goods for the department
- Services matters except those entrusted to Services and General Administration Department
- Administration of the agriculture related laws and the rules
- Matters incidental and ancillary to the above subjects

Services: The Agriculture Department is facilitating the farmers and agriculture sector through various services offered by its directorates categorically:

Directorate General Agriculture (Water Management)

- Water Users' Associations
- Watercourse Improvement
- Irrigation Schemes
- Precision Land Leveling
- High Efficiency Irrigation Systems
- Training and Capacity Building

Directorate General Agriculture (Field)

- Land Leveling
- Water Resources Development
- Soil and Water Conservation
- Research and Development
- Repair and Maintenance

Directorate General Agriculture (Research), AARI, Faisalabad

- Provision of Seed of New Varieties on Limited Scale
- Crop Production Technologies
- Testing of Soil, Water and Fertilizer Samples

- Analysis of Regulatory Fertilizer and Pesticide Samples
- Dissemination of Knowledge Through Farmers Advisory Services

Directorate General Agriculture (Extension and AR)

- Pre-Service and In-Service Trainings
- Transfer of Technology
- Production Technology Development
- Plant Clinics (Plant-wise)
- Biological Control Laboratories
- Subsidy on Seed and Fertilizer

Directorate General Agriculture (Pest Warning and Quality Control of Pesticides)

- Pest Scouting / Pest Surveys
- Forecasting of Pests and Crop Diseases
- Training of Farmers, Extension Workers and Pesticides Dealers
- Quality Control of Pesticides and Registration of Pesticides Distributors
- Screening / Standardization of Pesticides

Directorate General Soil Survey of Punjab

- Soil Surveys and Mapping (at Reconnaissance, Semi-detailed, Detailed and Ultra-detailed Level)
- Land Use Surveys
- Land Resources Inventories and Land Evaluation
- Land Use Planning
- Assessment of Agricultural Development Potential
- Physical / Chemical Analyses of Soils and Water (for Agriculture Purpose)
- Thematic Mapping (Land Use, Vegetation, Soil)
- Special Purpose Mapping (Suitability Maps)
- Training and Consultancy in Planning of Sustained Use of Land Resources
- Terrain Trafficability Studies

Punjab Agricultural Research Board

Funding Services

Directorate General Agricultural (Information)

- Technology Transfer through Print Media
- Technology Transfer through Electronic Media
- Technology Transfer through Social Media

Complaint Management: To resolve emergent problems of farmers, toll-free Punjab Agriculture Helpline 0800-15000 and 0800-29000 have been installed / activated in the Directorate, which remain active 12 hours daily from 08:00 a.m. to 08:00 p.m. The facility for

computerized recording of calls of farmer and display of callers ID with date and time at the helpline facilitates in locating callers for prompt feedback by technical experts on the same day. This facility has proved instrumental in improving knowledge and capacity building of farmers with enhanced confidence in Agriculture Department. The farmers are making best use of this helpline facility for seeking technical guidance about modern farm technologies as presently more than 12,000 telephone calls are being received / recorded annually to whom response / feedback from this Directorate is extended on the same day. Details are provided at their website <https://www.agripunjab.gov.pk/directory>.

Directorate of Agricultural Information, Punjab has also initiated SMS Helpline Service to extend technical guidance to the farmers using mobile phones. Farmers may send SMS at 0304-4000172 from any cellular network for seeking information and guidance to resolve their field problems. This initiative has further supplemented agricultural technology transfer efforts of the Directorate and farmers are becoming aware as how to benefit from scientific advancements, Government initiatives and facilities under development projects. Details are provided at <https://www.agripunjab.gov.pk/node/690>.

The extensive field network of the Department's Agriculture Extension Wing provides farmers with access to department staff for information, guidance, complaints and services.

Lahore Development Authority (LDA)¹²

Vision: To transform Lahore into a world class city through efficient public service delivery.

Mission: To provide quality public services through socio economic programs, urban development and efficient mobility while preserving cultural and heritage dynamics of Lahore.

Wings: The Lahore Development Authority was created under the LDA Act 1975 duly approved by Punjab Legislative Assembly. Previously it was Lahore Improvement Trust. There are three wings in Lahore Development Authority i.e., Traffic Engineering Planning Agency (TEPA), Urban Development, and Water and Sanitation Agency (WASA).

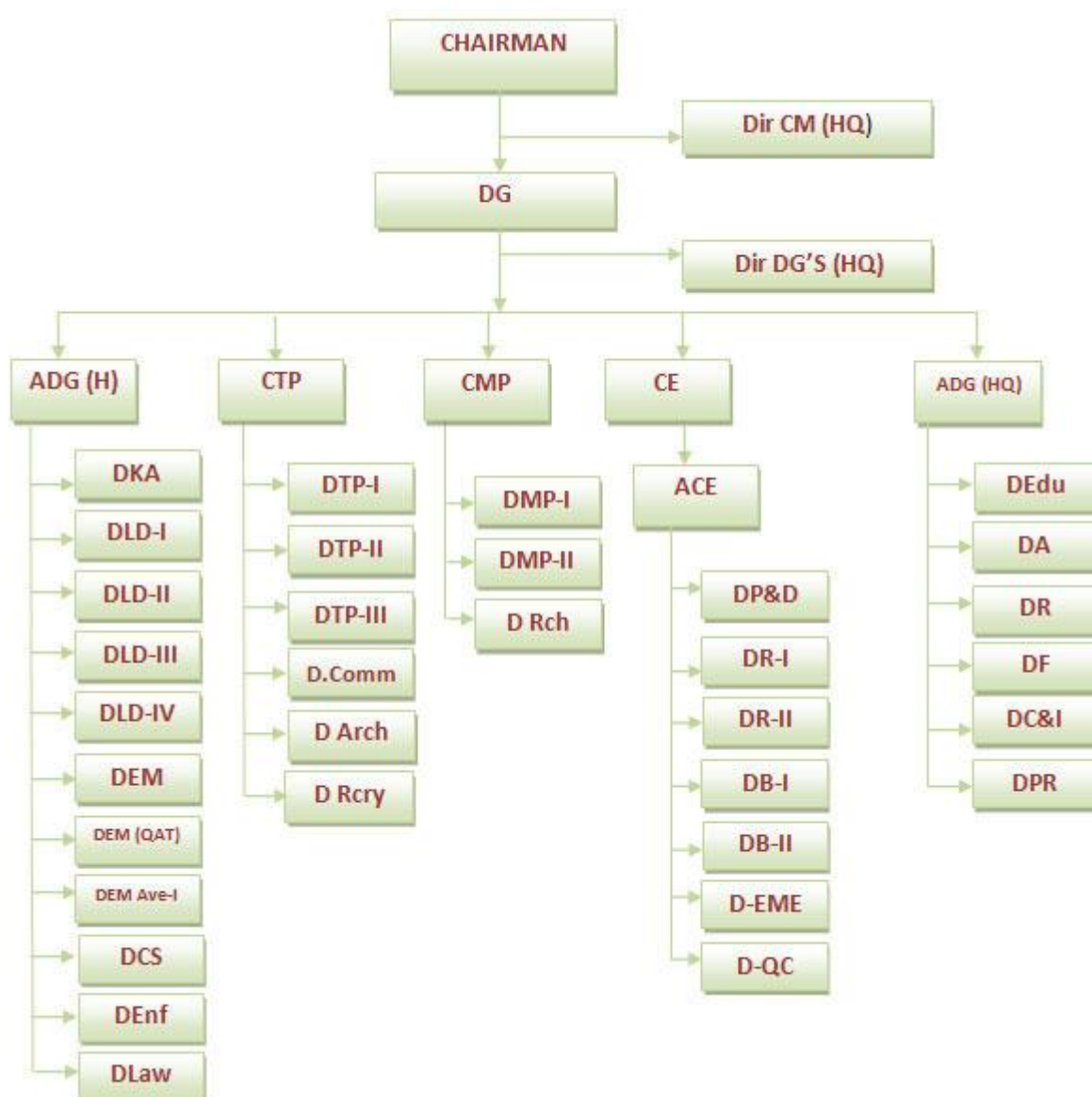
Goals and Targets: Following are the goals and targets of the LDA:

- Integrated and sustainable development
- Quality housing
- Provision for low-cost housing
- Rehabilitation of underdeveloped areas/slums
- Maintain natural and aesthetic beauty of the city
- Complaint redressal system (One Window Cell)
- Promote environmentally friendly activities through:
 - Potable drinking water
 - Sanitation and drainage
 - Wastewater treatment plants
 - Rainwater harvesting and water conservation
 - Promote urban forestry and plantation
 - Integrated traffic management system

¹² www.lda.gov.pk

Organogram: The organogram of LDA is given as under:

Complaint Management: LDA has an online Portal (<https://lda.gov.pk/ldaonline/login.html>) through which complaints can be filed to respective departments within the organization. For online complaints, an account is required for which a citizen must provide their CNIC number and mobile number.



Abbreviations:-

DG : Director General, ADG(H) : Additional Director General (Housing), DKA : Director Kachi Abadi, DLD : Director Land Development, DEM : Director Estate Management, DEM(QAT) : Director EM Quaid-e-Azam Town, DEM Ave : Director Estate Management Avenue, PDIT : Project Director Information Technology, DCS : Director Computer Service, DE : Director Enforcement, DHP : Director Hidden Properties, DLaw : Director Law , CTP : Chief Town Planner, DTP : Director Town Planning , DComm :Director Commercialization, D Arch : Director Architecture, D Rcry : Director Recovery, CMP : Chief Metro Politain Planner, DMP : Director Metro Politian Planning, D Rch :Director Recovery, CE : Chief Engineer , ACE : Add Chief Engineer , DDHQ : Deputy Director Head Quarter , DR : Director Revenue , DB : Director Building , DEME : Director , DP&D :Director Planning & Development, ADG (HQ) : Additional Director General (Head Quarter), DEdu : Director Education, DA :Director Admin, DF : Director Finance, DPR :Director Public Relation, C&I : Co-ordination & Implementation

Traffic Engineering & Transport Planning Agency (TEPA) LDA: It is the premier organization for developing Traffic & Transport infrastructure in the provincial capital, Lahore city. The scope of TEPA has been widened to Lahore Division (that now includes Sheikhupura, Nankana Sahib and Kasur). TEPA with coordination with JICA (Japan International Cooperation Agency) developed the first Comprehensive Study on Transport System in Lahore in 1991. The Transport Master plans were then revised in 2011, 2012 and Transport Master plan vision 2021.

Industries, Commerce, & Investment and Skills Development Department¹³

Overview: The Industries, Commerce & Investment and Skills Development Department is one of the major government institutions striving to promote industrial development, trade and investment in the province. The main focus of activity is promotion of trade and investment in the province. The Government of the Punjab is keen on creating a business-friendly investment climate in line with the Federal Government Policies and present the province of Punjab as an attractive investment destination for the entrepreneurs / investors.

The department works with the vision to encourage & promote industries including cottage industry for sustainable growth, credit facilitation including micro financing, facilitation in trade and business, availability of skilled manpower, consumer protection and co-ordination in price control.

Mission: The mission of the department is to create prosperity by strengthening the competitiveness of Punjab business environment, promoting trade by all means including competitive and efficient export supply chain and investment climate that helps develop the private sector and addressing issues related to fair trade and compliance with international trade regime.

Objectives: Enlisted below are some of the major objectives of the department:

- Promotion of industry and investment in the province
- Advocacy and implementation of trade and investment policies of the federal/ provincial governments
- Liaison with private sector, trade bodies and trade associations
- Creation of awareness about WTO related trade laws & standards
- Improve product competitiveness and export supply chains
- Conduct research and provide technical expertise to private sector
- Regulate prices of essential commodities

Functions: Following are the key functions:

- Legislation, policy formulation, and sectoral planning in respect of
 - Industries including industrial estates, small industries and handicrafts enterprises
 - Promotion of domestic, commerce, foreign and domestic investment
- Advocacy vis-à-vis federal policies that impact provincial economy/commerce and industry

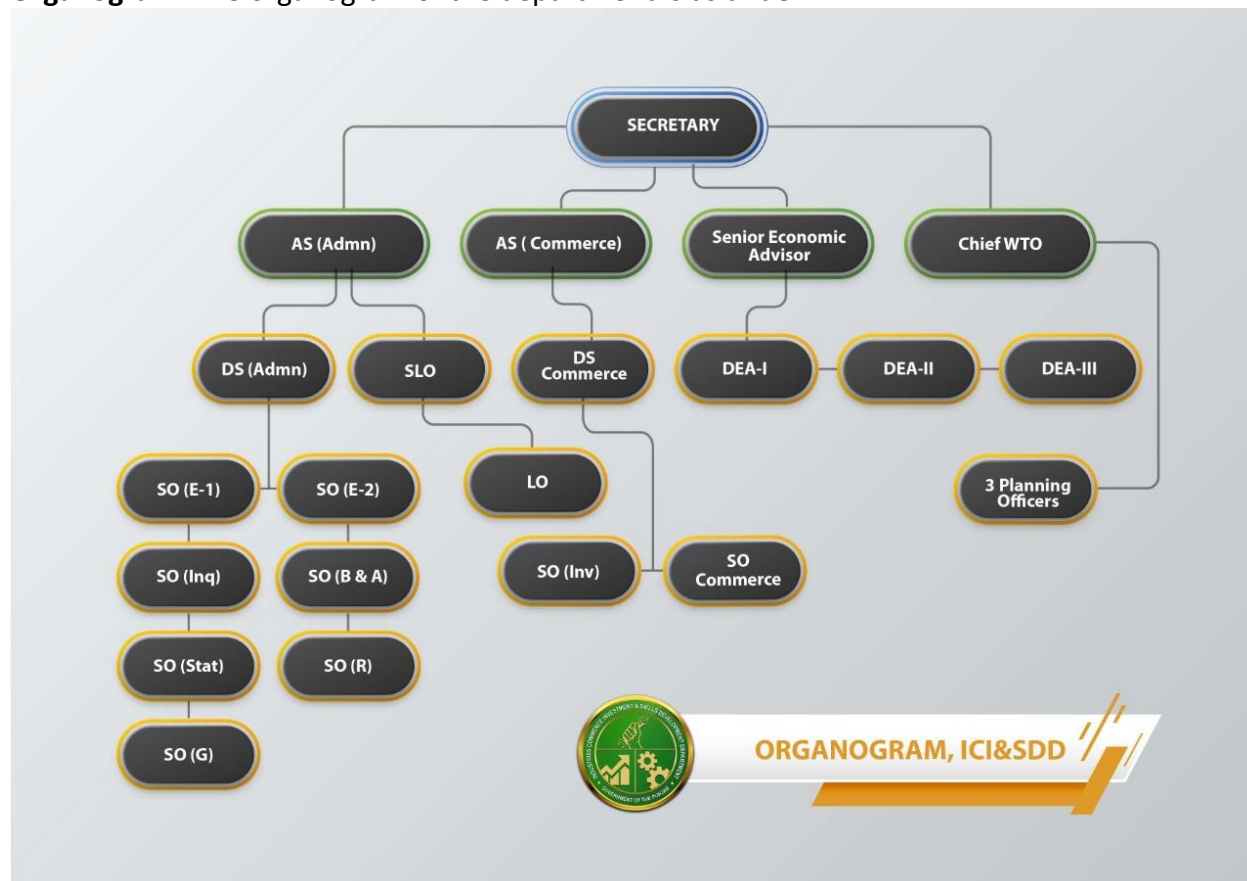
¹³ <https://icid.punjab.gov.pk/>

- Inter-provincial trade policy
- Coordination and facilitation of Chambers of Commerce & Industries and Trade Organizations in the province with respect to capacity building and business promotion through R&D
- Liaison with respective federal government agencies in matters pertaining to intellectual property rights
- Coordination with the federal government in matters relating to industrial development, commerce and investment
- Promotion of commerce & investment in the province through
 - Trade exhibitions
 - Seminars
 - Encouragement of foreign and domestic investment.
- Facilitation of foreign business delegations and investment
- Promotion of domestic commerce and trade within and outside the country through:
 - Facilitation of local exporters (abroad)
 - Representation of local industries in industrial fairs and exhibitions abroad
 - Facilitation of exports from the province, including programs to enhance competitiveness
- Promotion of industries through
 - Loans / Public Private Partnership
 - Industrial surveys
 - Pre-investment studies
 - Technical and business support for cluster development
 - Industrial fairs & exhibitions
- Land acquisition for industries and industrial estates
- Regulation of industrial location policy
- 14A Establishment of petrol pumps, CNG stations, LPG bulk storage decanting and incidental matters and matters relating to illegal sale or storage of petroleum products
- 14B Matters relating to weight and measures including:
 - Verification and inspection of petrol pumps, CNG stations, super inspections of lower tiers
 - Monitoring of all work related to weights and measures
 - Verification of calibrated tanks, flow meters, fillers and dip tanks
 - Verification and inspection of weights and measures at shops and commercial establishments
 - Verification and inspection of weights and measures in factories and other establishments
 - Matters relating to boilers and pressure vessels excluding those used in nuclear energy
 - Registration and regulation of companies, firms and societies
- Printing and stationery:
 - Establishment and budget of Provincial Printing Presses (Lahore & Bahawalpur)
 - Monthly income and expenditure statements
 - Printing and binding
 - Printing and Stationery Manual
 - Local purchase of stationery
- Matters relating to Technical Education and Vocational Training Authority (TEVTA)
- Policy formulation and administrative matters related to Attached Departments,

Autonomous Bodies and Special Institutions of the Department

- Budget, accounts and audit matters
- Purchase of stores and capital goods for the department
- Service matters except those entrusted to Services and General Administration Department
- Administration of the governing laws and the rules framed there-under

Organogram: The organogram of the department is as under:



Punjab Energy Efficiency and Conservation Agency¹⁴

Overview: The Government of Punjab, through its Energy Department, has established the Punjab Energy Efficiency and Conservation Agency (PEECA) for monitoring, implementing policies, regulations, running behavioral change, public awareness programs and training of energy management professionals. We work as designated agency under NEECA Act 2016 to create an environment which is conducive for the transformation of our society from an energy wasting to an energy efficient and conserving one. Simultaneously, to divert the market towards energy efficient solutions and to mitigate the existing inefficiencies in the system, market would be triggered through various energy efficiency and conservation initiatives by the Government of Punjab that would concurrently address the legal/policy/regulatory/financial/market issues to pave way for the private sector to mature and subsequently take the lead.

¹⁴ <https://peeeca.punjab.gov.pk/>

Functions

- Policy & Regulations formulation and the enforcement of the regulations in the Punjab province for conservation of energy in the light of the ENERCON's Vision and policies.
- Collaborate with the Federal Government to develop both regulatory and policy framework to enable conservation of precious energy by cutting down on line losses and other creative solutions. In addition to that, take necessary steps for the inclusion of net metering in the existing policy/legal framework regarding energy purchase and distribution.
- Develop and enforce creative metering solutions to conserve energy by mitigating theft, which is otherwise lost and takes its toll on national economy in the form of subsidy provided by the government.
- Specification and categorization of consumers in terms of their efficient use of energy.
- Synchronize efforts with ENERCON to standardize and label the energy efficient electrical equipment being produced and imported in Punjab.
- Impose regulations on the market to keep the inefficient electrical equipment off the market.
- Inter-agency co-ordination for the effective formulation and implementation of the regulations to help conserve energy e.g. Energy Conservation Building Codes.
- Build in-house capacity through training programs in regulations and policy development as well as in assessing the efficiency of the electrical equipment.
- Running promotional campaigns to increase awareness about the potential of saving energy through energy conservation.
- Institute provincial energy conservation/efficiency and management awards for various categories of energy consumers for the promotion and encouragement of energy conservation.
- Introducing and developing energy performance ratings considering the level of energy efficiency of buildings and projects, starting from one star for the least energy efficient building/project to five stars for buildings/projects that are self-sustainable and do not need electricity from the grid and conducting energy performance audits.

ANNEX 4: PROCEEDINGS OF THE CONSULTATION WORKSHOP

The consultative workshop for the Draft ESSA was conducted on 30th January 2025 from 3-6pm at the Planning & Development Board AKS Auditorium. The agenda of the workshop, feedback received, and list of participants is provided below.

Agenda:

Time	Session	Responsibility/Presenter
3:00pm to 3:30pm	Arrival and registrations	P&D
3:30pm to 3:45pm	Welcome by P&D	Ahmad Rajwana Project Director, PRMPU, P&D
3:45pm to 4:00pm	Introduction to PCAP and description of program activities – PforR and IPF	Shafick Hoossein Senior Environmental Specialist & TTL P-CAP World Bank
4:00pm to 4:30pm	Presentation on Draft ESSA Findings and Recommendations; and Stakeholder Engagement Plan	Mehrunisa Malik – Social Development Consultant World Bank Shafqat Khan – Environment Consultant World Bank
4:30pm to 5:15pm	Feedback from stakeholders	P&D and World Bank
5:15pm to 5:30pm	Closing Remarks	Ahmad Rajwana Project Director, PRMPU, P&D
5:30pm	Refreshments	

Feedback from Participants

Comments/Questions	Responses
Mariam Shah – Pakistan Air Quality Initiative (PAQI) PAQI is a CSO conducting research and awareness activities on air pollution in Pakistan. Our work is data centric and scientific. We appreciate that PCAP is giving attention to an inclusive design and stakeholder engagement. We have a set of recommendations (separately available) for PCAP that include the following: • Air Quality Monitoring & Data Transparency – Expansion of monitoring stations must be scientifically planned to ensure comprehensive coverage and real-time public access. • Regulation of Industrial & Vehicular Emissions – Punjab must establish clear, enforceable limits on industrial and vehicular emissions to reduce PM2.5 pollution effectively. • Public Awareness & Stakeholder Engagement – Ensuring that air quality data is real-time accessible	The recommendations from PAQI are appreciated and well received. Bringing all relevant stakeholders on board is important for PCAP and measures will be taken ensure this during the program. This will also be ensured through the Stakeholder Engagement Plan.

and actionable for both policymakers and the public is crucial for behavioral change. • Institutional Capacity for Air Quality Management – The Environmental Protection & Climate Change Department (EPCCD) must be strengthened with technical expertise, funding, and enforcement mechanisms. PAQI stands ready to support the Punjab Clean Air Program by: • Providing independent data analysis to validate emissions inventories and air quality trends. • Contributing to policy development for air quality monitoring regulations. • Designing and implementing community engagement programs to enhance public awareness of air pollution risks.	
Sughra Kashif - Social Specialist PGDP • The program has many environmental and social risks. How can it be categorized low risk? • Site selection of e-bus depots should be very carefully done to ensure minimal E&S impacts. • Learnings from E&S activities and performance of PGDP must be taken into consideration and built upon for PCAP. • Will the design of the program be inclusive for persons with disabilities? • Which World Bank ESF standards are applicable for PCAP? • It must be ensured that bus depots have separate toilets for women and persons with disabilities. • Must ensure that periodic TPV of ESSA is conducted.	All questions and comments were well noted and responded to by the ESSA team.
Adil Farooq Khan - Progressive Farmer and CEO of Dawood Agro Company I am the pioneer of rice mechanization initiative of Pakistan. For success and maximum impacts to air quality of super seeders, the initiative should be concentrated in targeted areas / hot spots such as Gujranwala Division and Lahore Division which grow 50% of the rice crop of Punjab. To include unskilled labor, PCAP should consider including rice transplanter machines. This will also benefit women.	Comments are well noted. PCAP will focus on identified hot spots for crop burning for maximum impact. Due to limited resources, transplanters are not supported by PCAP. However, they are part of the medium- and long-term plan under a project with ADB.

PCAP should provide incentives for e-bikes manufacturing as well, as e-2 and 3 wheelers have a very usage.	
Irfan Gondal - DD Planning Social Welfare Dept - Disabilities Act provides clauses on accessibility of public transport for persons with disabilities. The implementation needs to be ensured. - Will disability inclusion provisions under PCAP be expanded to general transport as well? - There is no inclusion of persons with disabilities in the consultations, which is disappointing.	SOPs for implementation of Disabilities Act in PCAP activities will be prepared and implemented. However, they will only apply to PCAP financed activities.
Asif Bajwa Chairman PAMIMA Govt of Punjab is supporting purchase of agricultural machinery by farmers at subsidized rates. Similarly solar systems are being provided to farmers. None of these benefits are extended to manufacturers of agricultural implements. Please consider this. Sales and income tax exemptions for manufacturers of super seeders may also be considered.	Comments are received and well noted. Solar units will be provided to industries under PCAP. Taxation is beyond the purview of PCAP.
Rashid Ahmad – WWF Pakistan Which department will prepare and implement the e-waste rules?	
Hassam Ali – WWF Pakistan There is a lot of activity on AQM by the government, private sector and NGOs. But they are all working in silos. How do we bring all of them together and focus on the hot spots that have been identified? E-buses – will there be a separate infrastructure for e-buses?	The SEP will include activities to bring together all stakeholders involved in AQM. PCAP may consider a platform or working group on this.

Participants List

Sr.#	Name	Designation/ Department
1	Ahmad Rajwana	Project Director PRMP
2	Mehrunisa Malik	Social Development Consultant World Bank
3	Shafqat-Ullah	Environmental Consultant- World Bank
4	Azher Uddin Khan	Environmental Consultant World Bank
5	Noureen Arif	Manager environment and social safeguards, Energy Department
6	Sughra Kashif	DD Social Safeguards PGDP
7	Hina Ishaque	Environmental Specialist Punjab Small Industries Co-operation (PSIC)

8	Rukhsana Shahid	Environmental Specialist PMDFC
9	Shahid Imtiaz	Consultant accounts PRMPU
10	Hira Ashraf	Assistant Manager (Program)/ PEECA
11	Abid Bodla	Member (ID) - P&D
12	Jamshed Bashir	Assistant Director Special Education Department
13	M.Irfan Gondal	DD(P), Social Welfare & Bait ul Maal Department
14	Aadil Farooq Khan	CEO Dawood Agro Company-Progressive farmer
15	Engineer Sajid Nasir	DGA(F)P, Agriculture Department
16	Rashid Ahmad	Manager, WWF Pakistan
17	Syed Hassam Ali	Senior Officer-Sustainability and Climate Change WWF Pakistan
18	M.Ubaid Iqbal	Director Environment LDA
19	Nasir Iqbal	Assistant Chief P&E cell, Agriculture Department
20	Wasim M.	Director (planning) Field Wing of Agriculture Department
21	Faisal Hafeez	AD (AFD project)
22	Farah Said	Department of Economics (LUMS)
23	Sadiq Ahmed	PO (Consultancy I) P&D
24	Azhar Hussain	Director Industries Department
25	Mariam Shah	Communication Specialist Pakistan Air Quality Initiative (PAQI)
26	Raja Akshaf	PO(ECA-IV), P&D
27	Saarya Haider	DS(Dev.), Transport Department
28	Hassan Ahmed	Assistant Manager/ PEECA
29	Asif Bajwa	Chairman - Pakistan Agricultural Machinery and Implements Manufacturers (PAMIMA)
30	Abid Omar	Founder - Pakistan Air Quality Initiative (PAQI)
31	Zeeshan Mustafa Maan	Environment Specialist PRIAT, Agriculture Department
32	Name Unknown	Environment Specialist PAHP
33	SPIU EPCCD	
34	Rizwana	Online participant
35	Noman Ashraf	Online participant

Photos from the Workshop





GOVERNMENT OF THE PUNJAB
PLANNING & DEVELOPMENT BOARD
EXTERNAL CAPITAL ASSISTANCE WING

List of Participants
30.01.2025 at 03:00 PM

Subject: INVITATION TO THE ENVIRONMENT AND SOCIAL SYSTEMS ASSESSMENT (ESSA) WORKSHOP FOR PUNJAB
CLEAN AIR PROGRAM IN LAHORE (JANUARY 30, 2025) (AKS Auditorium)

Sr. No.	Name	Designation/ Department	Office #/ Mobile #	Signatures
1.	Mehrunisa Malik	Social Dev. Cons. WORLD BANK	[REDACTED]	[Signature]
2.	SHAFQAT ULLAH	Env. Consultant WBS	[REDACTED]	[Signature]
3.	Abheruddin Khan	Env. Consultant WBS	[REDACTED]	[Signature]
4.	NOUREEN ARIF	Manager Environment & Social Safeguards	[REDACTED]	[Signature]
5.	Sugra Kabir	(D) Social Safeguards PSDP	[REDACTED]	[Signature]
6.	Hina Ishaque	Environmental Specialist Punjab Smart Industries Corporation (PSC)	[REDACTED]	[Signature]
7.	Rukhsana Shaukat	Environmental Specialist PMDFC	[REDACTED]	[Signature]

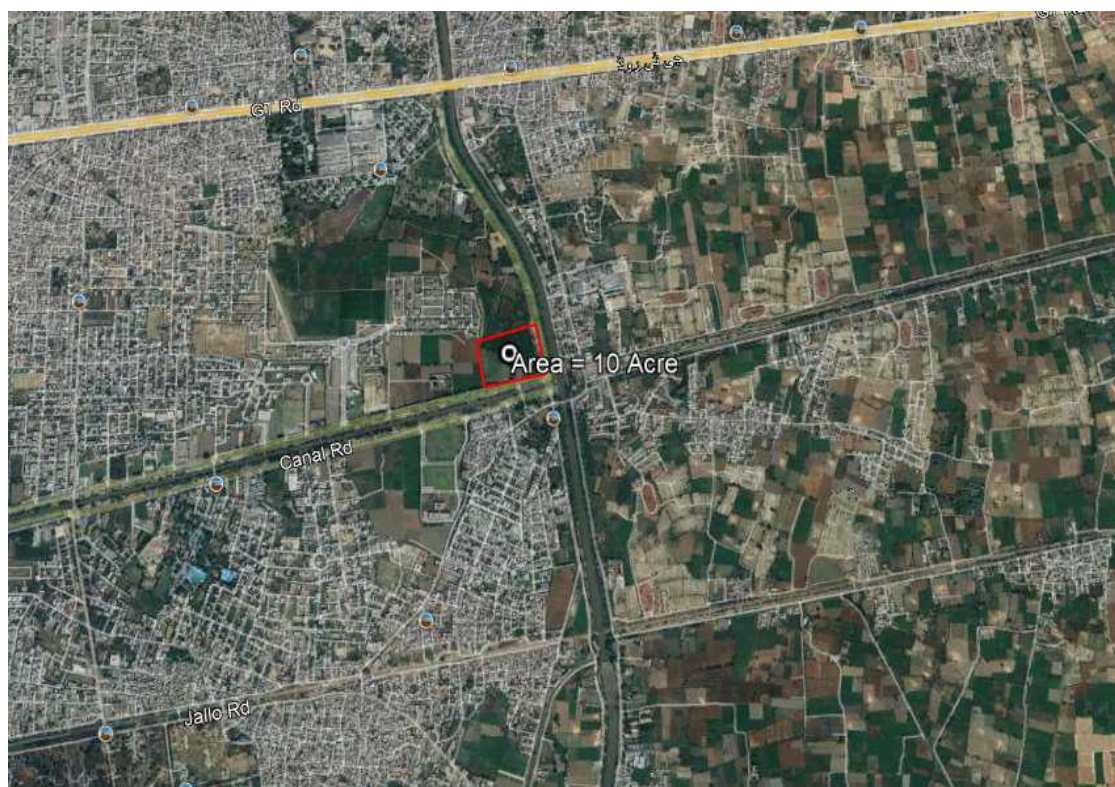
Sr. No.	Name	Designation/ Department	Office #/ Mobile #	Signatures
8.	Shahid Amir	(PRMPD) Consultant Accounts	[REDACTED]	[Signature]
9.	Hira Aswaf	Assistant Manager (Program) / PEECA	[REDACTED]	[Signature]
10.	Hassan Ahmed	Asst. Manager / PEECA	[REDACTED]	[Signature]
11.	Abid Booda	Member (ED), P&D	[REDACTED]	[Signature]
12.	Jamshed Bashir	Assistant Director Sped.	[REDACTED]	[Signature]
13.	M. J. Farhan Gondal	DD(P), SW & BM	[REDACTED]	[Signature]
14.	Aadil Fawaz Khan	C.E.O Dawood Agro Company / Progressive Farm	[REDACTED]	[Signature]
15.	Engr. Sajid Naveed	DGA(FP), Agri. Deptt.	[REDACTED]	[Signature]
16.	RASHID AHMED	Manager	[REDACTED]	[Signature]
17.	SYED HASSAM ALI	Senior Officer-Sustainability and CC	[REDACTED]	[Signature]
18.	M. Ubaid Iqbal	Dir. Environment L&A	0333-9116161	[Signature]

Sr. No.	Name	Designation/ Department	Office #/ Mobile #	Signatures
19.	Nasir Iqbal	Assistant Chief P&E Cell, Asst. Dept.	[REDACTED]	[Signature]
20.	Wasim M.	Director (Planning) Field wing of Asst. Dept	[REDACTED]	[Signature]
21.	Faisal Hafeez	AD (AFD Project)	[REDACTED]	[Signature]
22.	Farah Said	Department of Economics EIMS		Farah.
23.	Sadiq Ahmad	PO (Consultancy I) Fed	818 8825588	Sadiq
24.	Azhar Hussain	Director Industries Dept.		
25.	Nasim Shah	Communication Specialist		
26.		Anti Inequality Initiative		
27.	Raja Akshaf	PO (ECA- IV)		[Signature]
28.	Saarya Haider	DS (Dev), Transport Dept.		

ANNEX 5: PROPOSED/TENTATIVELY IDENTIFIED PRIVATE LANDS FOR BUS DEPOTS

Sr.No	Location	Area (Acre)	Coordinates	
			Latitude	Longitude
1	Near Khaira Bridge BRB Canal	10	31°35'15.05"N	74°29'57.20"E
2	Near Gajjumata Interchange along Hudhara Drain	9	31°23'23.51"N	74°21'16.99"E
3	Near Muridwal Village M-2 Link	8.97	31°29'21.52"N	74°13'45.87"E
4	Near Barki Village	5.88	31°28'21.94"N	74°30'58.37"E

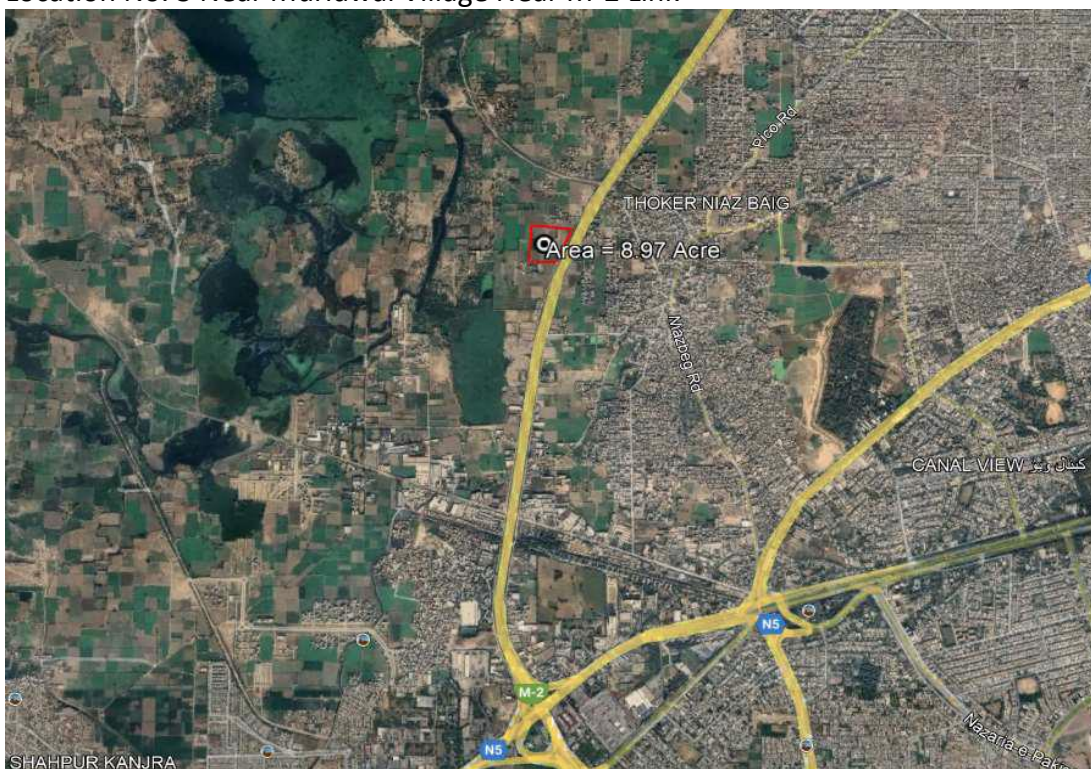
Location No. 1: Near Khaira Bridge BRB Canal



Location No. 2 Near Gujjumata Interchange Near Hudiara Drain



Location No. 3 Near Muridwal Village Near M-2 Link



Location No.4 Near Barki Village

